

Recent Facts about Photovoltaics in Germany

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Compiled by:

Dr. Harry Wirth
Division Director Power Solutions

Contact:

Sophia Judith Bächle
Communications
Telefon: +49 (0) 7 61 / 4588 5215
Fraunhofer Institute for Solar Energy Systems ISE
Heidenhofstrasse 2
79110 Freiburg, Germany
presse@ise.fraunhofer.de

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1 What purpose does this guide serve?

Germany is leaving the fossil-nuclear age behind, paving the way for photovoltaics (PV) to play a central role in a future shaped by sustainable power production. This compilation of current facts, figures and findings is regularly updated. It aims to help create an overall assessment of the progress in the PV expansion in Germany.

2 How much photovoltaics is needed for the energy transition?

In order to cover our entire energy demand from renewable energies (RE), a massive expansion of installed PV capacity is necessary, in addition to a number of other measures. Figure 1 shows the required nominal PV power according to a selection of studies and scenarios from the year of publication 2021 ([ISE3], [ISI], [DIW], [ARIA], [BDI], [ESYS], [Prog], [IEE], [HTW2], [ÜNB2], [AGORA], [Aurora]).

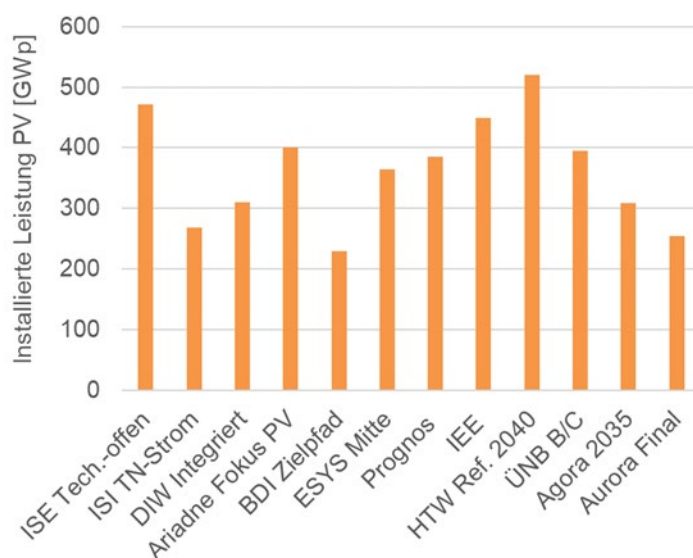


Figure 1: Installed PV capacity necessary to reach climate neutrality as determined by different studies. (The studies are named according to the institutions and, if applicable, the scenario).

The scenarios make different assumptions on boundary conditions, e.g., for energy imports, efficiency increases and acceptance. Some studies only consider the electricity system, others the entire energy system.

The EEG 2023 envisages a PV expansion to 215 GW_p by 2030 and to 400 GW_p by 2040. The annual net addition is to climb to a maximum of 22 GW_p by 2026. Increasingly, old installations also need to be replaced.

These replacement installations are currently of little significance, but they will increase to approx. 15 GW_p per year at full target capacity, assuming a useful system life of nearly 30 years. In the years 2013–2018, an average of only 1.9 GW_p/a was installed in Germany (Figure 2). With the deployment of **17 GW_p** in 2025, the installed capacity now is

118 GW_p DC [SPE], distributed across approximately **5.5 million systems** [BSW], including plug-in solar units. The approximately 8% lower installed AC rated power is relevant for grid feed-in.

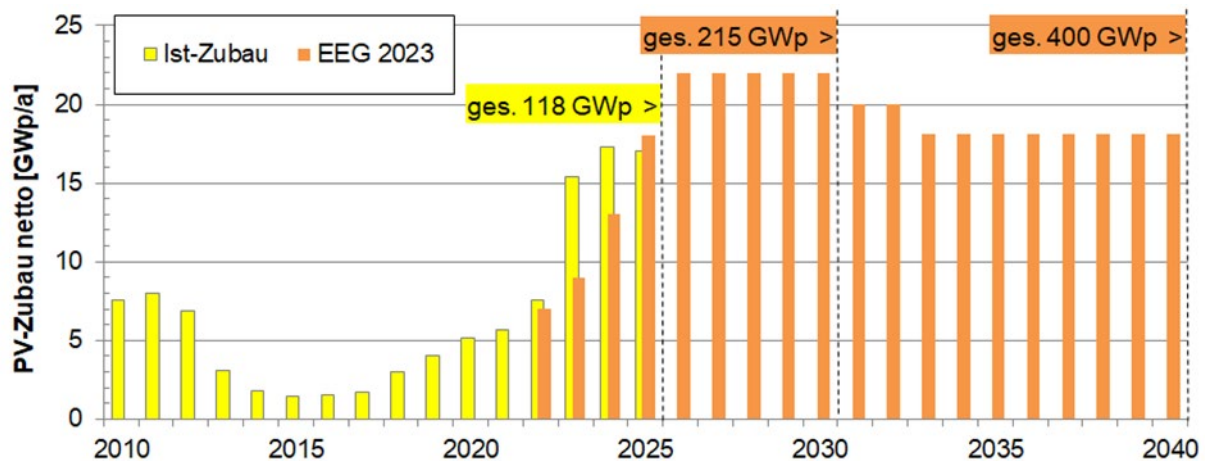


Figure 2: Annual net PV expansion: Actual values until 2025, expansion path to meet the legally binding goals; Actual values from [EEG2023], [ISE4].

3 Does PV contribute significantly to the power supply?

Yes.

With an electricity generation of **92 TWh** in 2025, photovoltaics covered **17 percent** of gross inland electricity consumption [BDEW2] in Germany (Figure 3). All renewable energies (RE) together covered a total of **55 percent**.

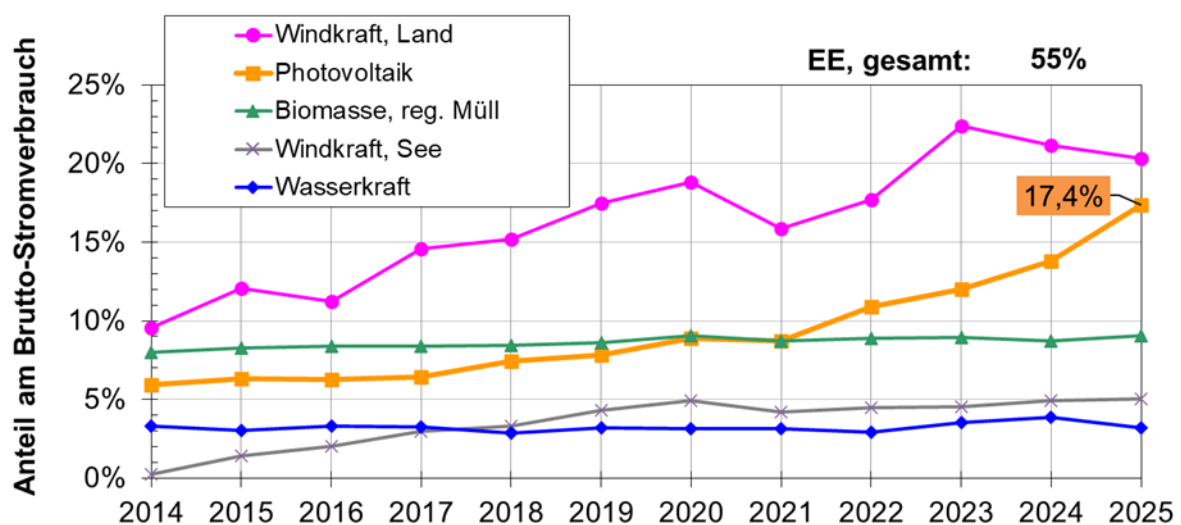


Figure 3: Development of the share of renewable energies in gross electricity consumption in Germany, Values from [ISE4], [AGEE].

Gross electricity consumption is the reference value for the statutory expansion targets of the energy transition and includes grid and self-consumption losses (Section 28.9). On sunny days, PV electricity can temporarily cover our entire electricity demand over several hours. Based on a forecasted gross electricity consumption of 658 TWh in 2030, the planned PV expansion to 215 GW_p would lead to a PV power share of about 30 percent, with renewable energies generally covering 80 percent.

4 Is PV power too expensive?

PV electricity was once very expensive.

If one compares the electricity production costs of new power plants of different technologies, PV comes off very favorably [ISE1]. Large PV power plants in particular produce unrivaled cheap electricity. However, the cost comparison is still considerably distorted as long as the supply costs for fuels are considered, but neither the full cycle costs with CO₂ recovery nor the follow-up costs of interrupted cycles, i.e., the costs of the climate crisis. The marginal costs for nuclear power are in the order of 1 ct/kWh, for coal-fired power 3–7 ct/kWh, for gas-fired power 6–9 ct/kWh, plus the fixed costs of the power plants (e.g., investment, capital). The marginal costs essentially cover the provision of the fuel, but not the neutralization of the radiating waste or environmentally harmful emissions (CO₂, NO_x, SO_x, Hg). So far, external costs and risks relating to environmental, climate and health damage have largely been disregarded in pricing ([UBA3], [FÖS1], [FÖS2]). The ignoring of these external costs represents a massive subsidization of the energy sources concerned (Section 5.2).

In order to promote the energy transition and to stimulate investments in PV systems of various sizes, the Renewable Energy Sources Act (EEG) came into force on 1 April 2000. It is intended to enable plant operators to operate economically at a reasonable profit with guaranteed power purchase. The aim of the Renewable Energy Sources Act is to continuously reduce the LCOE from RE by securing a substantial market for RE systems (see Section 4.1).

The development of PV generation capacity is only one part of the transformation costs associated with the energy transition. For a long time, this part was at the forefront of the discussion. In recent years, PV has become increasingly system relevant, which means that other transformation steps and cost types have come into focus. Besides the electricity generation from RE, other measures to enact are:

- Demand-side flexibility
- Electrification of the heat and transport sectors
- Expansion of grid-serving and system-serving storage (stationary batteries, heat storage, pump storage) and electrolyzers
- Expansion of electricity grids
- Transfer from coal to gas-fired power plants.

The associated costs are not caused by the PV expansion but belong, like the PV expansion itself, to the cost account of the energy transition. The costs of the energy transition are incurred by all energy consumers, for whom a sustainable energy supply must be created.

Without knowing the costs of an omitted energy transition, it is difficult to assess the costs of the transition.

4.1 Levelized Cost of Electricity

The levelized cost of electricity (LCOE) for a PV power plant is the ratio between the total costs of the plant (€) and its total electricity production (kWh) over its economic lifetime. The total costs for PV power plants are based primarily on:

1. purchase investments to construct and install the plant
2. financing conditions (return on investment, interest, plant lifetime)
3. operating costs over the lifetime of the plant (insurance, maintenance, repairs)
4. deconstruction costs

Investment costs are the dominant cost component of PV power plants. The price of the PV modules is responsible for about one fifth of the investment costs. This share is higher for large PV ground-mounted systems (PV FFA) than for small rooftop systems. History shows that the price development for PV modules follows a so-called "price-experience curve", i.e., when cumulative production doubles, prices fall by a constant percentage. Figure 4 shows the inflation-adjusted world market prices. Since the introduction of the EEG in 2000, PV module prices have decreased by 98 percent. In the long term, module prices [€/W_p] are expected to continue to fall, provided that major efforts continue to be made in the further development of products and manufacturing processes.

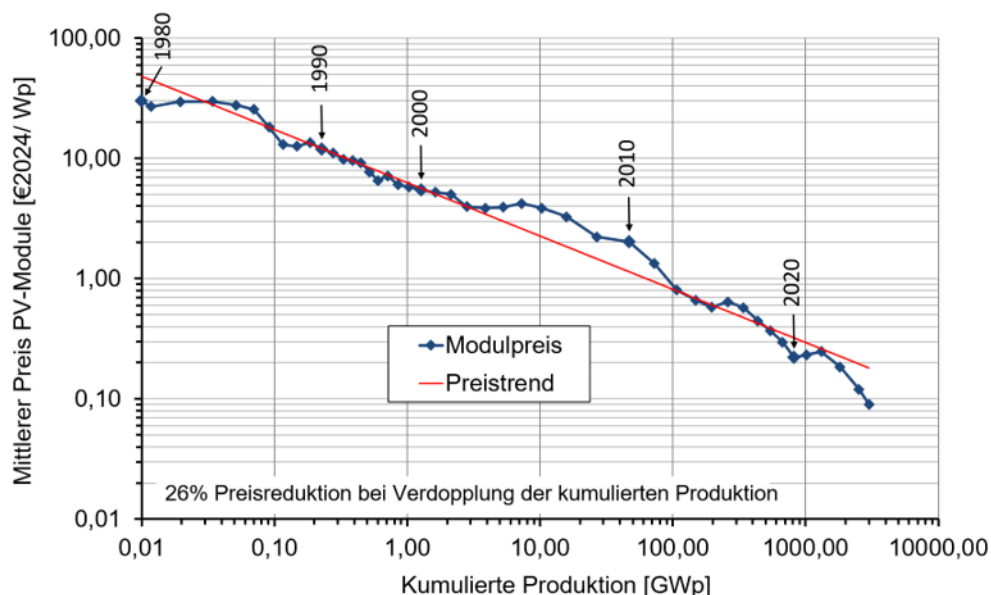


Figure 4: Historical development of prices for PV modules (PSE Projects GmbH/Fraunhofer ISE, data source: Strategies Unlimited/Navigant Consulting/EuPD).

Prices for PV power plants have also fallen, but not as much as for PV modules. Figure 5 shows the price development for rooftop systems of 10 to 100 kW_p nominal power in Germany. The annual operating costs of a PV power plant are comparatively low at approx. 1-2 percent of the investment costs.

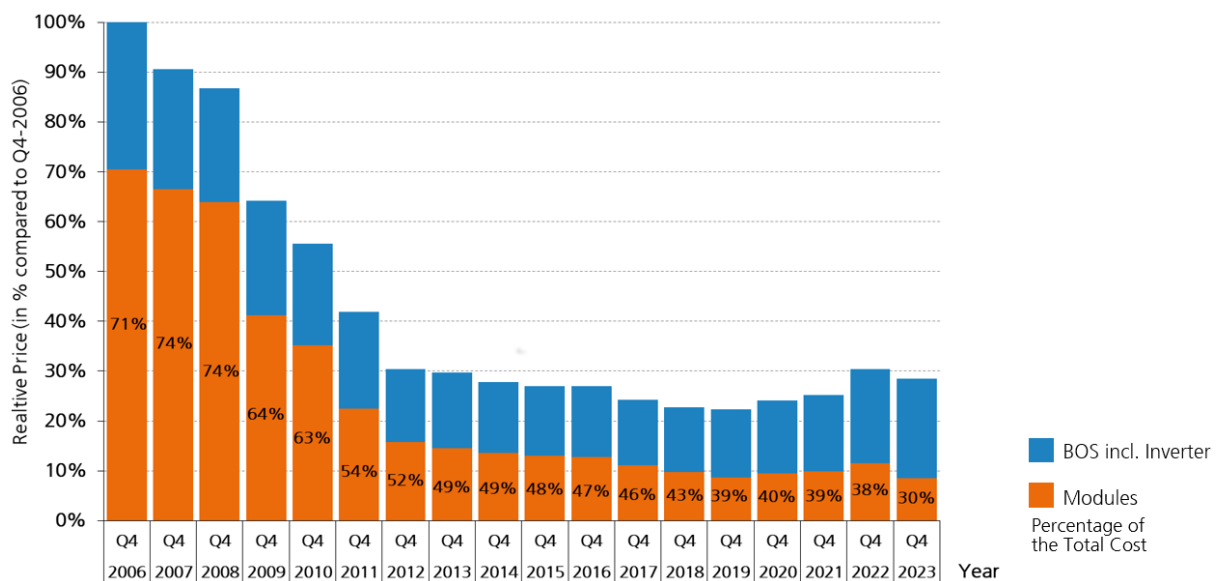


Figure 5: Development of the average end customer price (net system price) for installed rooftop systems with rated nominal power from 10–100 kW_p, data: BSW [ISE5].

New MW-sized power plants produce PV electricity at costs of approx. 4–7 ct/kWh. For small rooftop plants the costs range from 6–14 ct/kWh [ISE1]. These cost estimates always assume that the volatile electricity is completely purchased. Over the long term, further decreases in the LCOE can be expected. Due to the very high investment costs in the past, older PV power plants are significantly more expensive. An orientation value for the development of the electricity generation costs from new PV ground-mounted plants is provided by the tenders of the German Federal Network Agency (following Section).

4.2 Feed-in Tariff

The energy transition requires massive investments in renewable generation capacities for solar power, among other things. However, given the cost reduction dynamics to date (Section 4.1), there is a risk that investments will be postponed in anticipation of a continued trend (deflation effect). Moreover, since PV power plants largely produce electricity at the same time, the more expensive electricity from the power plant built today would no longer be competitive in the future (crowding-out effect). To nevertheless mobilize actors beyond the classic energy suppliers, in particular homeowners, businesses and small and medium-sized companies, a purchase guarantee for electricity that cannot be used by the investor is necessary for the economic service life of the power plant at a fixed, or at least a calculable, remuneration value. The basis for the expansion of PV has been the various versions of the Renewable Energy Sources Act since 2000. The various amendments have increasingly tried to promote and hinder PV expansion at the same time. The “breathing cap” (e.g., 1.5 GW_p according to the EEG 2017) limited the annual expansion, while low tendering volumes slowed down the free-field segment. In addition, there were a number of restrictions, e.g., with regard to area coverage, system sizes and self-consumption.

There are three basic EEG remuneration models, depending on the size and type of the PV power plant: fixed tariff [BNA3] or market premiums for direct marketing. For large-scale power plants, the remuneration is determined by the bidding rounds for the German Federal Network Agency's tender [BNA4]. The purpose of the market premium is to balance negative differences between the market price of solar electricity (monthly average day-ahead exchange electricity price) and a legally defined reference value [BNA3]. Surcharges are provided for certain cases of full feed-in or for landlord-to-tenant electricity supply.

The feed-in tariff for small rooftop systems with self-consumption that go into operation by Dec. 31, 2025 is up to **7.86 ct/kWh** for 20 years, depending on the size of the system. The Federal Network Agency's bidding round for the bidding deadline of 1 February 2018 determined the lowest average surcharge value to date of 4.33 ct/kWh, and the lowest individual surcharge of 3.55 ct/kWh dates from February 2020.

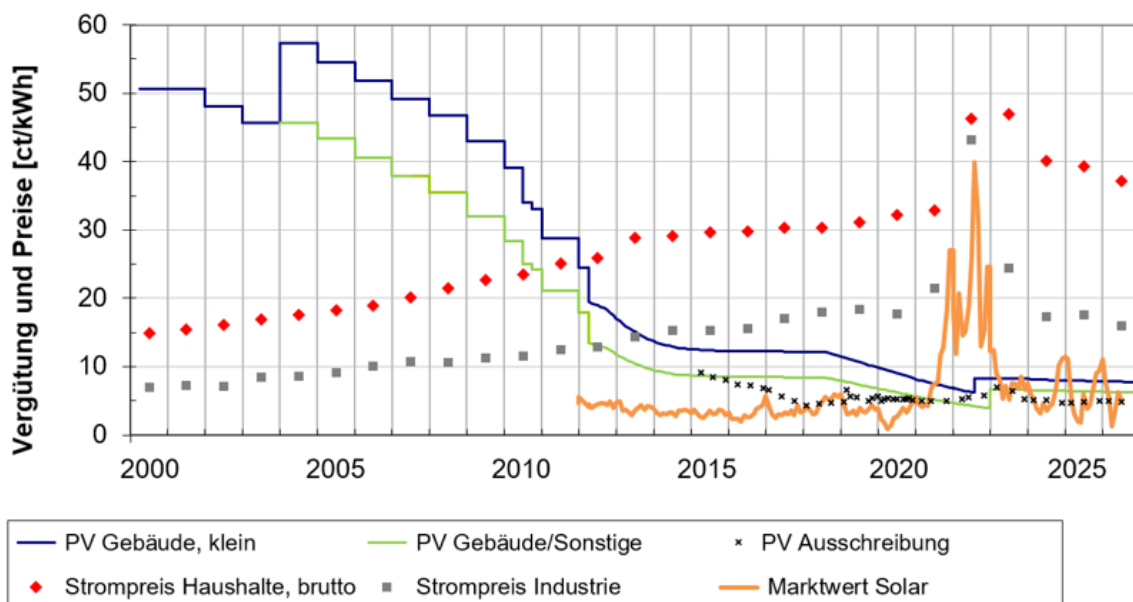


Figure 6: Fixed EEG feed-in tariff for PV power as a function of commissioning date according to system types "Building PV with up to 10 kW_p excess feed-in" and "Other systems up to 100 kW" from [Netz2] and [BNA3], average remuneration of the bidding rounds of the German Federal Network Agency [BNA3], electricity prices [BMWK1], [BDEW1] and average compensation for PV power [NETZ1].

For photovoltaics with storage, the tender, dated May 1, 2026, showed an average winning bid of 5.34 cents per kilowatt-hour. Since the introduction of the EEG in 2000, the nominal remuneration for PV electricity (without inflation adjustment) from new power plants has fallen by approx. 85–90 percent (Figure 6). Internationally, PV electricity at locations with high solar radiation has already been offered at rock-bottom prices of up to 1.12 €/ct/kWh (Portugal) and 0.87 €/ct/kWh (Saudi Arabia). For comparison: for the planned nuclear power plant "Hinkley C", which is to go into operation in the UK in 2025, a feed-in tariff of the equivalent of 12 ct/kWh plus inflation compensation has been guaranteed for a period of 35 years.

4.3 Pricing on the energy exchange and the merit order effect

PV electricity purchased under the EEG must be traded on the day-ahead market of the exchange. Because this turns it into “gray electricity”, the supplier cannot put a price on the sustainable quality of the green electricity. To estimate sales revenues from PV electricity, a mean electricity price (“market value solar” in Figure 6) is calculated based on the prices achieved on the European Energy Exchange (Figure 7).

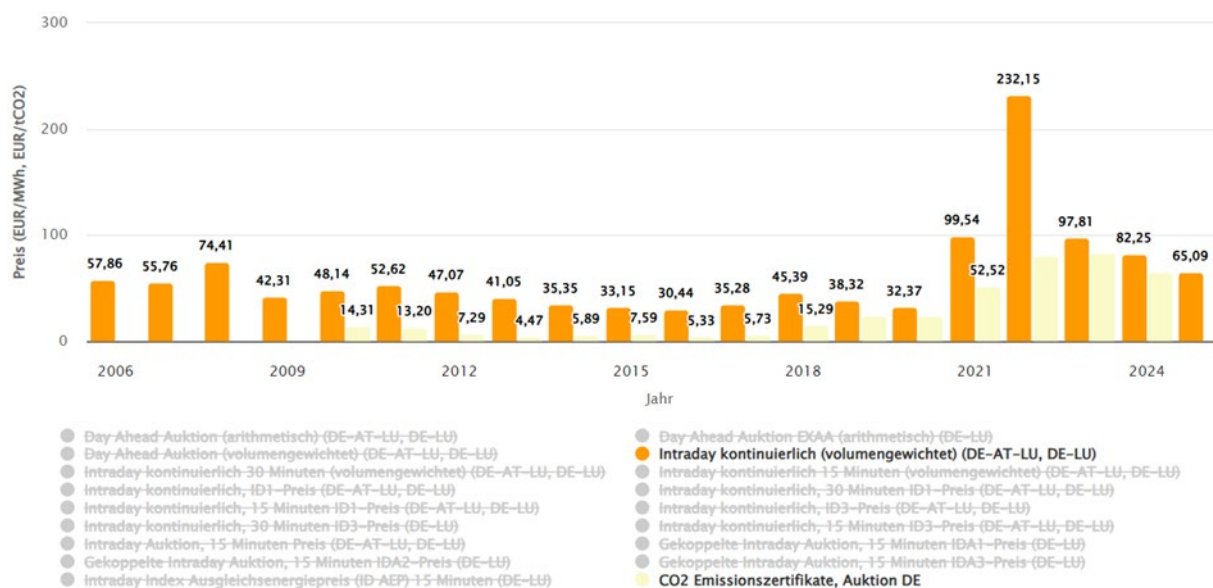


Figure 7: Exchange electricity and certificate prices in Germany [ISE4].

Ongoing price determination on the exchange takes place according to the “merit order” principle. The electricity producers’ sales offer for certain quantities of electricity, usually defined by the respective marginal costs, are sorted by price in ascending order. The electricity buyers’ purchase offers are sorted in descending order. The intersection of the curves results in the exchange price for the entire traded quantity. The most expensive offer that comes into play thus determines the profit margins of all cheaper suppliers.

The feed-in of solar electricity has legal priority, so it stands at the top of the series of offers. With fictitious marginal costs equal to zero, solar power always comes into play. But when solar power comes on, it comes on a massive scale in the core time of the day, when the load — and with it the electricity price in the past — reaches its midday peak. There it predominantly displaces expensive peak-load power plants (especially gas-fired power plants and pumped storage). This displacement lowers the total resulting exchange electricity price and leads to the merit order effect of PV feed-in.

Intraday kontinuierlich, 15 Minuten Durchschnittspreis (DE-LU) in Deutschland 2025

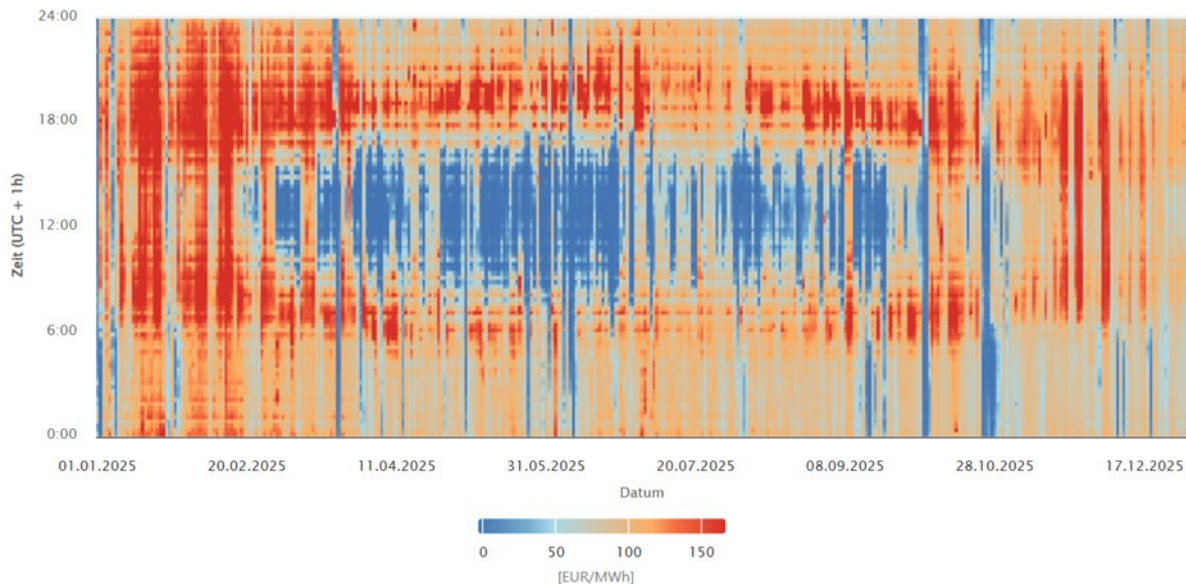


Figure 8 : Interday 15-minute average price in 2025 [ISE].

On sunny days, the power grid is sometimes unable to fully absorb the solar power fed into the system because thermal power plants do not curtail production, due to a lack of demand flexibility and storage capacity in the grid. As a result, the wholesale electricity price drops to levels near or below zero; in 2025, this occurred for nearly 600 hours. Feed-in solar power then also becomes worthless; in some cases, PV power plants must curtail production or receive no compensation under the EEG. With the prices, the revenues of all fossil electricity producers (gas, coal, oil) fall, but also the revenues for electricity from RE (solar power, wind power, hydropower). Furthermore, solar power reduces the utilization of the classic peak load power plants (gas, hydro) in particular.

As the share of volatile renewable energy sources increases, their market value factor—that is, the ratio of the spot market price achieved to the average spot market price over a given period—decreases. The reason: the supply of electricity increases during periods of high feed-in, and both solar and wind power exhibit a high degree of simultaneity.

With the increasing feed-in of RES-E, the Leipzig power exchange has become a residual power exchange. It generates a price for the demand-based supplementation of renewable electricity generation and no longer reflects the value of the electricity.

4.4 Determining the differential costs

The remuneration for PV electricity feed-in according to the EEG is determined annually by the transmission grid operators. The differential costs are intended to cover the gap between remuneration and revenues for PV electricity. The increasing feed-in of PV electricity and wind power lowers the market prices via the merit order effect and thus

paradoxically increases the calculated differential costs; the more PV is installed, the more expensive the kWh of PV electricity appears in the subsidy according to this method.

Figure 8 shows the development of the differential costs for the annual remuneration of the PV electricity generated. After a strong increase until 2014, the amount has stabilized between 9 and 10 billion euros.

The value of PV electricity is measured via the electricity exchange price. According to this method, its value was systematically underestimated: on the one hand, PV electricity has long since influenced the exchange price in the intended direction, namely downwards. On the other hand, the exchange price still largely ignores important external costs of fossil and nuclear power generation (Section 5.2).

A study conducted by the consulting firm Enervis on behalf of BSW-Solar analyzed the impact of photovoltaic systems on the exchange electricity price in 2024: Without PV electricity, the exchange electricity price would have been 15 percent higher on average. Thanks to PV, private consumers, businesses, and industry saved a total of €6.1 billion in electricity costs. [BSW2].

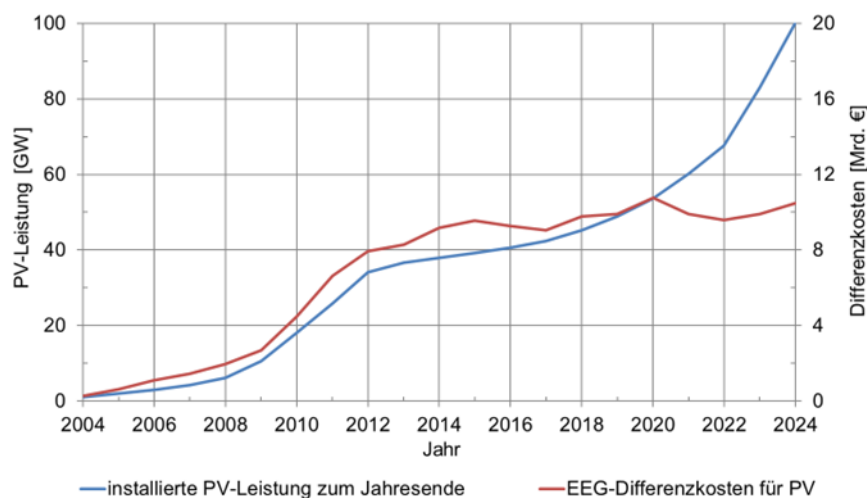


Figure 9: PV expansion and differential costs, data from [BMWK1], [BMWK3]. Medium-term prognoses of the ÜNB.

4.5 EEG levy

The difference between remuneration payments and sales revenues for RES-E, supplemented by other items, was compensated by the EEG levy until June 2022. The levy was borne by those electricity consumers who could not be exempted. Politicians had defined who had to finance the switch to renewable energies [BAFA]. It had decided to largely exempt energy-intensive industrial companies with a high share of electricity costs from the EEG levy. In 2021, 44 percent of industrial consumption was thus privileged. This comprehensive exemption increased the burden for other electricity customers, especially private households. Final consumers had to pay an additional 19 percent VAT on the levy. In 2021, there was a subsidy from the federal budget (Energy and Climate Fund, EKF) of 10.8 billion euros for the levy account for the first time. From July 2022, the EEG levy

was abolished, and the differential costs will be covered by the EKF and the German federal budget.

5 Subventions and Electricity Prices

5.1 Is PV power subsidized?

Yes, since 2021.

A subsidy is defined as a benefit from public funds. Up to and including 2020, however, the subsidy for PV electricity generation did not come from public funds, but from a selective consumption levy (Section 4.5), which was also partly levied on PV electricity produced and consumed by the consumer. Part of the energy consumers paid a compulsory levy for the necessary transformation of our energy system. This view was also confirmed by the EU Commission. The amount of the levy did not correspond to the total remuneration, but to the differential costs. On the cost side, the cumulative differential costs of the feed-in tariff for PV electricity amount to approximately 100 billion euros up to and including 2020 [BMWK3]. In 2021, there was a contribution from the EKF for the EEG account for the first time (Section 4.5). The revenues of the EKF come from emissions trading and federal subsidies, thus a subsidy will take effect from 2021. In 2020, EnBW built the “Weesow-Willmersdorf” solar park, the first large-scale PV power plant in Germany without electricity purchase via the EEG. This is a 187 MW_p project in Brandenburg [EnBW1].

5.2 Is fossil fuel power production subsidized?

Yes, the future costs of the subsidy are still difficult to estimate.

Policy makers influence the electricity price of fossil and nuclear power plants. Policy decisions define the price of CO₂ emission allowances, the conditions for smoke filters, the permanent storage of CO₂, the taxation, the insurance, and safety requirements for nuclear power plants. Policy makers thus determine the extent to which electricity consumers bear the elusive risks and burdens of fossil and nuclear power generation. These effects largely arise in the future through CO₂-induced climate crisis, the final disposal of nuclear waste and the perpetual loads of hard coal mining.

In 2005, an EU-wide emissions trading scheme (European Union Emissions Trading System, EU ETS) was introduced to make CO₂ emissions more expensive and to begin internalizing the costs. However, due to an oversupply of allowances, the price had collapsed by the end of 2017 and was thus practically ineffective (Figure 7). Across Europe, certificate trading also covers only 45 percent of greenhouse gas emissions, because important sectors beyond industry and the energy sector are excluded [UBA5]. A decision has now been made to expand the system to cover approximately 85% of emissions (EU ETS 2), with the changes set to take effect in 2027.

In Germany, a national emissions trading system for the heating and transport sectors was launched in January 2021 with the Fuel Emissions Trading Act (BEHG). Initially, only heating oil, natural gas, petrol, and diesel were considered, from 2023 coal and waste fuels will be included. Effective January 1, 2024, a fixed price of €45 per emission

allowance was set for the first time; for 2026, a price range of €55 to €65 per ton of CO₂ was established. These prices are offset by considerably higher costs caused by climate damage and the necessary recycling of CO₂ from the atmosphere:

- 1) The direct and indirect consequential costs of global climate change, which will also affect Germany, are difficult to estimate. According to calculations by the Federal Environment Agency, the emission of one ton of CO₂ causes damages of around **195–680 €/t**, depending on the higher or equal weighting of the welfare of current versus future generations [UBA3]. In Germany, almost 810 million tons of carbon dioxide and CO₂ equivalents were emitted in 2019, with corresponding damages of 157 billion euros or 551 billion euros, depending on the welfare weighting. For lignite-based electricity generation with an emission factor of 1075 g CO₂/kWh (Figure 9), the derived CO₂ price premiums are 21 and 73 ct/kWh, respectively. Considering the external effects, the costs to society as a whole for lignite-based electricity are thus many times higher than the pure electricity generation costs of 3.4–4.7 ct/kWh [FÖS2].
- 2) All climate protection scenarios assume that carbon capture and storage (CCS) will be necessary to contain the climate crisis. The costs of CCS consist of capture, transport, and storage and are estimated to range from 220 to 510 €/t between 2030 and 2035 [E4C]. Direct Air Carbon Capture and Sequestration (DACCS) with air filtration systems offers potential for permanent recapture without further intervention in the biosphere (Figure 10)
- 3)). In pilot plants, recapture via DACCS currently costs about 350-640 €/t, the cost potential with successful scaling is estimated at 210-330 €/t [ECTI]. Some questions about the final storage of the recycled CO₂ are still open.

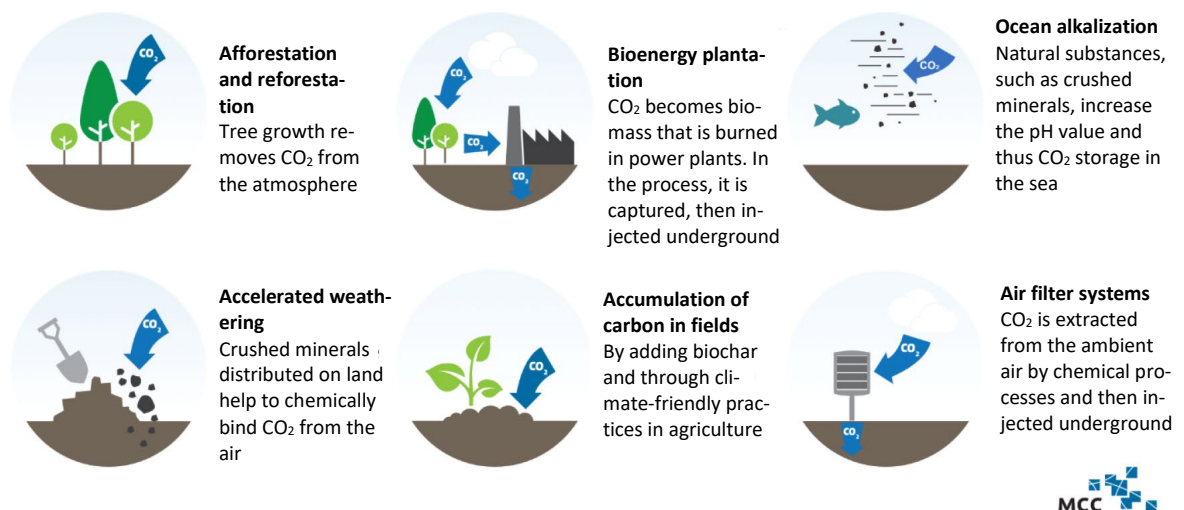


Figure 10: Process for removing CO₂ from the atmosphere and storing carbon [MCC].

A study by the International Monetary Fund estimates that global subsidies for coal, oil and natural gas, including external costs, will amount to 5.9 trillion US dollars in 2020 [IMF].

5.3 Is the PV expansion causing the electricity price for households to increase?

Not directly. Since the abolishment of the EEG levy in mid-2022, the electricity price does not include any components for the remuneration of PV electricity.

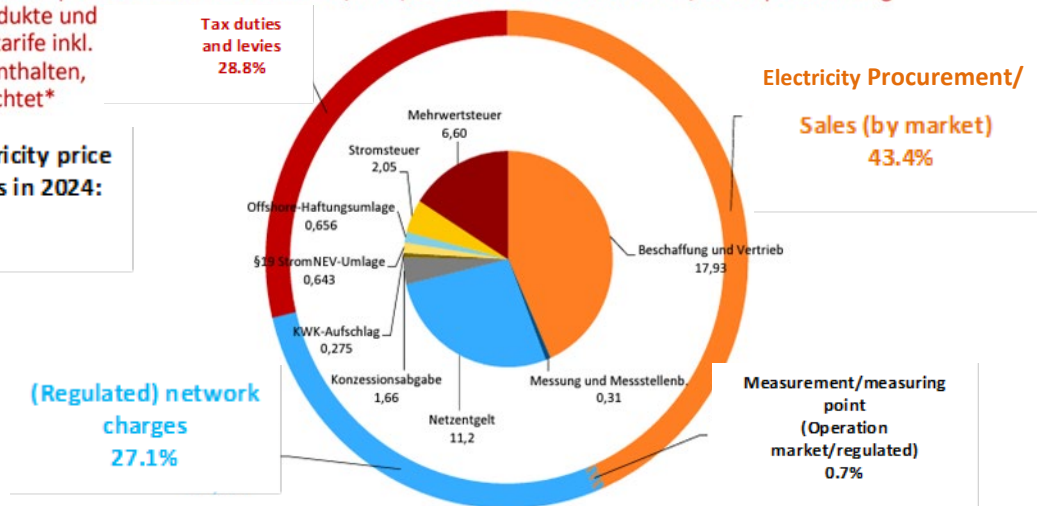
A sample household with an annual consumption of 3,500 kilowatt hours will pay an average gross electricity price of **37.2 ct/kWh** in 2026, including base price [BDEW1]. Figure 10 shows an exemplary price structure.

The electricity tax was introduced in 1999 in order to make energy more expensive through higher taxation; the revenues flow mainly into the pension fund. Private households pay value-added tax on electricity tax. The concession levy is a charge for the use of public roads. The Electricity Grid Charges Ordinance (Section 19 Strom-NEV) is designed to reduce the burden on electricity-intensive industrial companies at the expense of other final consumers. The Combined Heat and Power Act (KWKG) promotes the operation of combined heat and power plants for electricity generation. Grid charges have risen sharply over the last few years. They also include the cost of local grid upgrades that benefit PV deployment.

Strompreis Haushalte 2024

Durchschnittlicher Strompreis für einen Haushalt in ct/kWh, Jahresverbrauch 3.500 kWh, Grundpreis anteilig enthalten, Tarifprodukte und Grundversorgungsstarife inkl. Neukundentarife enthalten, nicht mengengewichtet*

Average electricity price for households in 2024:
41.35 ct/kWh



Quelle: BDEW; Stand: 07/2024

* ausführliche methodische Erläuterung zur Durchschnittsbildung s. Folie 2

Figure 11: Composition of the average household electricity price in 2024 [BDEW1].

The electricity price for private households in Germany is about 50 percent higher than the European average (source: stromreport.de, reference year 2020), but the purchasing power per inhabitant is also 60 percent higher here (source: statista.de, reference year 2019). If electricity prices and purchasing power are taken into account, Germany is in

the middle of the European field. Added to this is the high security of supply. In low-price countries, power outages are commonplace.

6 Are we exporting large amounts of PV power to other European nations?

No.

The monthly values in the Energy Charts (www.energy-charts.de) for 2025 show that energy exports were lower during the summer months than in the winter (Figure 11).

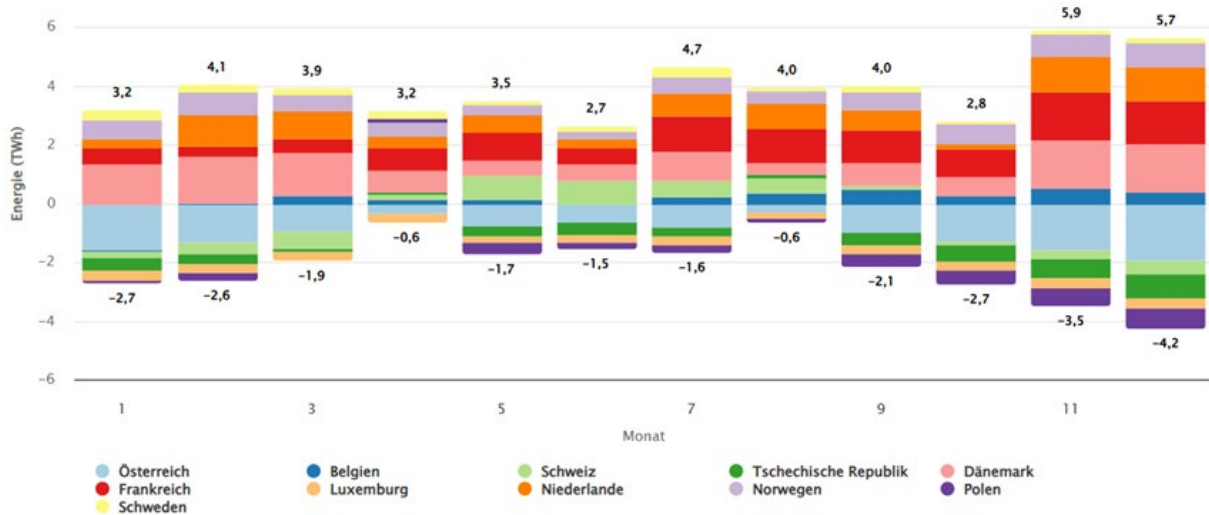


Figure 12: Monthly cross-border electricity trade in 2025 for Germany; negative values indicate exports [ISE4]

7 Can small PV systems generate attractive returns?

Yes.

In principle, small PV systems can generate income via the EEG remuneration for feeding to the grid and via the reduction in electricity consumption thanks to self-consumption. Systems without self-consumption ("full feed-in") receive higher remuneration than systems with self-consumption ("partial feed-in", Figure 12). Attractive returns are possible due to the drop in prices for PV modules, the sharp rise in electricity consumption costs and the increased remuneration for full feed-in.

The greater the difference between the purchase costs for electricity and the electricity production costs of the PV system, the more worthwhile self-consumption becomes. In systems without storage, the potential for self-consumption depends on the coincidence between the generation and consumption profiles. Depending on the size of the system, households achieve 20–40 percent self-consumption in relation to the electricity generated [HTW1]. Larger systems increase the degree of coverage of the total electricity

demand with PV electricity but reduce the share of self-consumption. Commercial or industrial consumers achieve particularly high self-consumption values if their consumption profile does not drop significantly at the weekend (e.g., cold stores, hotels and restaurants, hospitals, server centers, retailers). Energy storage and transformation technologies offer considerable potential for increasing self-consumption (see Section 22.3).

The yield of a system is higher in sunny regions. In fact, the regional difference in the annual sum of irradiation is not transferred 1:1 to the specific yield (kWh/kW_p, Section 28.5) because, for example, the operating temperature of the modules, pollution effects or the duration of snow accumulation also play a role. To obtain a rough estimate of the discounted levelized cost of electricity LCOE (not adjusted for inflation, see Figure 12: Estimation of the LCOE for small PV systems under different irradiation conditions, plus remuneration according to [EEG2023]), the following assumptions were made:

- optimal orientation of module with regard to annual yield (approximately 30° south)
- average annual total of horizontal global irradiation in Germany 1100 kWh/(m²a)
- Performance Ratio of 85 percent (see Section 28.7)
- Annual yield degradation of 0.5 percent
- Lifetime of 20 years
- Annual operating costs of 1 percent (of plant price)
- Inflation rate of 2 percent
- Nominal imputed interest rate of 4 percent (average of own and borrowed capital investments)

The Levelized Costs of Electricity (LCOE) are estimated on the basis of the net present value method. The current expenditure and the LCOE are discounted to the time of commissioning using the specified interest rate. In the case of full financing through equity, the imputed interest rate corresponds to the achievable return. For comparison: the Federal Network Agency has set the return on equity for investments in electricity and gas grids for new plants at 6.91 percent before corporate income tax [BNA1].

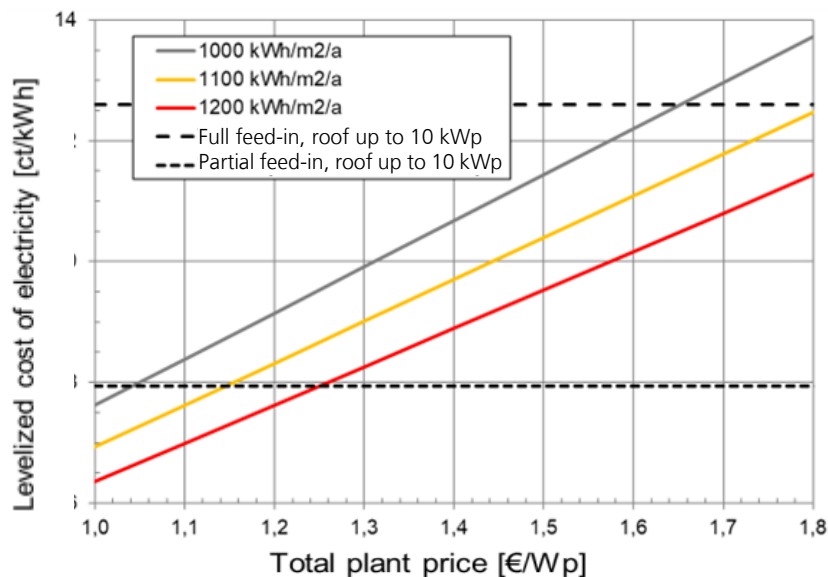


Figure 13: Estimation of the LCOE for small PV systems under different irradiation conditions, plus fixed remuneration according to EEG.

The return on a PV system is not risk-free during the EEG remuneration period. Neither manufacturer guarantees nor plant insurance reduce the investor risk to zero. The utilization of electricity from the 21st year of operation is regulated for the first time by the [EEG2021]. Self-consumption capability plays a major role in calculating the continued operation of “de-subsidized” plants [SCBW].

8 Is the supply of solar components ensured?

No, China has built up a critical monopoly.

The complete PV value-added cycle based on silicon wafer technology (Figure 13) begins with the production of high-purity polysilicon and continues with the crystallization of silicon ingots and the sawing of silicon wafers. This is followed by cell production and module production. If the coverage of more than one stage is to be emphasized, one speaks of (vertically) integrated PV production.

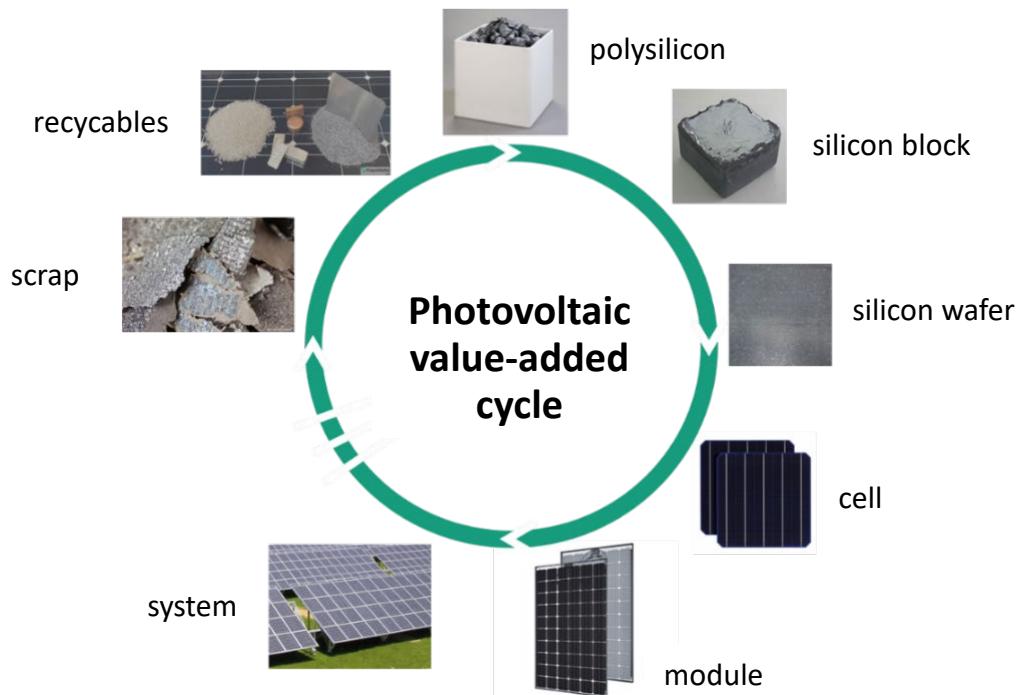


Figure 14: Value Chain for Photovoltaics.

Many material and component manufacturers are part of the extended PV value cycle. These include suppliers of silver pastes for solar cells, special films, wires, solar glass, and junction boxes for solar modules. Other players complete the cycle with additional components for power plants through to recycling:

1. Material production: solar silicon, metal pastes, connector wires, plastic foils, solar glass, glass coating
2. Manufacture of intermediate and final products: modules, cables, inverters, mounting structures, tracker systems
3. Mechanical engineering for cell and module production
4. Installation (especially trade)
5. Power plant operation and maintenance
6. Recycling

In 2023, China was dominating all value chain stages with market shares of more than 80 percent, as shown in Figure 14.
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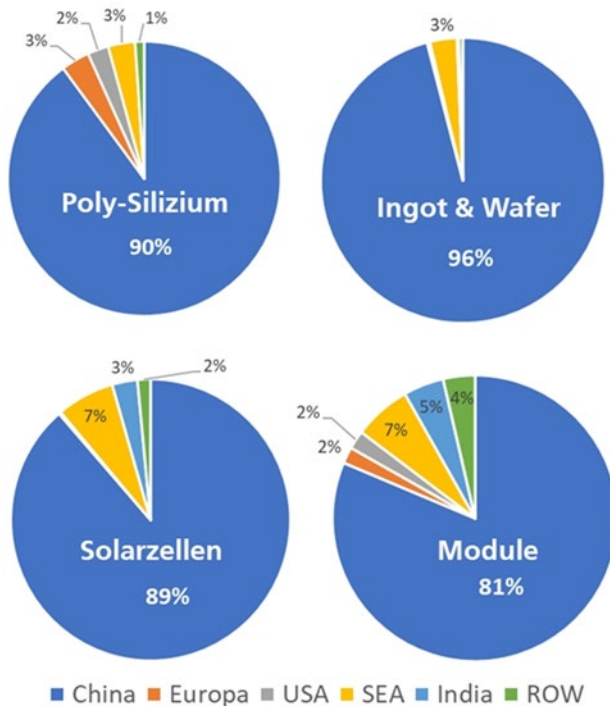


Figure 15: Production capacity according to value chain stage and country, data: Wood Mackenzie, 2023

While there was still a complete PV supply chain in Germany and Europe around 2010, the production of some feedstocks has been discontinued due to the regional demand that has decreased in the meantime. Notable shares of the global market are held by inverter manufacturers and silicon manufacturer Wacker.

In Europe, there are small module production facilities, however, hardly any cell or wafer production. A slump in trade with China would seriously jeopardize the expansion of PV in Germany and also severely impair local module production due to a lack of intermediate products. Compared to modules produced in China, integrated European production would result in additional costs of 10.3 €/t/W_p (+100%) for modules, 10.8 €/t/W_p (+20%) for large PV power plants, and 0.75 €/t/kWh (+14.5%) for PV electricity production costs [SPE2].

9 Does installing PV only create jobs in Asia?

No, but Germany has lost many jobs in the PV production industry in the last decade.

In 2025, there were 84,700 jobs (gross employment) in the photovoltaic industry. At its peak in 2011, the figure was 142,700 [Bert]. Germany lost many jobs in the 2010s due to companies closing and experiencing bankruptcy. First the PV cell and module production were affected, then companies involved in PV machine engineering followed.

According to calculations by Fraunhofer ISE, a vertically integrated 10 GW production from silicon ingot via wafer and cell to module creates approx. 7500 full-time jobs [ISE8]. According to a study by EuPD Research based on figures from 2018, around 46,500 full-

time employees are needed to install 10 GW of PV [EuPD]. In 2022, the solar industry (PV and thermal) employed 84,100 people [BMWK press report from January 24, 2024].

The hope that the combination of the Renewable Energy Sources Act (EEG), investment subsidies in the new federal states and research funding would be enough to establish Germany as a world-leading production location for PV cells and modules seemed to be fulfilled as recently as 2007, when a German company topped the international rankings in terms of production volume. Since then, German manufacturers have dramatically lost market share because of the decisive industrial policy in the Asian region and the massive investments in production capacities generated there. Labor costs play a subordinate role in this development, as PV production has reached a very high degree of automation. Turnkey production lines that deliver very good PV modules have been available “off the shelf” for several years, which has enabled a rapid transfer of technology.

Effective laws on feed-in tariffs have triggered massive investments in PV power plants in Germany and Europe. However, there was a lack of economic policy support to ensure that production capacities remained competitive. On the other hand, China and other Asian countries have succeeded in mobilizing many billions of domestic and foreign capital for the construction of large-scale production lines by creating attractive investment and credit conditions.

Despite the high import quota for PV modules, a large part of the value added associated with a PV power plant remains in the country. In the long term, falling manufacturing costs of PV modules on the one hand and rising freight costs and long freight times on the other will improve the competitive position for module production in Germany.

10 What funding is being directed to PV research?

In 2024, the German federal government invested 60 million euros in supporting photovoltaic research (funds disbursed, Figure 15), with the largest share going to production technologies.

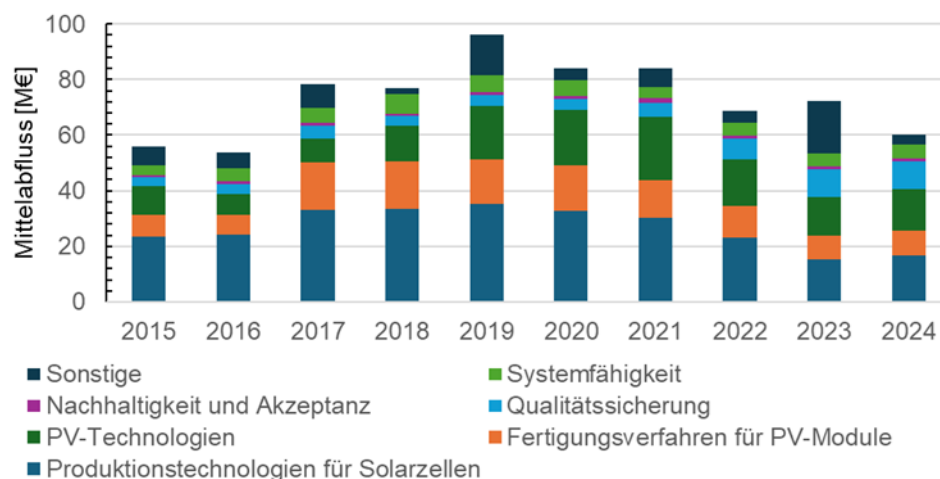


Figure 16: Federal funding for photovoltaic research in € million [BMWK2].

11 Does PV power overload our energy system?

At times, yes, as the energy system is lacking flexibility.

The energy transition is driving change in many different areas, but they are all moving at different speeds. Photovoltaic expansion was on track over the past years, but the restructuring of the energy system and the adjustment of the electricity market have not been keeping pace. In many areas, there was or still is a lack of (sufficiently effective) price signals and technical and/or regulatory requirements, among others, for:

- the stabilization of solar power production, as almost all PV power plants are facing directly south
- the grid- and system-serving expansion and operation of stationary battery storage systems – (There are strict rules governing who can operate storage systems where, and how, as well as which costs are allocated to storage. Also there are disincentives (a single electricity price zone, rigid fees) and incentives are lacking for home storage systems.
- load flexibility and the installation of smart meters (There are not enough price signals and intelligent measurement systems to active flexible loads.)
- the grid- and system-serving bidirectional operation of charging infrastructure for e-mobility (The ramp-up of e-mobility, especially bi-directional) is behind schedule.)
- the electrification and grid- and system-serving operation of the heating sector (The ramp-up of the heat pump is behind schedule.)
- the grid- and system-serving flexibilization of electricity production from biomass. (Incentives for flexible use were introduced late.)
- flexibilization of fossil fuel-based electricity generation (The transition from coal to gas is long overdue.)
- curtailment of generation peaks. (Due to misguided regulatory incentives, many controllable PV power plants continue to feed electricity into the grid even when electricity prices are negative. Due to numerous failures, there are occasional bottlenecks in the feed-in of PV electricity to the German grid, resulting in many hours with electricity exchange prices near or below zero and, after sunset, unnecessarily high electricity exchange prices.

Grid-serving operation addresses the dominant constraints in the grid, thus helping to reduce grid costs and the need for grid expansion. System-serving operation focuses on frequency control, voltage control, and restoring power supply at the transmission grid level. The primary responsibility for system stability lies with the transmission system operators. A nuanced examination of grid- and system-serving operation is provided by the [FfE] analysis titled “What Is Grid-Oriented Operation?”

The necessary measures to deter this are explained in section [22.3 Transformation Steps](#).

11.1 Transmission and distribution

Most solar power plants in Germany are connected to the low-voltage grid; Figure 16 illustrates how they are distributed according to plant size. Many systems generate solar

power decentralized and close to consumption; they hardly place any demands on the expansion of the transmission or medium-voltage grid and do not require long connection lines. On sunny days, a high density of PV systems in a low-voltage grid can lead to electricity production exceeding electricity consumption locally due to the high simultaneity factor. Transformers then feed power back into the medium-voltage grid. With very high system densities, the transformer station can reach its capacity limit. Large-scale PV power plants or local clusters of systems in sparsely populated areas sometimes require an enhancement of the power grid and/or transformer stations, or the use of stationary battery storage systems. Battery storage systems allow generation peaks to be leveled out at the feed-in point and the feed-in volume to be reduced through increased self-consumption.

A demand-driven distribution of PV installations across grid sections reduces the need for additional capacity for grid feed-in. For example, Brandenburg or Mecklenburg-Western Pomerania have four to five times more PV capacity installed per capita than North Rhine-Westphalia or Hesse [ISE4].

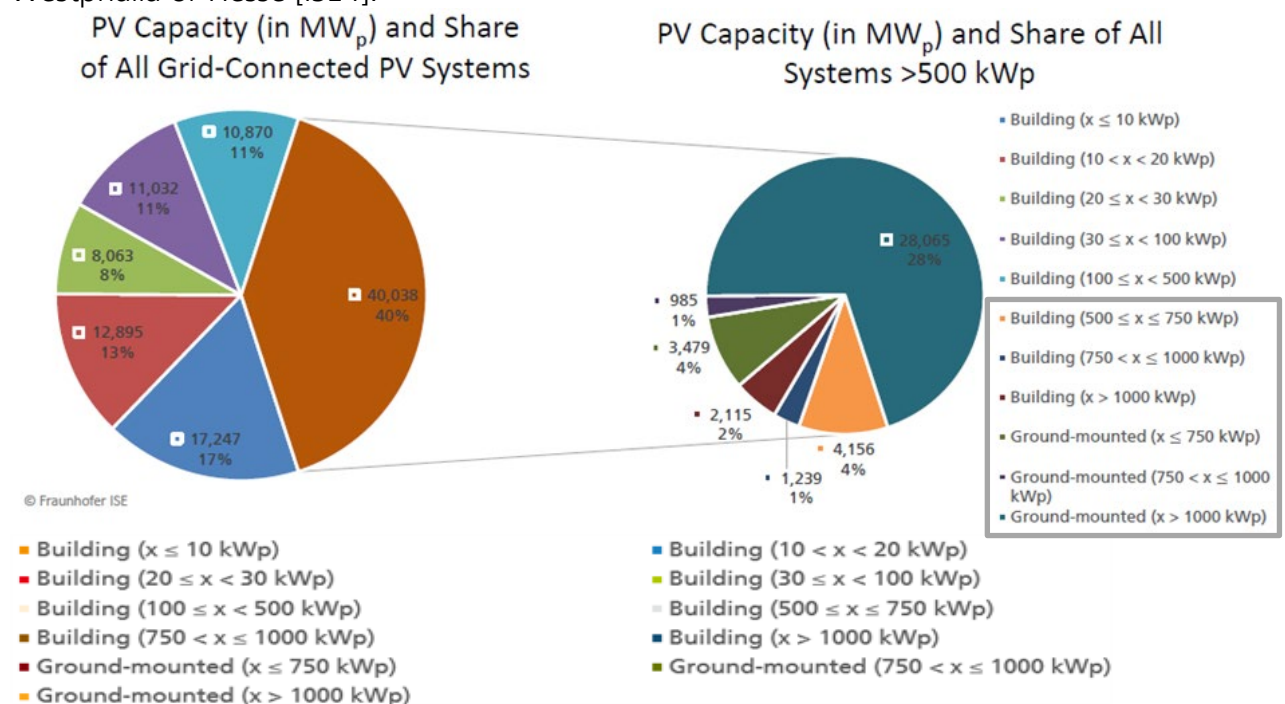


Figure 17: Distribution of installed PV power according to plant size, Dec. 2024 [ISE5].

Since solar radiation and wind often occur in a complementary way (11.2.3), it makes sense to connect them to the same feed-in point. Hybrid power plants, i.e. the combined installation of solar and wind power plants, ideally with a battery storage system, can significantly reduce the requirements for grid-expansion and transformer capacity. This combination ensures an improved utilization of the feed-in infrastructure.

11.2 Volatility

11.2.1 Can solar power production be predicted?

Yes.

Reliable [performance forecasts](#) based on satellite data can be created for individual power plants or regional clusters, with local sky imagers or monitoring stations being added as needed. On a national level, the generation of solar electricity can also be planned very well thanks to reliable national weather forecasts and yield models (Figure 17).

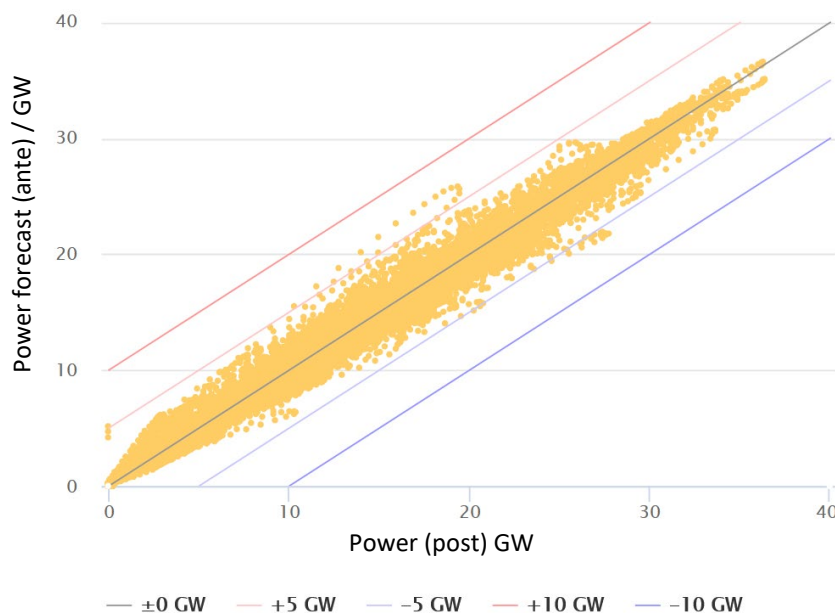


Figure 18: Actual and predicted hourly generation of power in 2021 [ISE4].

Due to decentralized generation, regional changes in cloud cover cannot lead to serious fluctuations in Germany-wide PV electricity production. Even the effects of a solar eclipse with a high, trans-regional simultaneity factor can be reliably forecasted and compensated.

11.2.2 Do solar and wind power complement each other?

Yes.

Due to the climate, solar radiation and wind strength in Germany correlate negatively on time scales of hours to months.

On a **quarter-hourly basis**, with an average installed capacity of approx. 57 GW_# PV and 63 GW_P wind power at the end of the year, in 2021 practically never more than 60 GW of capacity (i.e., 50 percent of the nominal PV + wind capacity) entered the electricity grid (Figure 18). Viewed on an **hourly basis**, less than 1 percent of the electricity generated would have been lost if the combined PV + wind capacity had been limited to a maximum of 40 GW (i.e., 33 percent of the nominal PV + wind capacity).

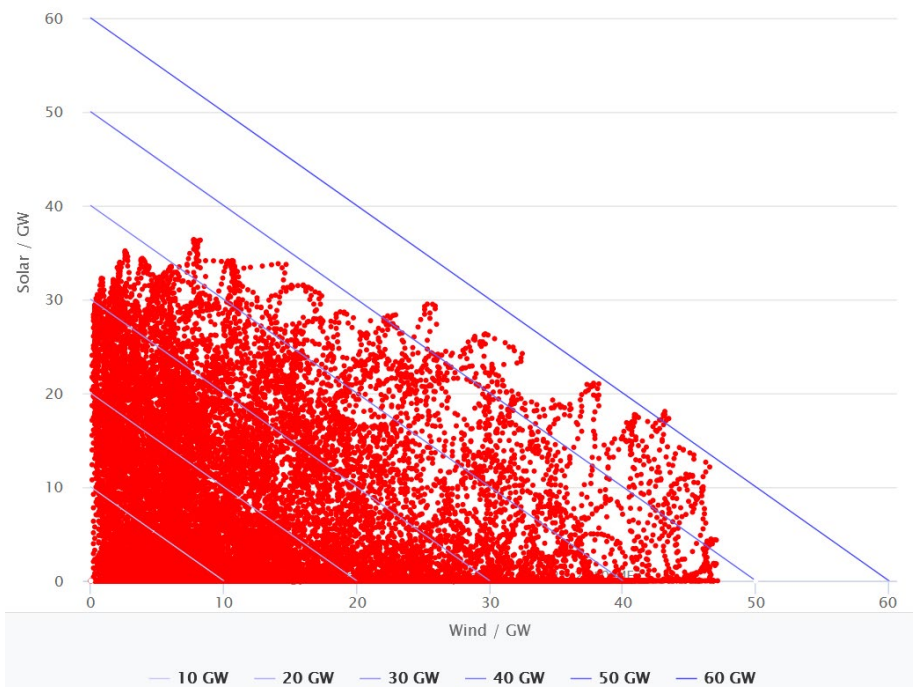


Figure 19: Average power for the supply of solar and wind power in 2021, 15-minute values [ISE4].

The combination of PV and wind power also leads to a stabilization of the yield on a **daily basis**. While the relative mean absolute deviation of daily electricity production from the arithmetic mean in 2021 was 55 percent for PV and 56 percent for wind, the value for PV + wind was only 37 percent.

Figure 19 shows the **monthly totals** of electricity production from PV and wind power, as well as the moving annual averages. The average relative deviation of the monthly values from the moving annual value is 53 percent for PV and 30 percent for wind. For the sum of PV and wind the deviation is 14 percent and thus significantly more stable than the sectors alone.

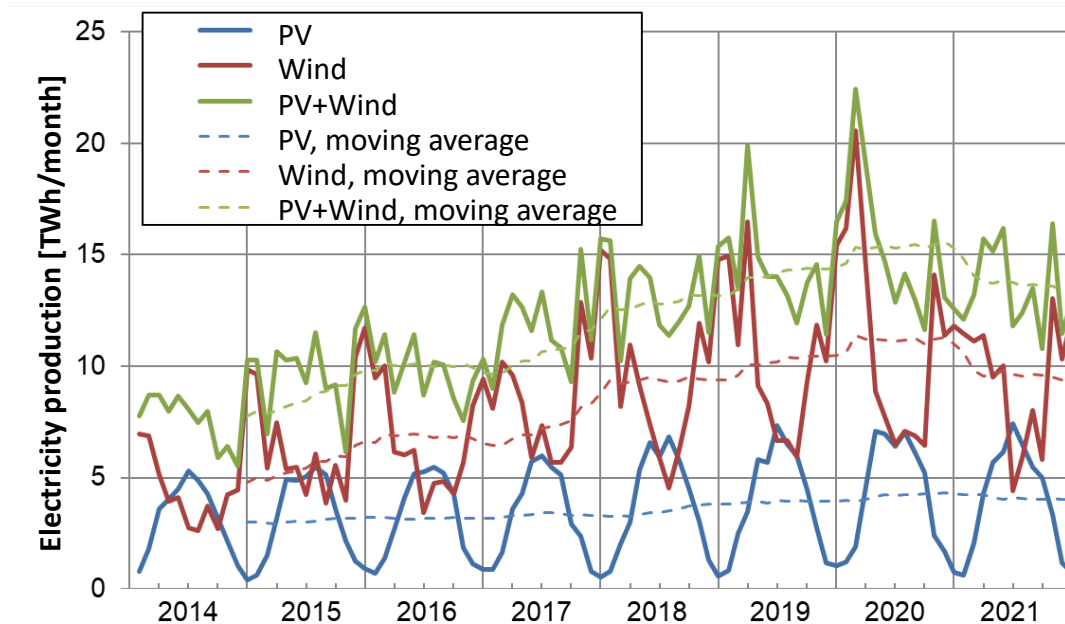


Figure 20: Monthly totals of PV and wind power production, data from [ISE4].

This approach requires that solar and wind power can be integrated into the grid in a timely manner. On a national scale, this is only partially feasible due to the concentration of wind farms in the north and solar power plants in the south, as well as the weak north-south connection in the transmission grid.

11.3 Does the grid feed-in reach the installed capacity?

No, at least not across Germany.

Information on the installed PV capacity of an individual power plant or a region refers to the DC side of the PV system, i.e. to the nominal power of the modules (<https://www.clearingstelle-eeg-kwkg.de/haeufige-rechtsfrage/221>). However, inverters are often undersized for better utilization, so that their nominal AC power is significantly lower than the nominal DC power of the connected PV modules. Across Germany, for example, the average DC rated power is just under 10% higher than the AC rated power [ISE4]. For large PV power plants with a southern orientation, the AC/DC ratio is typically around 85%. When batteries are added on the DC-side, this ratio can fall to 50%. As a rule, small rooftop systems must be reduced to 70% capacity at times, or sometimes even 60%. The systematic undersizing of inverters, or also curtailment, already creates a significant gap between installed capacity (DC) and maximum feed-in.

This gap is widened by technically induced losses with performance ratio values below 90% (see Section 28.7 Performance Ratio) and often inconsistent weather conditions across Germany. Self-consumption also reduces feed-in, even if its potential for load flexibility has not yet been fully exploited. The same applies to the charging of DC-coupled batteries. Figure 20 shows the trend in recent years in the ratio of maximum feed-in to the public grid to installed PV capacity. In 2024, the peak PV feed-in across Germany was 53%, just over half of the installed PV capacity. Even with the addition of PV power plants away from the dominant south-facing, flat-roofed locations, the ratio will continue to fall.

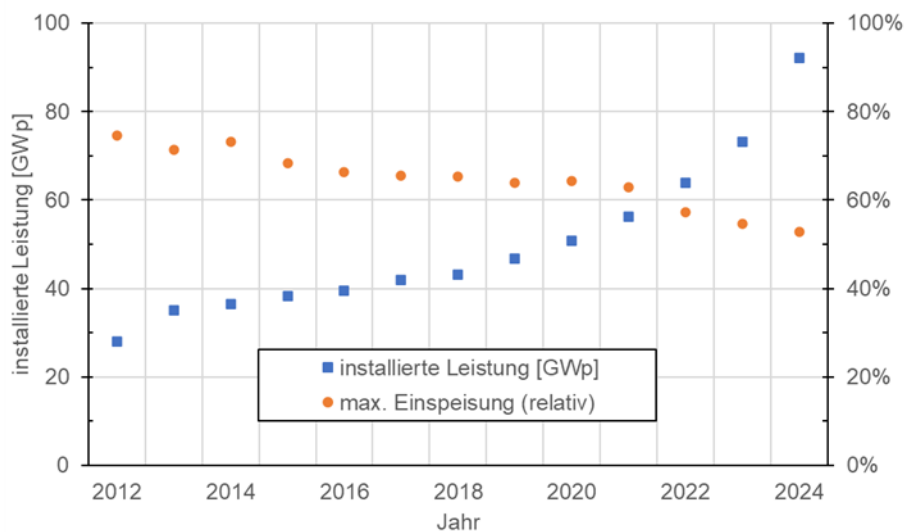


Figure 21 Annual maximum of the grid feed-in compared to installed PV capacity at the end of the corresponding month, data from [ISE4].

11.4 Is a significant amount of PV electricity lost through redispatch?

No, but the amount lost has clearly increased in 2024.

In the event of bottlenecks in the transmission or distribution grid, generation plants on one side of the bottleneck must be throttled down and those on the other side ramped up in order to serve the load (redispatch). In 2024, 1.4 TWh of PV electricity was curtailed in redispatch [SMARD], which is 1.9% of the PV grid feed-in [ISE4] for that year.

For the most part, curtailment (7.9 TWh) mainly affects wind power, which is generated more cheaply in the north but cannot be transported to southern Germany due to insufficient transmission capacity. For years, high wind speeds have resulted in significant costs due to payments for curtailed electricity in the north and the procurement of the shortfall in electricity in the south (redispatch measures).

11.5 Conflicts with slow-response power plants

The generation profile of PV electricity with peaks around noon matches the load profile of the power grid well. However, conflicts are increasing with sluggish power plants that can only follow a fluctuating residual load to a very limited extent for technical and economic reasons. Older coal-fired power plants, especially lignite-fired power plants, cannot contribute balancing power in an economically justifiable way. Run-of-river hydro and most biomass power plants also offer little flexibility. In principle, however, volatile generators with their negligible marginal costs must be given priority.

These unresolved conflicts occasionally lead to significant overproduction and negative wholesale electricity prices, as the example in Figure 21 shows.

During hot spells in the past, fossil and nuclear power plants caused critical heating of the rivers used as cooling reservoirs. Photovoltaics installed in Germany have eliminated this problem and can also ease such situations in neighboring countries such as France, because they fundamentally reduce the load on fossil and nuclear power plants, especially on summer days.

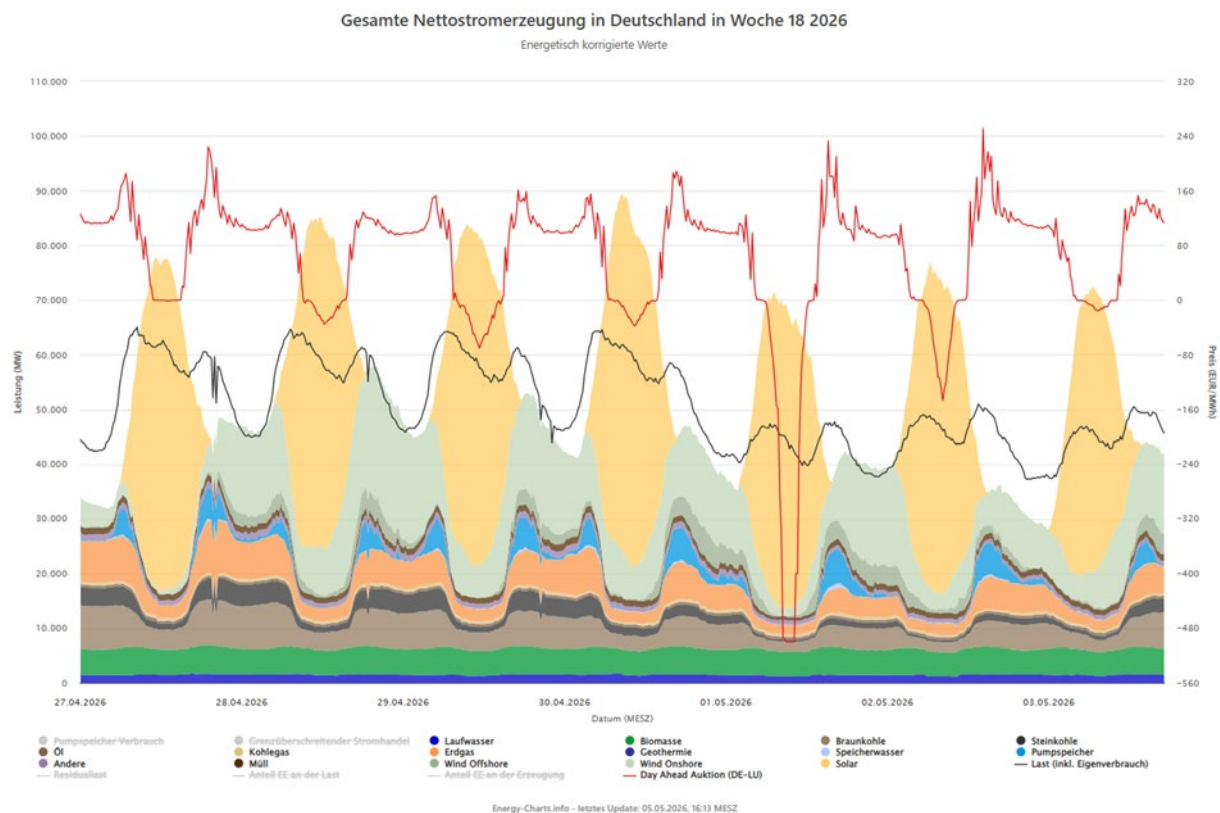


Figure 22: Example showing the course of electricity trading price, conventional and renewable electricity in the 18th calendar week in 2026 [ISE4].

11.6 Does storage facilitate the PV expansion?

Yes, electrical energy storage systems significantly facilitate self-consumption and grid integration of PV electricity.

Today, battery storage systems have become so affordable, and their applications and business models so diverse, that the option of on-site battery storage should always be considered when planning new PV systems of any size (see Section 22.3.7.1).

11.7 Does volatile solar power endanger security of supply?

Not yet, however, it is becoming increasingly important to reduce midday peak production through load shifting, storage, or curtailment. As PV output increases, it will increasingly be called upon to act as a stabilizing control variable.

The security of supply for end consumers has actually improved with the massive increase in installed photovoltaic capacity (Figure 22).

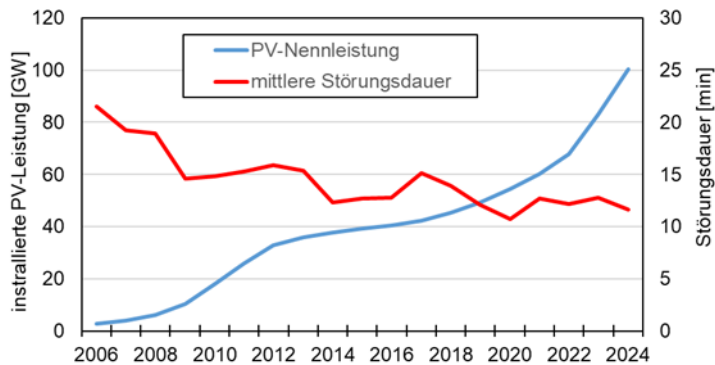


Figure 23: System Average Interruption Duration Index (SAIDI) for low and medium voltage grid. Data from [BNA2].

12 Is there enough space for PV in Germany?

Yes, and without any significant conflicts with agriculture or nature conservation. An important concept for the development of significant land potential is integration. Integrated photovoltaics (www.integrierte-pv.de) enables double land use, additional land consumption for new PV power plants is significantly reduced or avoided altogether. PV systems specially tailored to the application are combined with agriculture, erected on artificial lakes, used as the shell of buildings, car parks, traffic routes and vehicles, or they provide ecosystem services on re-naturalized biotopes and moorland (Figure 23).



Figure 24: Applications for the integration of photovoltaics.

Relying exclusively on one or a few of these possible applications today does not do justice to the urgency of the energy transition. We no longer have the time to try out options one after the other. All technically and socio-economically promising application options must be tested quickly and promoted in a dedicated start-up phase to explore optimization and cost reduction potentials through scaling and learning effects. Only after this broad start-up phase can well-founded decisions be made in favor of certain applications and technologies.

In the following analysis of potentials, a distinction is made between a theoretical, a technical and an economic-practical or realizable or developable potential. The **theoretical potential** considers the maximum possible implementation of a technology based on the entire supply (physical rough calculation). The **technical potential** is lower because it already considers basic technical boundary conditions (technical rough calculation). The **regulatory potential** considers additional, non-monetary legal restrictions and exclusion criteria. The **economic-practical potential** considers all relevant boundary conditions, in particular economic (incl. infrastructure), sociological (incl. acceptance), and competing uses (e.g. solar thermal and PV on roofs). Different sources draw slightly different boundaries between the categories.

The area used for agriculture in Germany is just under 17 million hectares (Figure 24). [Agrivoltaics](#) (Agri-PV) uses land simultaneously for agricultural crop production (photosynthesis) and PV electricity production (photovoltaics). Agri-PV covers a broad spectrum in the intensity of agriculture and in the additional effort required for PV system construction. It ranges from intensive crops with special PV mounting systems to extensively used grassland with marginal adaptations on the PV side and high potential for eco-system services. Agri-PV increases land efficiency and enables a massive increase in PV power, while preserving fertile soils for agriculture or in combination with the creation of species-rich biotopes. Agri-PV is already being used on a GW scale worldwide, but there are only a few systems in Germany.

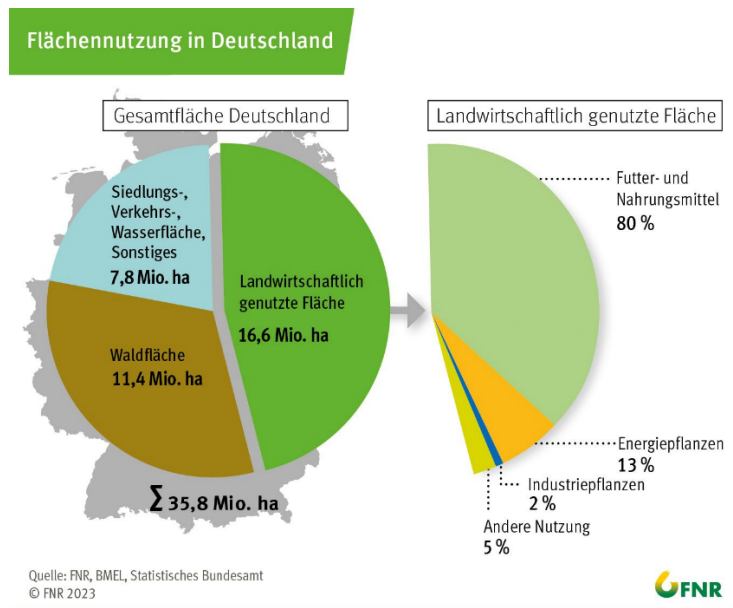


Figure 25: Land use in Germany [FNR].

Agri-PV with high-mounted modules allows cultivation in partial shade under the modules. A number of crops show hardly any yield losses with reduced irradiation, some even benefit. If permanent crops (e.g., orchards and vineyards) are considered in their entirety and one-third of arable land (without maize) as technical potential, an occupancy density of 0.6 MW_p/ha results in a technical potential of **1.7 TW_p**. Modules mounted close to the ground with wide row spacing enable cultivation between the rows. At an occupancy density of 0.25 MW_p/ha with vertical modules, the cultivation of forage crops on permanent grassland alone opens up a technical potential of a further **1.2 TW_p**.

Energy crops are grown on 13 percent of agricultural land, especially for the production of biogas, biodiesel, vegetable oil and bioethanol [FNR]. The land efficiency is significantly below what would be possible with Agri-PV plants (Section 16). Energy maize alone is cultivated on 1 million ha; this area corresponds to **600 GW_p** of nominal capacity if converted to APV with suitable crops (or in **biodiversity PV**, see Section 4).

Opencast lignite mining has destroyed an area of 1773 km² [UBA4] in Germany, more than three times the area of Lake Constance. Parts of this mining area have already been flooded or are still being flooded, and there are many more artificial lakes. In total, this opens up a technical potential of **45 GW_p** for **floating PV** (FPV, [ISE9]). In the case of active quarry lakes, the PV can contribute to the self-sufficiency of the production facilities and use existing grid connection points, if available. Worldwide, there are already several gigawatts of floating PV installed.

In Germany, there are about 40 million buildings (Figure 25). Building envelopes, i.e., roofs and façades, offer a technical potential in the order of **1000 GW_p** [Eggers]. The analysis only includes areas that receive a minimum of 500 kWh/(m²·a) of irradiation and have a connected minimum size. Today, less than 10 percent of the rooftop potential and less than 1 per mille of the façade potential is used.

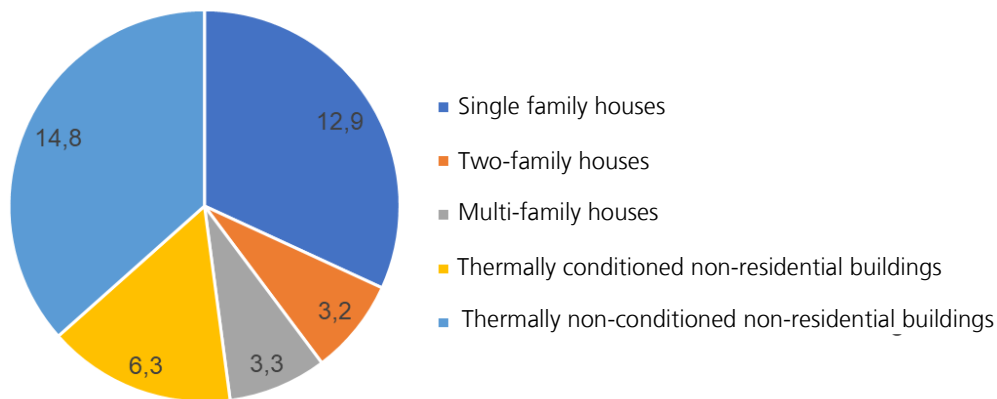


Figure 26: Building stock in millions, data from [DENA], [IWU].

As yet, PV use is largely limited to rooftop systems, even though numerous building-integrated products (BIPV, "[building-integrated PV](#)") are certified and commercially available. BIPV products include PV panels and PV tiles for pitched roofs, lightweight PV systems for roofs with low load-bearing capacity, PV systems for green roofs, PV modules for cold façades (curtain-type, rear-ventilated façades), external thermal insulation composite systems (ETICS) with PV, opaque and semi-transparent PV insulating glass and PV solar control louvres (Figure 26). BIPV offers material-efficient solutions for high-pitched rooftops, enabling surface areas to be used that would not be suitable for roof-mounted systems for aesthetic reasons. The advantage of roof-mounted systems is that they are less tied to construction and renovation cycles.

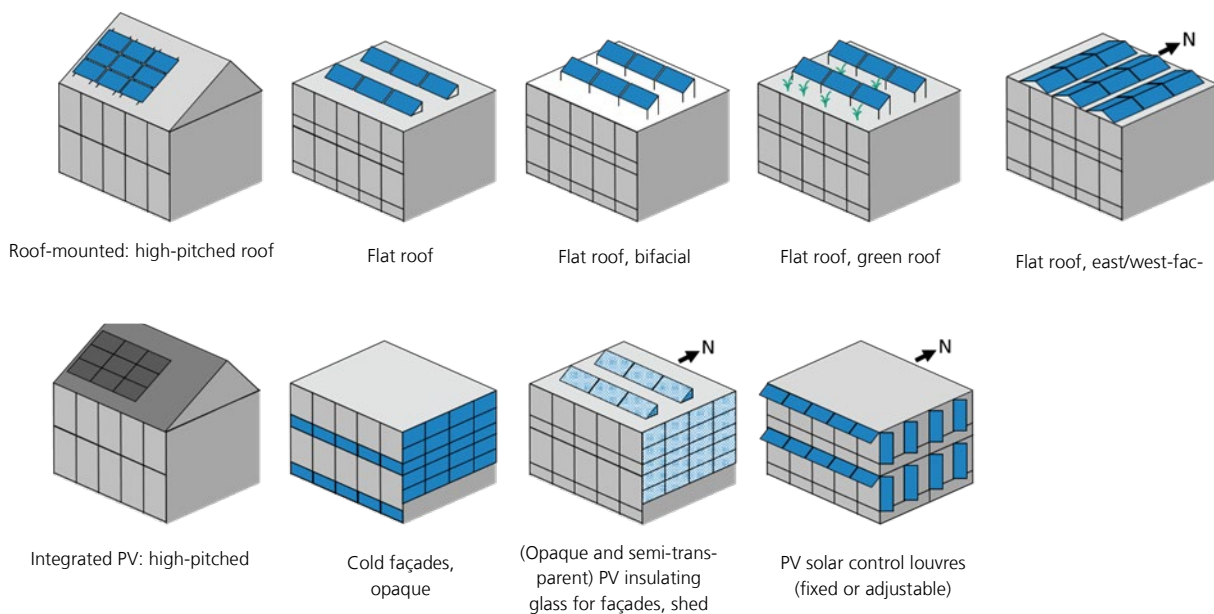


Figure 27: PV solutions for roofs and façades (top row: roof-mounted systems, bottom row: integrated PV).

A study by the Federal Environment Agency assumes 670 km² of sealed settlement areas [UBA10]. This includes built-up settlement areas, but not building areas or traffic areas such as roads or railways. Part of this area can be covered with PV modules to provide shade or can be covered with PV modules that can be walked on (UPV, "[urban PV](#)"). The more than 300,000 large car parks in Germany alone would open up the technical potential of **59 GW_p** if covered with PV modules.

Further potential on a GW scale is offered by the integration of [PV into traffic routes](#) (RIPV, from "Road Integrated PV"), including PV noise barriers, horizontal surfaces (as PV canopies or road surfaces) and rail tracks. PV canopies are particularly suitable for tunnel entrances and for highly emitting motor roads in urban areas. With the switch to electric mobility, the envelope surfaces of electric vehicles are being added as [vehicle-integrated PV](#) (VIPV).

Which part of the aforementioned technical potential can be used practically depends on complex economic, regulatory and technical boundary conditions, as well as questions of acceptance. In principle, integrated PV, which merges with the envelope of buildings, traffic routes and vehicles, uses land together with agriculture or occupies water surfaces in flooded opencast mines, will have somewhat higher electricity generation costs than simple open-space power plants. On the other hand, integrated PV avoids conflicts of use and creates synergies, for example by replacing a building façade, using the substructure of a noise barrier, or increasing the range of electric vehicles.

From a current energy law perspective, the available potential for ground-mounted PV includes verges along motorways and railway lines, conversion areas and, if a federal state uses the state opening clause of the EEG, also disadvantaged agricultural areas. In Baden-Württemberg alone, the restriction-free area suitable for PV FFA according to these criteria amounts to 3850 km² (<https://www.energieatlas-bw.de/sonne/freiflachen/potenzialanalyse>). This is predominantly permanent grassland and arable land according to the state-specific "Open Space Ordinance" (FFÖ-VO). With an occupancy density of 0.6 MW_p/ha, this area absorbs **230 GW_p** PV, for example as agri-photovoltaics or [biodiversity PV](#) (Section 14). Current figures for the whole of Germany are not yet available. A study commissioned by the Federal Ministry of Transport and Digital Infrastructure with figures from 2014 estimated the expansion potential of restriction-free open spaces for PV at 3164 km² [BMVI]. In its brochure from 2020, the German Golf Association (DGV) estimates the area of golf courses in Germany at 48,000 hectares. In a report, the German Federal Environment Agency estimated that the total area used for ground-mounted PV systems was around 32,000 hectares by the end of 2021.

13 Is PV electricity a privilege reserved for homeowners?

No.

There are various operator models for PV systems that are suitable for co-owned roofs of condominium buildings, individual owners and construction companies [EAFR]. Red tape has been reduced and the EEG has created incentives for landlord-to-tenant electricity supply. In addition, tenants can install plug-in solar panels (balcony modules), on their balcony balustrades, walls or terraces. These solar modules can be plugged into the

tenant's household grid to supply electricity for self-consumption, lowering the tenant's electricity bills (<https://www.pvplug.de/>). Depending on the intended location of the module, the landlord's permission may be required. In 2024 more than 780,000 plug-in solar systems were registered. It is estimated that the actual number is two or three times higher.

If a building is not suited for a PV installation, community energy cooperatives provide the opportunity to contribute to the construction of PV power plants. Anyone who is not interested in investing can still choose a utility company with a sizable portfolio of PV power plants, for example, Elektrizitätswerke Schönau (EWS) or Energie Baden-Württemberg (EnBW).

14 Do PV systems destroy ecologically valuable areas?

No, quite the opposite, they usually promote renaturation.

If an area is taken out of intensive agriculture (energy crop cultivation) and converted into grassland on which an open space PV plant (OSPP) is installed, then biodiversity basically increases [ESD]. Photovoltaic systems with a biodiversity net gain are called biodiversity-friendly photovoltaics, or Biodiv-PV. Fertilizer is no longer used, and less demanding plants are given a chance to thrive. Fencing around the OSPP protects the area against unauthorized access and free-roaming dogs, which is advantageous for ground-nesting birds, for example.

Further improvements can be achieved through minor adjustments to the [Biodiv-PV](#) system. Increasing the row spacing between the module racks, raising the modules slightly, planting wildflower mixtures instead of a grass monoculture, and practicing careful landscaping allow for a biotope to develop. The rows of modules create a sequence of microhabitats with varying levels of solar radiation, temperature, and humidity. Wider row spacing allows for a steeper module tilt, resulting in higher electricity yields during the winter months, when solar electricity has a higher market value. The steeper tilt also contributes to lower yield losses due to soiling and snow cover. Biodiv-PV is suitable for areas with poor soil quality, while agriculturally high-quality soils are better suited for agrivoltaics (APV). According to the Federal Agency for Nature Conservation, peatland soils in Germany cover 1.4 million hectares, of which about 50% is used as grassland and 25–30% as arable land. Draining peatlands and repurposing them for intensive agricultural use dramatically increases the CO₂ emissions of the area. Just under 7% of Germany's greenhouse gas emissions currently originate from drained peatlands (Source: Federal Environment Agency 2022). PV plants can assist in restoring degraded peatlands as part of a rewetting process ([Peatland PV](#)). The partial shading provided by the PV panels counteracts drying, therefore reducing the breakdown of peat. Revenue from electricity production helps to offset the yield revenue lost during the transition from extensive agriculture to sustainable forestry and agriculture on rewetted peatlands. In Deutschland there are approximately 334,500 ha of arable land on degraded peatlands. Considering a system

density of 0.4–0.8 MW_P/ha, this represents a technical potential of 134–268 GW_P of PV capacity [ISE10].

15 Do PV power plants find acceptance in the population?

Yes.

The free scalability of PV power plants enables decentralized expansion, down to small balcony module or plug-in PV systems with a nominal power of a few hundred watts. At the end of 2024, 4.8 million PV systems were installed in Germany, of which 66 percent are small systems with outputs below 10 kW_P [ISE5]. This high number shows that the technical possibilities are being used extensively. Photovoltaics is ideal for implementing the concept of community energy and promoting acceptance of the energy transition through participation.

According to a representative survey by Lichtblick, solar systems are among the most popular power plants. Figure 27 shows the distribution of answers to the question “When you think about the construction of new energy generation plants in Germany, on which types of plants should the focus be?”

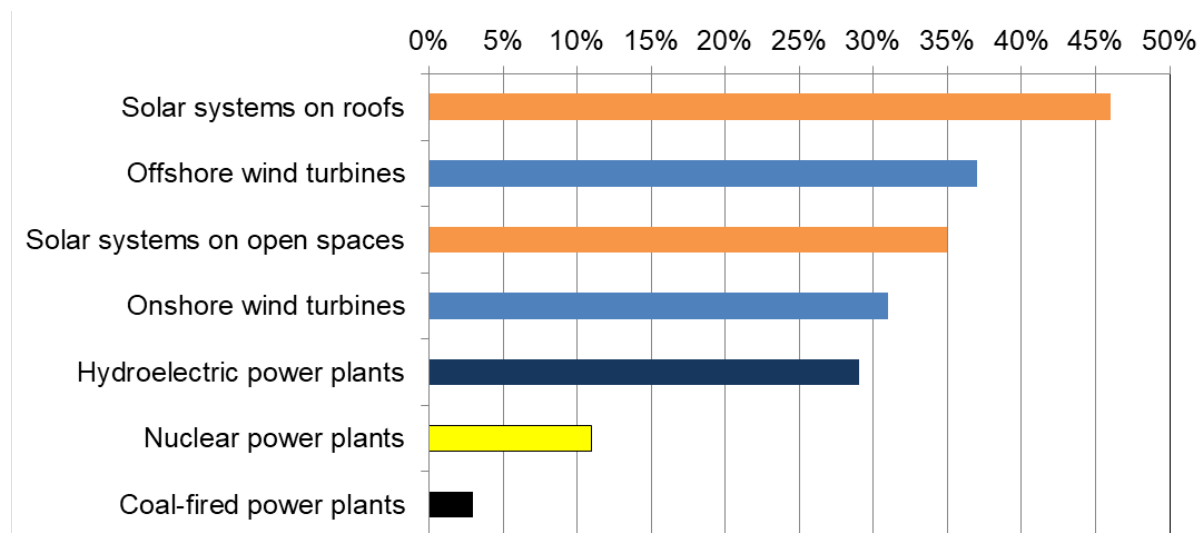
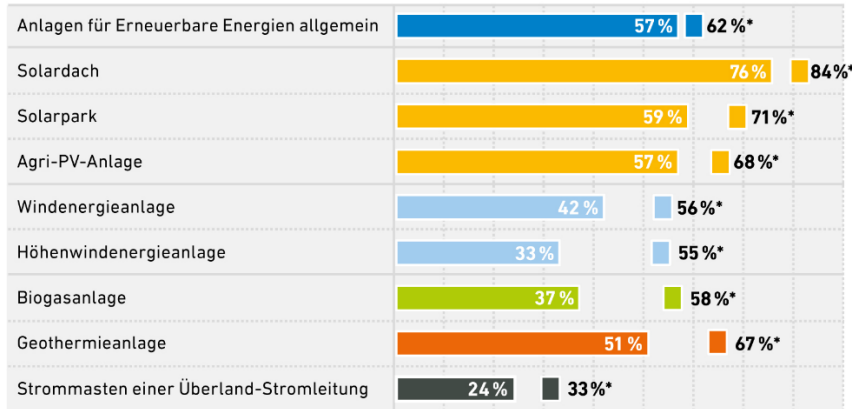


Figure 28: Survey results on the construction of new power plants, data from [Licht2].

Zustimmung zu Erneuerbare-Energien-Anlagen in der Umgebung des eigenen Wohnorts

Zur Stromerzeugung in der Nachbarschaft bis 5 km finden eher gut bzw. sehr gut...



* Befragte mit entsprechenden Anlagen in der eigenen Nachbarschaft.

Quelle: Umfrage von YouGov im Auftrag der
Agentur für Erneuerbare Energien, n=1.012; Stand: 11/2023
© 2023 Agentur für Erneuerbare Energien e.V.



Figure 29: Survey results on the acceptance of different types of power plants [AEE].

PV power plants are also by far the most popular power plants from the point of view of residents, as a survey by the Renewable Energy Agency shows (Figure 28). The popularity increases when such power plants can be practically experienced in one's own neighborhood.

16 Are PV plants in Germany efficient?

That depends on what kind of efficiency you mean.

In terms of the solar energy incident on the panels, the actual efficiency of energy conversion in PV modules is only around 18–19% for newer systems, but the sun shines for free, so there are no fuel costs.

More relevant parameters are the relationships between the amount of electricity generated on the one hand and the expenditures on the other, e.g., costs, space requirements, CO₂ emissions, material and energy usage, grid connection capacity, etc. These parameters provide targets for technology optimization and allow comparisons with other energy technologies.

The amount of electricity generated by a PV power plant (annually and over its entire service life) depends on numerous factors, in particular:

- Irradiation and temperature
 - Annual total irradiation at the location
 - Orientation of the PV modules
 - Irradiation on the rear side of the module (only for bifacial modules)
 - Reduction in irradiation due to shading, soiling, and snow cover
 - Ambient temperature and heat dissipation

- PV modules
 - Nominal efficiency
 - Change in efficiency during operation (module temperature, low light, oblique light, inhomogeneous irradiation)
- Inverters and cables
 - Nominal efficiency
 - Change in efficiency during operation (thermal curtailment behavior, partial load operation)
 - Power and line losses
- Operation and maintenance
 - Degradation of PV modules and inverters
 - Downtime due to technical malfunctions
 - Curtailment of electricity production due to infrastructure and market bottlenecks

According to figures from the German Weather Service, the average annual total horizontal global radiation in Germany for the years 2001–2020 is **1102 kWh/(m²·a)**, with a linear trend of +0.3%/annum between 1991 and 2020. In the period from 1981 to 2010, the average value was still 1055 kWh/(m²·a). Global radiation varies in Germany by approximately $\pm 10\%$ depending on the location.

In open-space installations, PV modules are often mounted rigidly at an angle of 20° - 25° to the horizontal and oriented towards the south to optimize the annual electricity yield per installation area. This increases the total irradiation on the module by approximately 15% relative to the horizontal total irradiation, resulting in a geographical average for Germany of approximately 1270 kWh/(m²·a). To achieve even higher irradiation, modules are tracked in one or two axes according to the position of the sun or the brightest section of the sky.

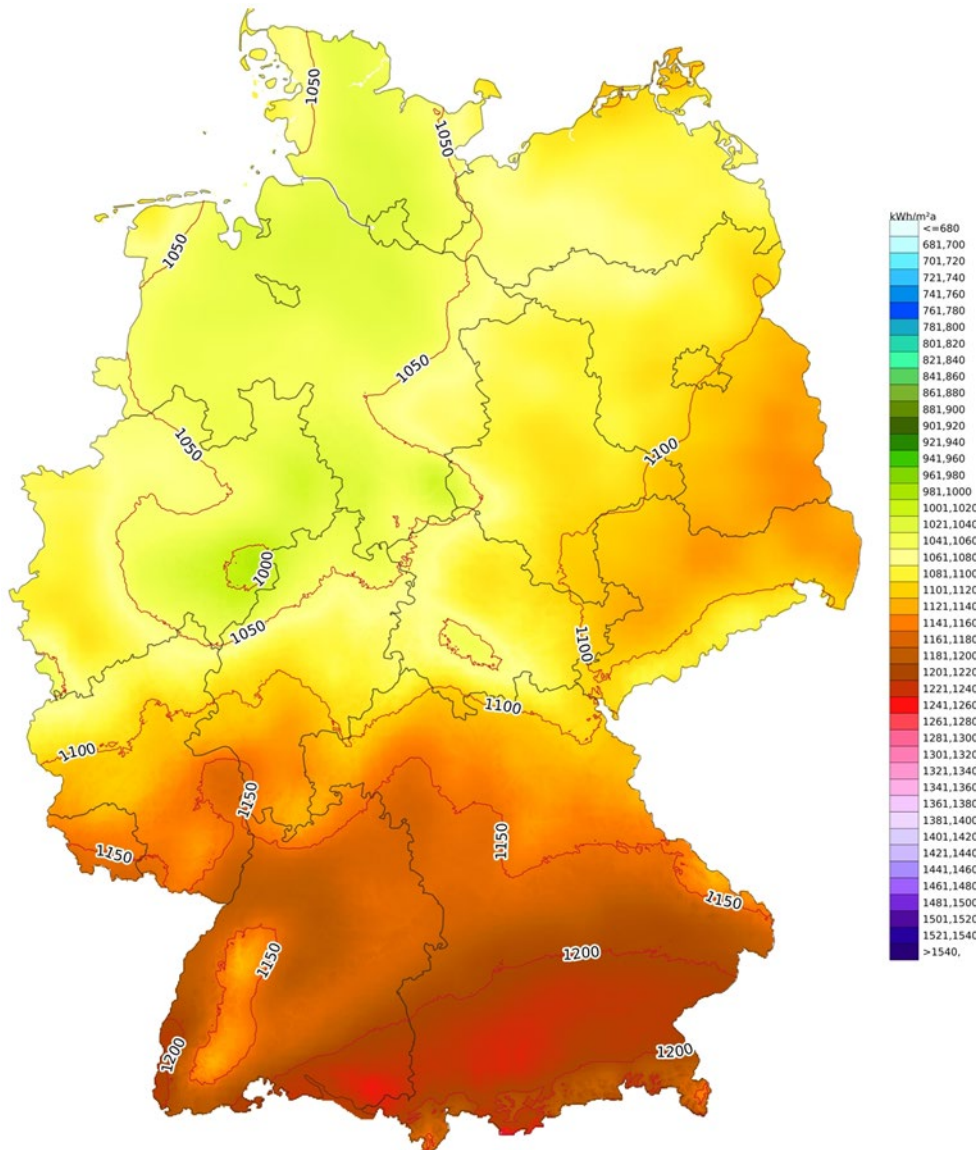


Figure 30: Annual horizontal global radiation in Germany, averaged over the period 2001 –2020, Data source: [DWD].

The nominal efficiency (see Section 28.1) of commercial wafer-based PV modules (i.e., modules with solar cells based on silicon wafers) from new production has increased in recent years by an average of approx. 0.5 percentage points per year to reach **22 percent** in 2024 [ISE5]. This corresponds to a nominal output of 220 W per square meter of module. The efficiencies of the best modules are approx. 10 percent higher. However, PV systems do not operate at the nominal module efficiency, because of operational losses and degradation effects (Section 16.2). These effects are summarized in the so-called performance ratio (PR). A PV system installed today achieves PR values of **80–90 percent** on an annual average (typical value), including all losses due to increased operating temperature, unfavorable irradiation conditions in terms of intensity, spectrum and angle of incidence, soiling, shading, snow accumulation, line resistance, conversion losses in the inverter, moderate derating (DC/AC ratio typically > 1) and, in some cases,

downtimes due to malfunctions. Additional yields increase the PR of bifacial modules through rear-side irradiation, as the reference value of the PR only factors in front-side irradiation. [Yield assessments](#) provide information on the performance and yields that can be expected from PV power plants for concrete locations, components and system designs based on historical irradiation data.

The DC current delivered by the modules is adapted by inverters for grid feeding. The efficiency of new PV inverters is around 98 percent. Ground-mounted PV systems usually show slightly higher PR than systems on pitched roofs, thanks to better convective cooling, optimal orientation, better maintenance, less of shading, more efficient inverters, and possibly bifacial additional yields. In terms of irradiation, newly installed PV power plants therefore operate with average efficiencies of around **19 percent**.

The average electricity consumption of a two-person household for electrical appliances, lighting, hot water (hygiene purposes) and space heating was 3.5 MWh in 2021 [DESTA-TIS]. The per capita values for one-person households are higher, for multi-person households it is lower. On average, rooftop PV systems achieve **922 hours of full** use in 2021 in the trend scenario [ÜNB1], (compare Section 16.4). Ten modules á 400 Wp on a house with a roof area of 18 m², facing roughly south and with a moderate slope, are sufficient to generate the electricity that corresponds to the average annual electricity demand of a two-person household (3.5 MWh).

On flat roofs and in open spaces, modules are elevated to increase their yield. Because of the spacing required for this, they occupy several times their own area when oriented to the south, depending on the angle of installation. Today, PV FFAs are usually built with reduced tilt angles (approx. 20°–25°) and row spacing, resulting in an occupancy density of around 1 MW/ha at module efficiencies of 21 percent. In 2010, this value was still 0.35 MW/ha [ZSW]. With a view to optimal development of biodiversity, larger row spacing is advantageous (Section 13).

By way of comparison, the efficiency of electricity generation from energy crops based on irradiation is well below 1 percent; for electricity generation from energy maize, for example, it is 0.2 percent. The efficiency related to irradiation for the conversion of fossil organic matter such as coal, oil or natural gas into electricity is likely to be in a similar order of magnitude. However, corresponding combustion power plants normally only relate their efficiency figures to the conversion of the chemical energy already present in the fossil energy source. For coal-fired power plants in Germany, for example, an average efficiency of around 38 percent is given.

When biofuels are burnt in vehicles, modest efficiencies are achieved in relation to the energy radiated and the land use. A passenger car with a diesel combustion engine that consumes 5.5 l of biodiesel per 100 km can travel about 32000 km with the annual yield of a 1-hectare rapeseed field of 1775 l/(ha*a) [FNR]. With the annual yield of a new PV system (1 MW_P/ha, 980 MWh/MW_P) on the same area, a battery electric vehicle (e-car, consumption 16 kWh per 100 km) travels approx. 6.1 million km, the range is higher by a **factor of 190** (Figure 30). Even an agri-PV system (Section 11), which allows simultaneous agricultural use of the land, would take an e-car a **factor of 116** further. This comparison does not take into account charging losses for e-cars, energy consumption for rapeseed cultivation or energy supply from by-products of rapeseed use.

If one compares the efficiency of land use for electricity production, then agri-PV performs better than maize by a **factor of 32**, for example. Silage maize, which is cultivated on an area of approx. 0.9 million ha in Germany, yields 18.7 MWh_{el}/ha of electricity [FNR], whereas the figure for high altitude APV (Section 11) is approx. 600 MWh_{el}/ha. This comparison does not consider waste heat utilization (CHP) from methane combustion.

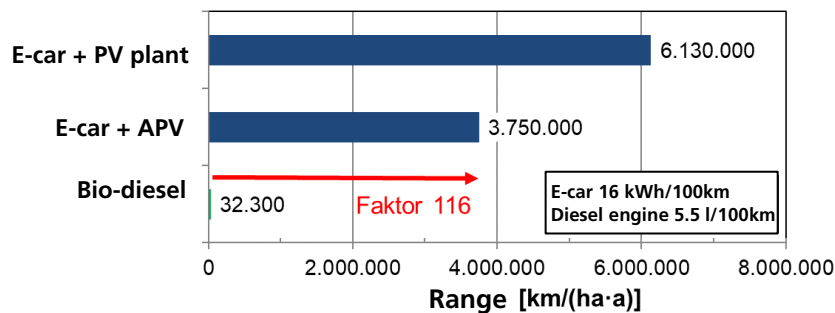


Figure 31: Range of electric and biodiesel vehicles per hectare of land used.

In southern Spain or North Africa, specific yields of up to 1600 kWh/kW_p can be achieved, but long transmission lines to Germany would lead to energy losses and cost surcharges. With 800 kV extra-high voltage lines, power losses can be reduced to about 0.5 percent per 100 km. Lines for high-voltage direct current transmission (HVDC) reduce transport losses to just under 0.3 percent per 100 km, plus conversion losses. A 5000 km HVDC line would thus have about 14 percent pure line losses.

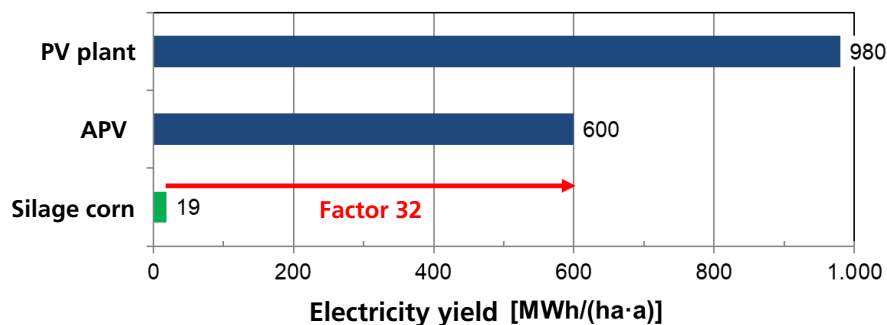


Figure 32: Electricity yields of PV power plants and silage maize per hectare of land used.

16.1 Are PV systems only viable on roofs that are optimally oriented?

No, it is worth considering the overall profitability.

Figure 32 shows the relative annual yield potential in Freiburg in relation to the orientation of the roof or façade area. The maximum yield with a south orientation and an inclination of 40° shows a flat line, with small deviations having little effect on the yield. Installations located in more northern latitudes require a steeper inclination to achieve their maximum yield.

		Orientierung																		
		Ost		Südost			Süd			Südwest			West		Nordwest			Nord		
		-90°	-75°	-60°	-45°	-30°	-15°	0°	15°	30°	45°	60°	75°	90°	105°	120°	135°	150°	165°	180°
Horiz.	0°	84%	84%	84%	84%	84%	84%	84%	84%	84%	84%	84%	84%	84%	84%	84%	84%	84%	84%	84%
	10°	83%	85%	87%	89%	90%	91%	91%	91%	90%	89%	87%	85%	83%	81%	79%	77%	76%	75%	75%
	20°	82%	86%	90%	92%	95%	96%	96%	96%	94%	92%	89%	85%	81%	77%	73%	70%	67%	66%	65%
	30°	81%	86%	90%	94%	97%	99%	99%	98%	96%	93%	89%	84%	79%	74%	68%	63%	59%	57%	56%
	40°	78%	84%	90%	94%	97%	100%	100%	99%	97%	93%	88%	82%	76%	69%	63%	56%	51%	48%	47%
	50°	74%	81%	87%	92%	96%	98%	99%	97%	95%	91%	85%	79%	72%	65%	57%	50%	44%	40%	39%
	60°	70%	77%	83%	88%	92%	94%	95%	94%	91%	87%	81%	75%	68%	60%	52%	45%	38%	33%	31%
	70°	64%	71%	77%	83%	86%	89%	89%	88%	85%	81%	75%	69%	62%	54%	46%	39%	32%	27%	26%
	80°	57%	64%	70%	75%	79%	81%	81%	80%	77%	73%	68%	62%	55%	48%	40%	33%	27%	23%	21%
Vert.	90°	50%	56%	62%	66%	69%	70%	71%	70%	68%	64%	60%	54%	48%	41%	34%	28%	23%	19%	17%

Figure 33: Relative yield potential, no shade, located in Freiburg, calculated online using https://re.jrc.ec.europa.eu/pvg_tools/en/.

16.1.1 High-pitched rooftops facing east/west

As seen in the direct comparison in Figure 32, PV on a west-facing roof with an inclination of 45°, for example, annually produces about 26 percent less electricity than a south-facing roof with the same angle of inclination. The levelized cost of electricity is accordingly increased. A PV installation on an east or west-facing roof can still be worth-while, in particular if electricity prices are high and self-consumption is employed.

The question of whether an east or west-facing roof should be used in addition to a south-facing roof can be answered by reviewing marginal costs. Additional PV modules can be expected to come at comparatively lower installation costs, and their orientation towards the west and late-afternoon generation profile should increase the self-consumption potential ().

16.1.2 Vertical south-facing façade

A PV system on a vertical south-facing façade produces about 29 percent less electricity per year than an installation with a module inclination of 45° on a south-facing roof (Figure 32). The same applies to vertically mounted, south-facing plug-in solar panels. Compared to a horizontal installation, the seasonal fluctuations in the energy yield of a vertical installation is considerably smaller (Figure 47), which makes incorporating self-use or system integration easier. The lower yield of vertical systems occurs during the sunny half of the year, when the “overproduction” of installations on south-facing roofs leads to a lower share of self-consumption, or rather when the market value factor of solar energy decreases (Section 4.3). The financial viability of a façade PV installation is therefore only marginally affected by the reduced annual electricity yield from vertical facades.

16.2 Do PV modules degrade?

Yes, but normally very slowly.

Wafer-based PV modules age so slowly that it can be a challenge for scientists to even prove power losses. A study by Fraunhofer ISE on 44 larger, quality-tested rooftop systems in Germany showed an average annual degradation of the nominal power of the modules of approx. 0.15 percent [ISE2]. Manufacturers' guarantees for a maximum power loss of their PV modules of 10–15 percent over 25–30 years of operation are common.

These values do not take into account failures due to production defects. Faults in the material composition or defects that arise during production, transportation or mounting may accelerate degradation or even lead to the modules failing altogether. Degradation may affect the appearance of the modules (e.g., delamination), their yield and/or their electrical safety. A [damage analysis](#) serves to quantify deviations, identify causes and create forecasts for further degradation dynamics.

The declared nominal power of modules usually refers to operation after initial degradation. Depending on the material of the solar cells, light-induced degradation (LID) of 1–2 percent occurs in the first days of operation, as extensive measurements at Fraunhofer ISE have shown.

For systems with quality assurance for component selection, design, installation and operation, the frequently made assumption of 0.5 percent average yield loss per year seems very conservative in this context.

16.3 Can PV modules become soiled?

Yes.

In many cases, this keeps the resulting yield loss during the modules' service life at a manageable level. Local effects such as leaf litter, bird droppings, dust from construction sites or agriculture, soot from nearby chimneys, regional effects such as pollination or trans-regional effects such as Saharan dust events may cause problematic soiling. While heavy rain will mostly clean the modules, soiling does accumulate over the years, increasing yield losses. This especially applies to regions experiencing longer periods without rainfall as a result of the climate crisis.

Modules with a very flat inclination (less than 15°) are particularly affected. The bottom edge and especially the bottom corners of modules are the areas that collect the most dirt. When solar cells are connected in series, extreme soiling of partial surfaces can cause yield losses that are much higher than should be the case in relation to the surface area. Whether, when and how frequently it pays to have the modules cleaned varies depending on many influential parameters.

16.4 Do PV power plants seldom operate at full capacity?

Yes, but new technologies allow for significantly higher availability.

The parameter "full-load hours," (FLH), also known as "full utilization hours" or "specific yield" of a power plant, is the amount of annually generated electricity multiplied by the nominal power plant output on the AC side (kWh/kWp).

The higher the FLH of a power plant, the more efficiently the downstream infrastructure is used, especially grid connections. Nuclear, coal, and gas-fired power plants can produce at rated capacity almost continuously, if necessary, provided that sufficient fuel and cooling water are available. Controllable power generators such as gas power plants (biogas, biomethane, natural gas) are used to cover the residual loads, where moderate FLH is advantageous for the power system.

For volatile power generators such as solar and wind, the FLH is an important measure of availability. However, temporary deficits in the usability of volatile electricity, whether due to grid bottlenecks or a lack of flexibility in demand and storage, force curtailment and thus reduce the FLH of power plants. We therefore distinguish between available FLH, which refers to the amount of electricity provided, and usable FLH, which includes forced curtailment on the demand side.

In addition to controlled curtailment, PV power plants often also have an internal, permanent power limitation due to the software and, in some cases, the hardware of the inverter (DC/AC ratio > 1). This power limitation reduces the plant's nominal power on the AC side but can have a negative effect on the electricity produced, depending on the storage capacity and the module orientation. The effect of the FLH on the nominal power output prevails, however, so that power limitation increases the FLH.

16.4.1 Available Full-Load Hours (FLH)

Due to irradiation conditions, PV modules only operate for slightly less than half of the total 8,760 hours per year, and then mostly at partial load. The size of a PV power plant has no direct influence on the number of full-load hours, unlike wind power plants, where hub and tower heights are decisive.

A new, flat south-facing PV power plant with an irradiation at the module level of 1270 kWh/(m²·a) and a performance ratio (PR, see [Section 28.7](#)) of 85% would achieve approx. 1080 FLH without any curtailment.

By tracking the modules, the available FLH can be significantly increased through higher annual yields (Sections [22.3.1](#), [22.3.7.1](#)). With a standard tracking system with a horizontal axis and axis-parallel module arrangement, the FLH can be further increased by providing a larger DC/AC ratio for the inverter (e.g., 1.5). This allows 2000 FLH to be approximated with low internal curtailment losses. With additional support from stationary storage, the DC/AC ratio of the inverter can be further increased, for example, to 2, which brings the available FLH to around 2500.

Even with conventional PV-FFA with south-facing modules, high DC/AC ratios with low internal curtailment losses can be achieved in combination with batteries, which almost doubles the FLH of the power plants.

The sharp price drop in stationary batteries is making hybrid power plants the new standard. They have significantly higher FLH than pure PV power plants and higher electricity yields thanks to a shifting in grid feed-in to the evening hours.

Very high FLH (in the order of 2000) can also be achieved with bifacial PV modules in a vertical east-west arrangement with a high DC/AC ratio of the inverter. Storage is not necessary here because there is no dominant generation peak at midday. This design was made possible by the availability of PV modules with a high bifaciality factor (> 90%).

Many yield-enhancing measures and developments based on the influencing factors mentioned at the beginning (section 16) also increase the available FLH.

16.4.2 Useful Full-Load Hours (FLH)

Due to often uniform weather conditions across regions and the predominant south-facing orientation of PV modules, PV electricity production has a high simultaneity factor, with a pronounced peak at midday.

Since only a few battery storage systems are connected to the grid or operated in a grid-friendly manner today, periods of high solar power production can lead to grid bottlenecks and, due to a lack of flexible reserves on the demand side, also to sales bottlenecks for solar power. It is these sales bottlenecks that, in turn, lead to very low or even negative prices on the electricity exchange.

As a protective measure in critical situations, many controllable PV power plants are curtailed, which reduces the amount of electricity generated and the full-load hours of the power plant.

The losses from flexible curtailment are an important reason for transmission system operators to assume, in their trend scenario for 2025, only 864 FLH for ground-mounted PV systems in Germany in 2026, and 895 FLH for rooftop systems, with a further downward trend [TSO]. For onshore wind energy, 1904 FLH are expected in 2026 in the trend scenario, and 3125 FLH for offshore wind energy.

17 Does PV make relevant contributions to climate protection?

17.1 Do anthropogenic CO₂ emissions danger the climate?

Yes.

Increasing global warming has been proven beyond doubt [IPCC]. Compared to the pre-industrial era, the average global temperature has risen by 1.1 °C, and by as much as 1.6 °C over the land surface. The vast majority of scientists are convinced that anthropogenic emissions of CO₂ and other greenhouse gases are causing the increase in atmospheric greenhouse gas concentrations and thus the global temperature rise.

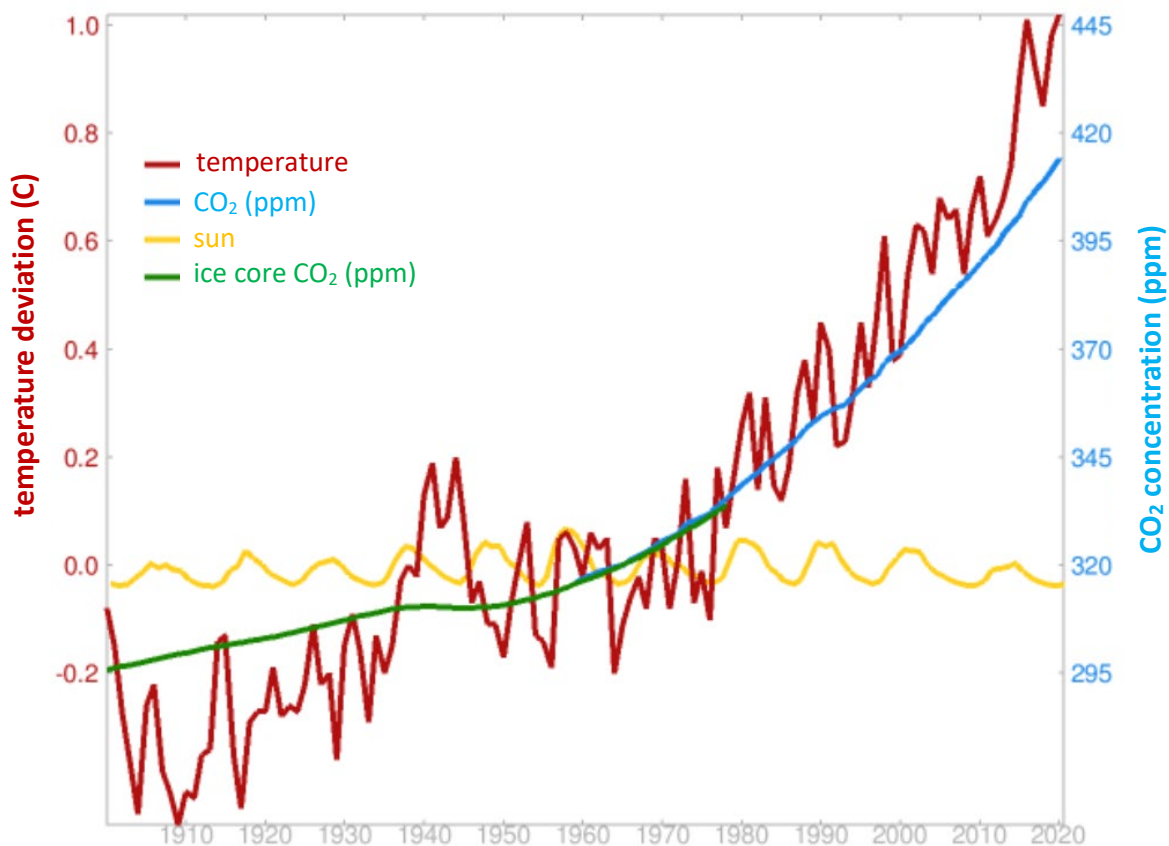


Figure 34 Development of atmospheric CO₂ concentration, mean global temperature change and solar activity (<http://herdsoft.com/climate/widget/>).

In 2025, the concentration of CO₂ in the atmosphere reached its highest level in the last two million years [WMO]. Figure 33 and Figure 34 show the development of the atmospheric CO₂ concentration alongside the global, or rather, the Antarctic temperature.

A rapid rise in the global temperature endangers the stability of the global climate system to a hitherto little understood extent, with effects on the lives of people threatened by extreme weather conditions, the world population's primary food production, infrastructures, populated coastal zones, adding additional pressure on biodiversity and the preservation of biotopes.

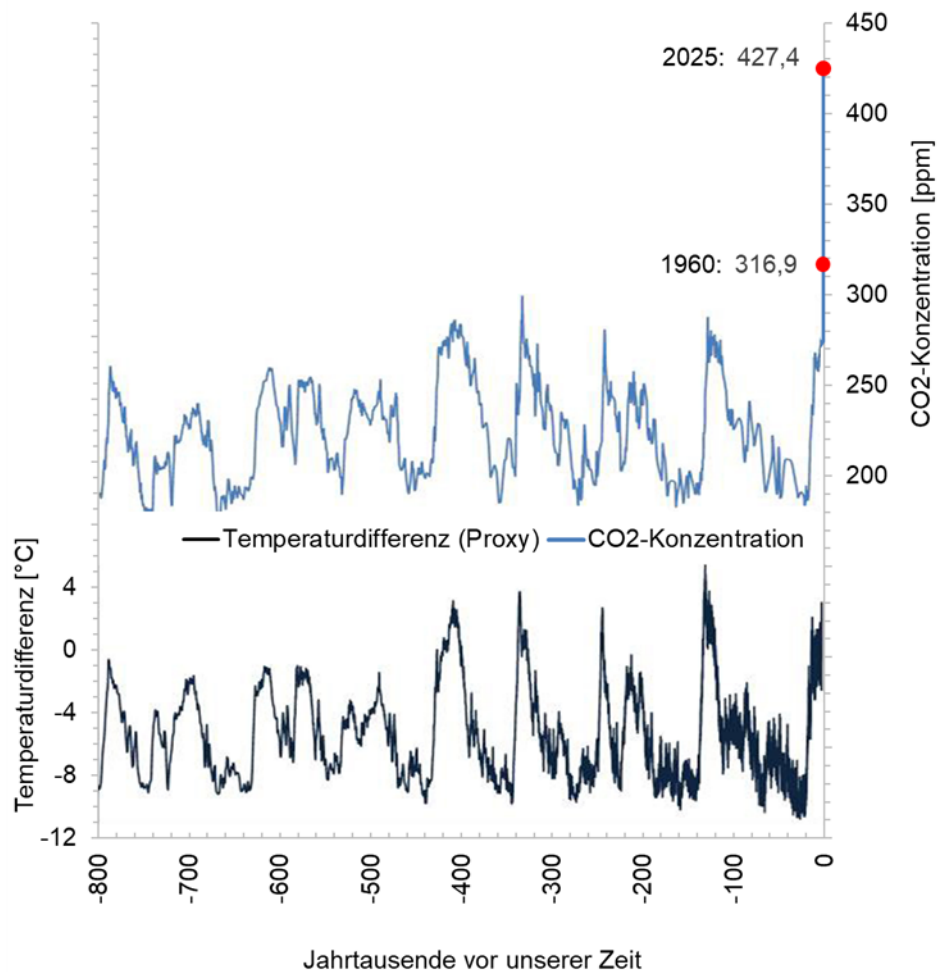


Figure 35: Estimates of atmospheric CO₂ concentration and temperature difference in Antarctica based on ice cores [EPA]; red: CO₂ measurements from Mauna Loa Observatory [https://www.esrl.noaa.gov/gmd/ccgg/trends/data.html].

17.2 Does PV make a significant contribution to CO₂ emission reduction?

Yes.

The CO₂ equivalent (GHG) emission factor for the electricity mix in Germany, including upstream emissions, has dropped from 860 g CO₂ e/kWh in 1990 to about **498 g CO₂/kWh** in 2022 [UBA6]. The expansion of RE has been a major contributor to this reduction.

To estimate the overall emission factors of individual fuels for electricity generation, it is necessary to differentiate, amongst other factors, between the country of origin, the materials handling technology used, the primary energy concentration, the primary energy content of the raw material deposits, methane leaks during natural gas production, as well as the total (for combined heat and power) and the effective electrical efficiency of power plants.

While PV systems do not release any CO₂ during operation, a holistic assessment must also take into account the production and disposal of the system. The specific yield [kWh/(kWp·a)] and thus the annual irradiation sum at the operation site play a major role,

as does the lifetime of the PV system. Thanks to advances in production and efficiency, the carbon footprint of PV electricity has been dramatically reduced in recent decades. It is therefore important to consider life cycle assessments (LCAs) of current technologies when evaluating new systems.

The IEA PVPS Task 12 has found that a large ground-mounted PV system in Piacenza in Northern Italy (GHI 1,368 kWh/(m²·a) with 85 MWP, modern PERC technology and 21.2 percent module efficiency, has an emission factor of 25.7 g CO₂-eq/kWh [IEA5]. In Germany, only about **30–35 g CO₂-eq/kWh** could potentially be achieved due to the somewhat lower solar radiation (approx. 20 percent lower on average).

The German Federal Environment Agency has compared the greenhouse gas potential of PV electricity from ground-mounted and pitched roof mounted PV systems based on older data and only found little difference between the two ([UBA9], see Figure 35). PV modules produced in Europe together with their feedstocks are particularly favorable because the electricity mix here contains a higher share of renewable energy and the transport distances are significantly shorter. Glass-glass modules are favorable due to their lower degradation and the absence of aluminum frames [LCA]. With the continued increase in efficiency and the share of RE in production, GHG emissions per kWh of PV electricity will continue to decrease.

Roof-mounted and ground-mounted systems, solar radiation 1,200 kWh/(m²·a)

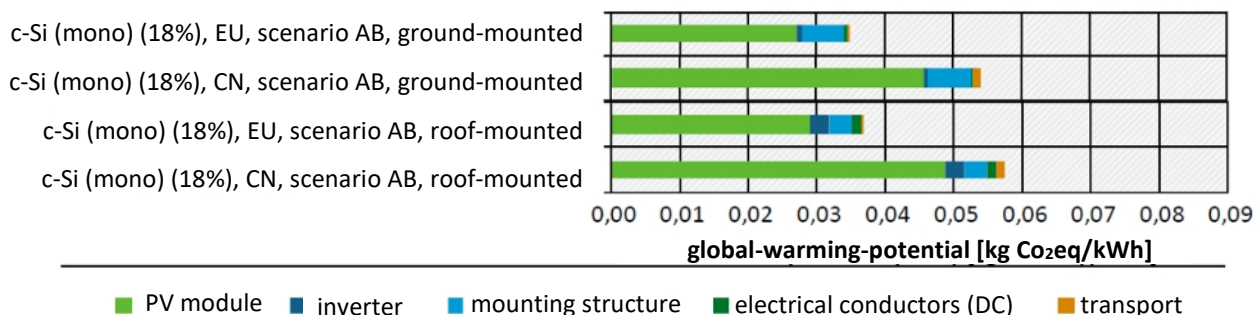


Figure 36: Global warming potential of electricity generation with mono c-Si PV for roof-mounted and ground-mounted systems [UBA7].

Figure 36 shows the GHG potential of PV compared with coal, natural gas and nuclear power. The emission factor of power generation from biogas with energy crops is around 160–184 g CO₂-Eq./kWh [UBA9].

In 2022, the use of PV in Germany avoided a net 41,7 million tons of greenhouse gas emissions (Figure 37). In the calculations, the emissions from the production of the PV system components were taken into account approximately.

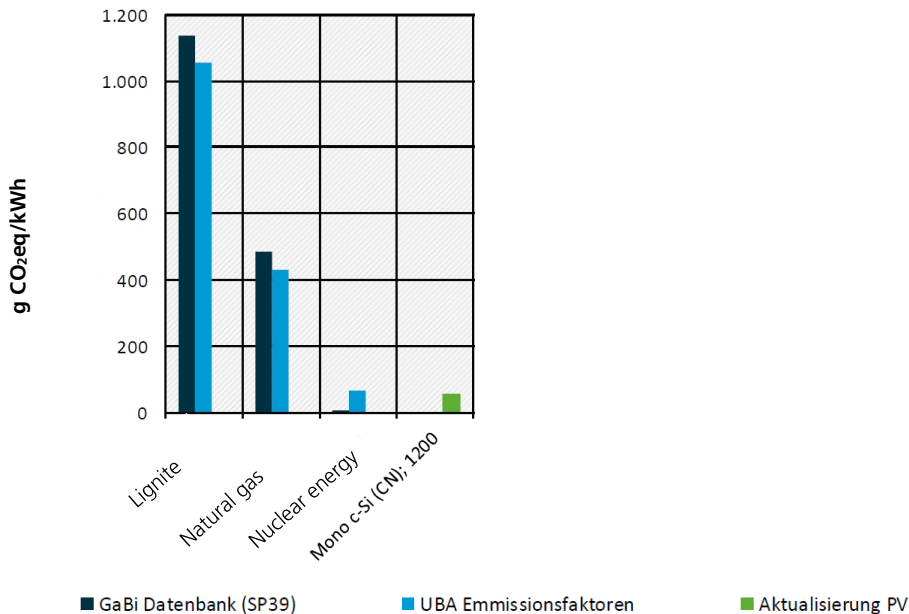
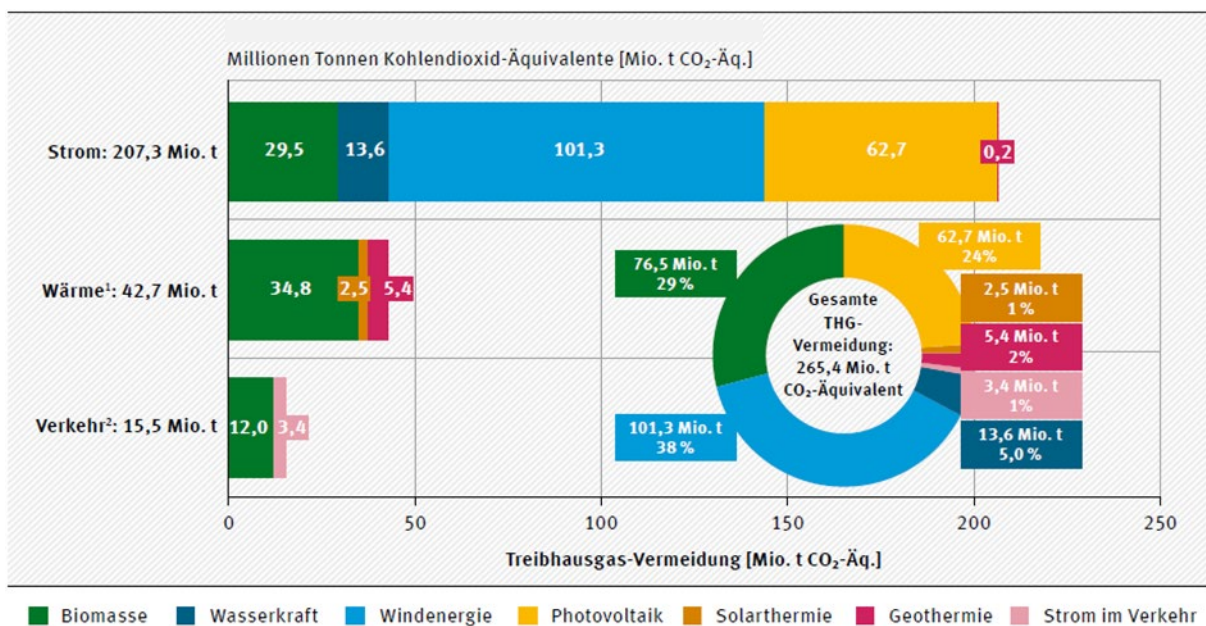


Figure 37: Global warming potential of different electricity generation technologies [UBA7].



¹ ohne Berücksichtigung des Holzkohleverbrauchs

² ausschließlich biogene Kraftstoffe im Verkehrssektor (ohne Land und Forstwirtschaft, Baugewerbe sowie Militär), basierend auf vorläufigen Daten der Bundesanstalt für Landwirtschaft und Ernährung (BLE) für das Jahr 2024 sowie den fossilen Basiswerten gemäß § 3 und § 10 der 38. BImSchV

Source: Umweltbundesamt (UBA)

Figure 38: Avoided greenhouse gas emissions through the use of renewable energies in 2025 [AGEE].

German energy policy also has a high international relevance. With a production volume of 171 Mt in 2016, Germany was number one internationally in the extraction of lignite, ahead of China. Germany accounts for less than 3 percent of global electricity consumption, and the trend is still downwards. However, German policy has played a pioneering role in the development of instruments to promote RE, first and foremost the EEG. The

EEG instruments have received considerable international attention and have served as a model for similar regulations in dozens of countries. China has now become a pioneer in PV expansion and has overtaken Germany many times over in terms of annual installed capacity. In its country report "Germany 2013", the International Energy Agency (IEA) praised the Renewable Energy Sources Act (EEG) as a very effective expansion instrument that has significantly reduced the costs of generating renewable energy in recent years [IEA1]. The Germans' departure from nuclear energy has also attracted international attention. Other European countries have decided to phase out nuclear power (e.g., Belgium, Switzerland, Spain) or have already done so (Italy, Lithuania). However, the EEG achieved the highest impact in terms of CO₂ avoidance in its early years through a "side effect": by creating the largest and most secure international sales market for PV over several years, it has significantly accelerated global scaling, technology development and price reduction (Figure 38). PV reduces the consumption of fossil raw materials for electricity generation worldwide.

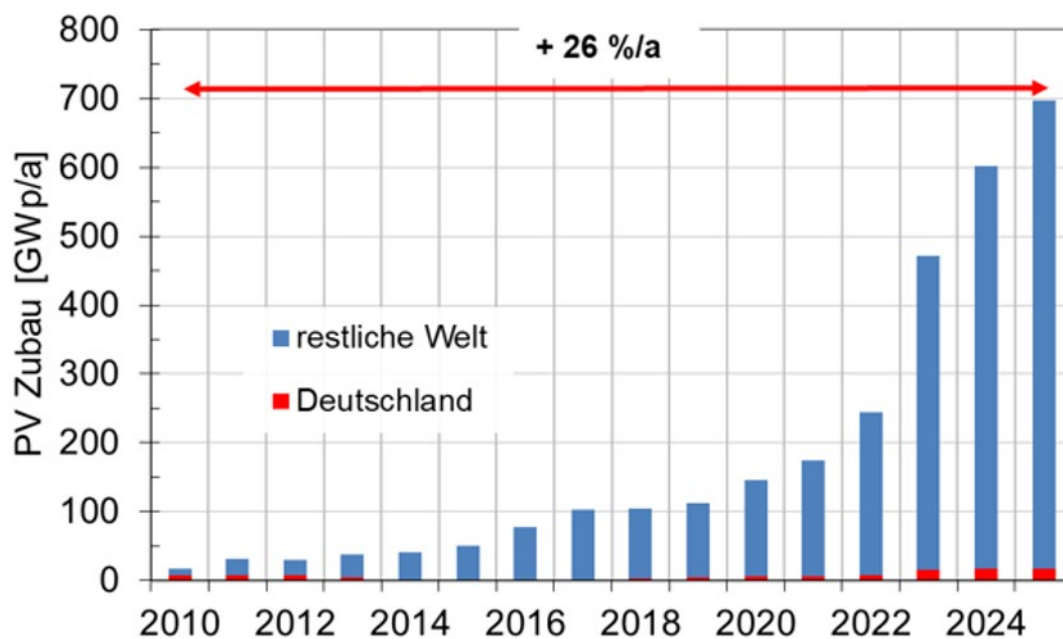


Figure 39: Development of annual PV additions [IEA3] and average annual growth rate.

The German EEG has thus made PV electricity more affordable for many people in developing countries. From this perspective, the EEG is also "probably the most successful development aid program of all time in this area" (Bodo Hombach in Handelsblatt 11.1.2013), which also saves considerable amounts of CO₂ in developing countries. According to IEA estimates, PV will become the world's second-largest source of renewable electricity in 2026, behind hydropower.

17.3 Does the production of PV modules use more energy than they can deliver during operation?

No.

The Energy Payback Time (EPBT) indicates the period of time that a power plant has to be operated in order to replace the primary energy invested. The harvest factor (Energy Returned on Energy Invested, EROEI or EROI) describes the ratio of the energy provided by a power plant and the energy expended for its life cycle.

The energy return time and harvest factor of PV plants vary with technology and plant location. An analysis commissioned by the Federal Environment Agency has determined EPBT for PV power plants for a plant operation in Germany (assumed average annual irradiation sum at the module level of 1200 kWh/(m²-a)) of 1.6 years for multi- and 2.1 years for monocrystalline Si modules [UBA7]. With a lifetime of 25-30 years and an annual yield degradation of 0.35 percent, this results in harvest factors of 11-18. Calculations by Fraunhofer ISE based on the latest production data indicate an EPBT of under **1.3 years** for systems with commercially available monocrystalline Si modules in Germany [ISE5]. Component production in Europe further reduces the EPBT due to the higher share of green electricity compared to imported components from China.

17.4 Are there other environmentally harmful gases released during the production of PV?

Yes, for some thin-film technologies, but their market share is below 5 percent.

In the production of thin-film PV, nitrogen trifluoride (NF₃) is still sometimes used to clean coating systems. Residual amounts of this gas can escape into the atmosphere. NF₃ is more than 17,000 times more harmful to the climate than carbon dioxide. Current emission levels are not known, but NF₃ emissions will be determined in 37 countries from 2013 in accordance with the amended Kyoto Protocol.

18 Do PV modules heat up the earth?

Light-colored surfaces reflect a large portion of the incident solar radiation while dark surfaces absorb more. As a result, they heat up more and radiate more heat into the surrounding air. The **solar albedo** (solar reflectance) of a surface indicates the percentage of incoming solar radiation that is reflected. The rest is absorbed or transmitted. In addition to the visible spectrum, the solar albedo also includes the UV and IR parts of the spectrum. Albedo values found in literature vary widely from surface to surface. According to the Swiss Federal Office of Meteorology and Climatology MeteoSwiss, arable land has an albedo of 7–17 percent, grassland 12–30 percent and forests 5–20 percent. The Thuringian Ministry of Environment estimates roof tiles to reflect 10–35 percent, light-colored roofs 60–70 percent.

18.1 Local effects

18.1.1 Do PV modules heat up their environment?

That depends on the reference material.

As most materials, PV modules absorb a large portion of the incident solar radiation. However, because they convert part of the absorbed radiation into electrical energy rather than heat, their effective albedo and thus the local generation of heat is similar in magnitude to that of typical surrounding materials or vegetation.

For example, when new PV modules convert solar radiation into electrical energy with an actual efficiency of 18 percent, and also reflect approx. 5–10 percent of the irradiation, they generate as much local heat as a comparable surface with an albedo of 23–28 percent. When it comes to PV modules in operation, this is referred to as **effective albedo**. In the example above, it ranges from 23 to 28 percent. Only when PV modules are not supplying electricity does their effective albedo drop to the value of their solar reflectance of 5–10 percent. Comparing the effective albedo of PV modules with the albedo of the surface area they cover helps determine whether the installation increases or reduces waste heat generation. Another important influential parameter is evaporative cooling, which plants use to cool themselves and their surroundings. This is true at least as long as the plants can draw enough water from the soil. After that, evaporative cooling stops and the plants dry out. Conversely, partial shading of plants by PV modules can reduce the water demand of the plants and the soil remains moist longer. This effect argues for a combination of PV with, for example, agriculture, green roofs and degraded moorland soils that are to be rewetted (Sections 12 and 14).

18.1.2 Do PV modules reflect solar radiation into the surrounding environment?

Only very little.

The latest designs of conventional PV modules have a solar reflectance in the order of 5–10 percent, depending on the choice of material and module design. Light-colored frames and back sheet foils increase reflection. PV modules are optimized to absorb as much solar radiation as possible in the active layer. Thermal insulation glazing, or better yet, solar control glazing, reflects many times more (in the order of 10–30 percent). If one compares a glass building façade with a PV façade, the PV façade reflects significantly less solar radiation downwards to the street level. Although PV modules can cause glare (Section 19), they reflect very little solar radiation.

18.1.3 Do PV modules store solar heat?

Only very little.

The heat storage capacity of ordinary PV modules is low compared to a solid concrete wall, for example. As a result, the PV module heats up much faster than a concrete wall under solar radiation, but also cools down faster in the evening. The actual impact of PV modules on the urban climate, compared to other building materials, depends on numerous factors and may need to be analyzed on a case-by-case basis.

18.2 Global effects

18.2.1 Does solar power production contribute to the greenhouse effect?

Yes, but much less than the electricity production of fossil fuel power plants and it will stop once the production of PV components is fully powered by green electricity.

In particular, PV electricity replaces electricity from fossil fuel power plants, which release up to 1.1 kg CO₂/kWh_{el}, depending on the fuel. The CO₂ is released into the atmosphere, where it acts as a greenhouse gas for an unlimited period of time (Section 17.1). Compared to fossil power plants, electricity generation via PV massively reduces the release of CO₂ and thus effectively slows down the greenhouse effect (Section 17).

18.2.2 Do PV modules heat up the earth's atmosphere?

That depends on the albedo of the area they cover.

When PV modules are installed, their effect on local warming corresponds to the change in albedo. Dark PV modules with an actual efficiency of 18 percent and an albedo of 5 percent covering a light-colored tile roof with an albedo of 30 percent (unfavorable combination) generate 400 Wh additional waste heat, or 40 percent of the produced electricity. To compare, thermal power plants (coal, nuclear) produce around 2 kWh of waste heat per kWh of electricity, or 200 % of the produced electricity. Solar power production generates less waste heat than thermal power production, even when the modules are installed on relatively light-colored surface areas.

19 Can PV modules cause glare?

Yes.

Similar to window glass, the cover glass of PV modules can cause glare, depending the sun's position, the orientation of the module, glass texture and the observer's field of vision. The effect will only occur at certain times of the day and on certain days in the year. A [glare analysis](#) can provide details. When it comes to PV modules, it is usually not about a measurable impairment of vision (disability glare) but a sense of disturbance due

to high luminance (discomfort glare). The cover glass of PV modules only has a one boundary surface that may cause glare, which usually has a slight surface texture and an additional antiglare (AR) coat, the maximum luminance of its reflection is significantly lower than that of windows with up to 6 smooth boundary surfaces that lack AR coating. In particularly critical cases, for instance, when a PV system is located near an airport, PV modules with special, more strongly textured cover glass can reduce the risk of glare.

20 Do PV modules emit irradiation that is harmful to health?

No, provided the components are certified and have been mounted properly.

The term “electromagnetic pollution” includes static, electric and magnetic fields as well as electromagnetic (EM) fields that are generated by technological devices.

Static electric fields are caused by charge imbalances (potential) between two objects, for instance between module string cables, or between live and grounded components. The field strength depends on the voltage and the distance between the live objects. A module string typically provides voltages between 400 – 1000 V, depending on the number of modules connected in series. There is no knowledge of cases of damage to health from static electric fields associated with PV systems.

Static magnetic fields surround electrical currents, for instance in live cables. A rooftop PV module typically produces a current in the range of 10 – 15 A. While magnetic fields can magnetize certain materials, the magnetic field strength decreases rapidly with increasing distance from the cables and is further weakened by the usual paired arrangement of the wires carrying currents in opposite directions. The live wires of a PV system should not be routed through living or bedrooms. There is now knowledge of cases of damage to health from static magnetic fields associated with PV systems.

Electromagnetic fields (“waves”, “radiation”) are created by accelerating or decelerating electrical loads. Solar cells produce a direct current (DC) that does not generate EM fields, neither within the PV modules nor in the DC power lines. Inverters convert the direct current from the module string or individual module to 50 Hz alternating current (AC) for grid feeding. Most small PV systems use a central inverter, though some use module-integrated inverters located on the rear side of the modules.

Currents in the kilohertz frequency range flow through such inverters, and these alternating currents generate EM fields. The law requires manufacturers to shield inverters so that the EM value near the device stays below the threshold value. Inverters must be mounted outside living and bedrooms, with the distance and shielding effect of the material between the inverter and the dwelling area being the determining factor. During operation, inverters can feed a portion of high-frequency electricity back into the DC circuit, and therefore also into the PV modules. Transformerless inverters transmit low-frequency (e.g., 50 Hz or 150 Hz) voltages to the DC side. There is now knowledge of cases of damage to health from electromagnetic fields associated with PV systems. A study commissioned by EnergieSchweiz [BFE] provides numerical values and comparisons with typical domestic electrical appliances. Low- and high-frequency currents in inverters may also produce audible sound during operation and although this may cause disturbance depending on the inverter’s location, this is not EM radiation.

21 Are PV systems capable of replacing thermal power plants?

No.

Although PV power plants reduce the consumption of fossil fuels, energy imports and CO₂ emissions, they do not replace controllable generation capacities, not even in combination with wind power, battery storage and pumped storage, which can only be expanded to a limited extent in Germany. Windless nights and especially windless, cloudy winter weeks without significant PV or wind power production are critical. Only by adding flexible thermal power plants, fueled by natural gas, biogas and, in the future, hydrogen, can generation gaps be covered all year round.

22 Are we capable of covering a significant proportion of our energy demand with PV power?

Yes, to the extent that we adapt our energy system and the energy-related structures to the requirements of the energy transformation.

22.1 Starting point: energy demand and energy supply

The traditional energy industry extracts fossil and nuclear energy sources (primary energy), converts them and prepares them for end consumers (Figure 39).

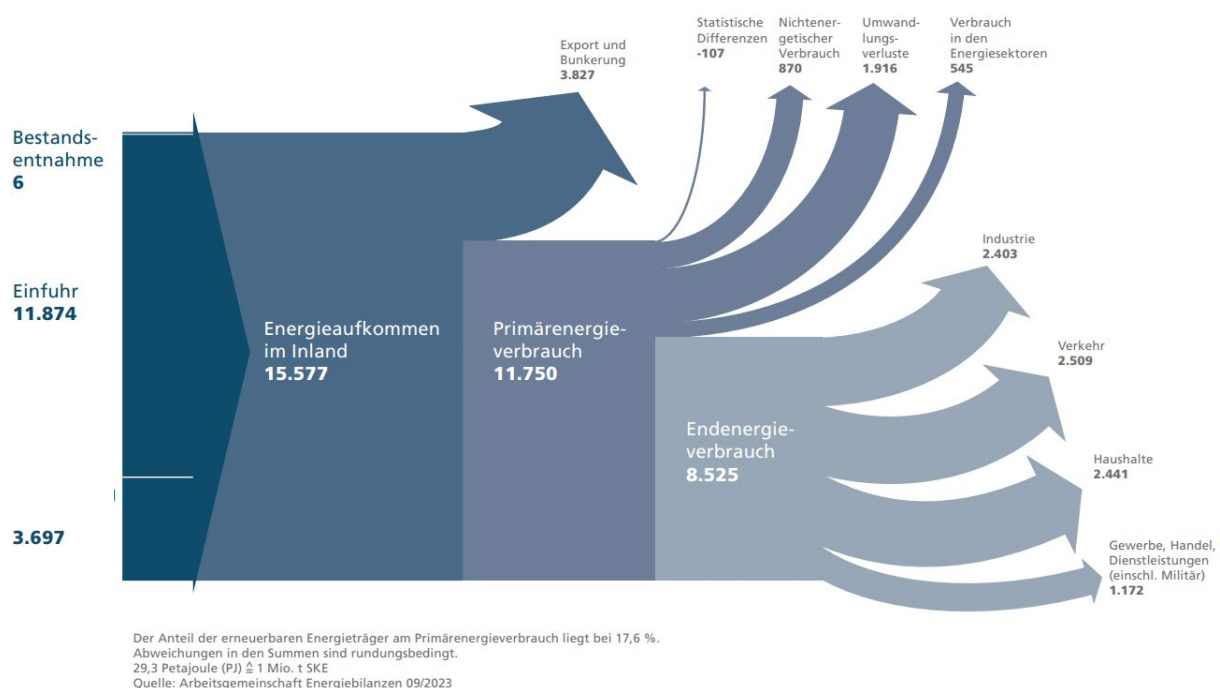


Figure 40: Energy flow diagram 2022 for Germany, figures in petajoules [AGEB2].

There are dramatic efficiency deficits in conversion and final energy consumption (cf. Section 22.3.3). Our future energy demand is by no means the same as today's primary energy consumption, neither in terms of quantities nor in terms of energy carriers.

Up to now, Germany has been highly dependent on energy imports, associated with the risk of volatile prices, political influence by producing and transit countries and the risk of disruptions in raw material logistics, e.g., pipeline blockages or low water in rivers.

The costs of fossil energy imports are in the order of 40-135 billion euros annually (Figure 40). A large part of the money went to supporting autocratic regimes.

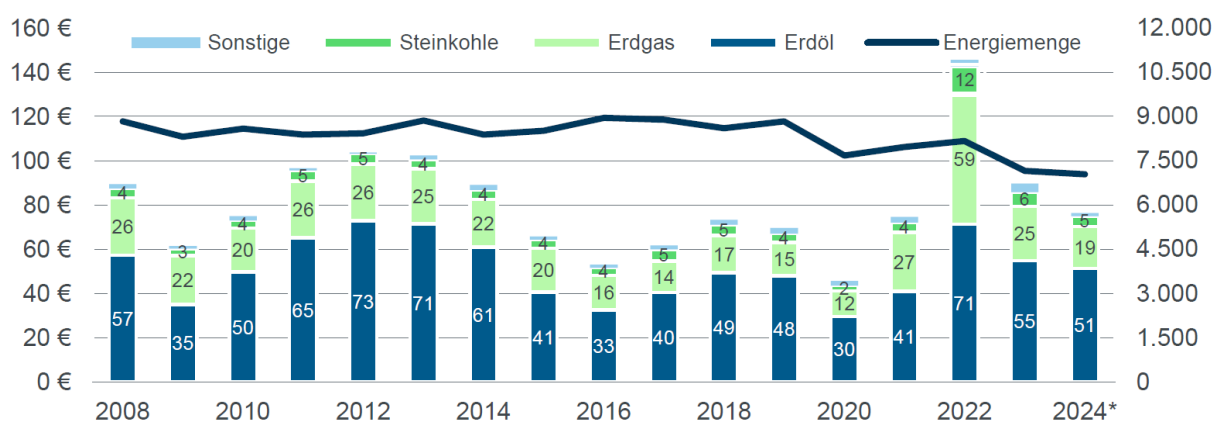


Figure 41: Value and energy volume of net imports of fossil fuels (left axis: value in EUR billion, nominal; right axis: energy volume in PJ) [KfW].

Most final energy (36 percent) is used to generate mechanical energy ("power") for transport and in stationary engines (Figure 41). Approximately 800 TWh of final energy is used annually for space heating and hot water [BMWK1].

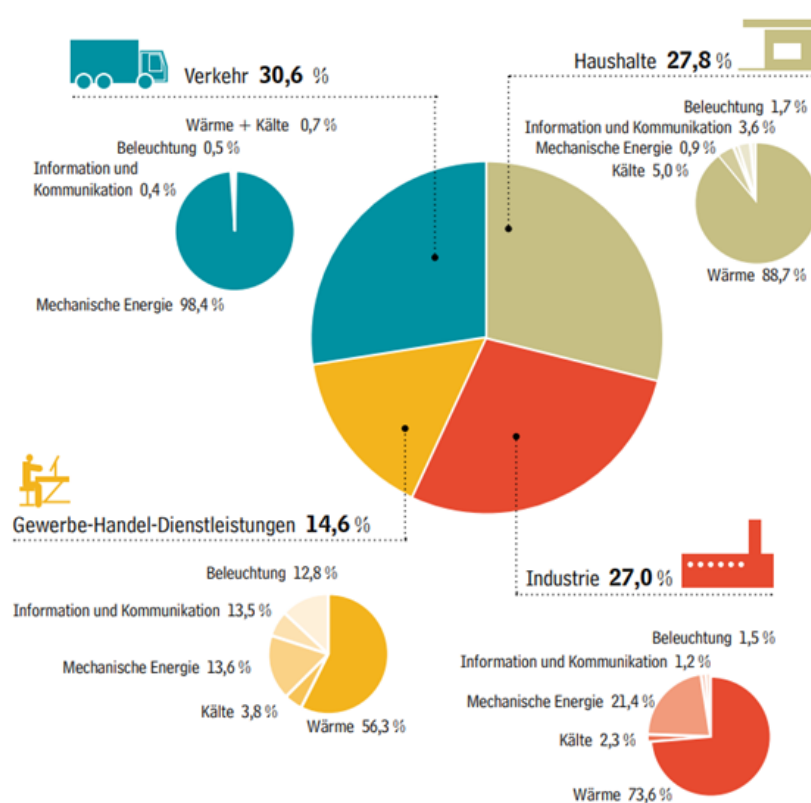


Figure 42: Final Energy Consumption in Germany by Sector and Application in 2024, figures from [AGEB3]

The electricity load fluctuates periodically so that less electricity is needed at night, on the weekends and during holidays. Electricity suppliers distinguish in the load profile between base, medium and peak load, cf. Section 27.8. The base load is the load share around 30–40 GW, which hardly changes over 24 h. The medium load fluctuates slowly, the peak load comprises the rapidly changing load share above the base load. The medium load fluctuates slowly, the peak load comprises the rapidly changing load share above the base and medium load. Electricity consumption and energy demand for water heating are slightly lower in summer than in winter. Mineral oil sales (petrol and diesel fuel) show very little seasonal fluctuation [MWV]. The heating demand correlates negatively with global radiation, with the highest coincidence in spring.

22.2 Energy scenarios

Our current energy system in Germany, based on fossil and nuclear generation, is a discontinued model. There is a plethora of energy scenarios for the coming decades, and they increasingly count on RE [UBA, ACA, ISE3]. Researchers at the Fraunhofer Institute for Solar Energy Systems ISE have investigated different transformation paths to a climate-neutral energy system for Germany in simulations based on hourly time series with consideration of sector coupling (Figure 42). In an economically optimal scenario, PV must contribute an installed capacity of 471 GW [ISE3].

Energiesystemmodell REMod

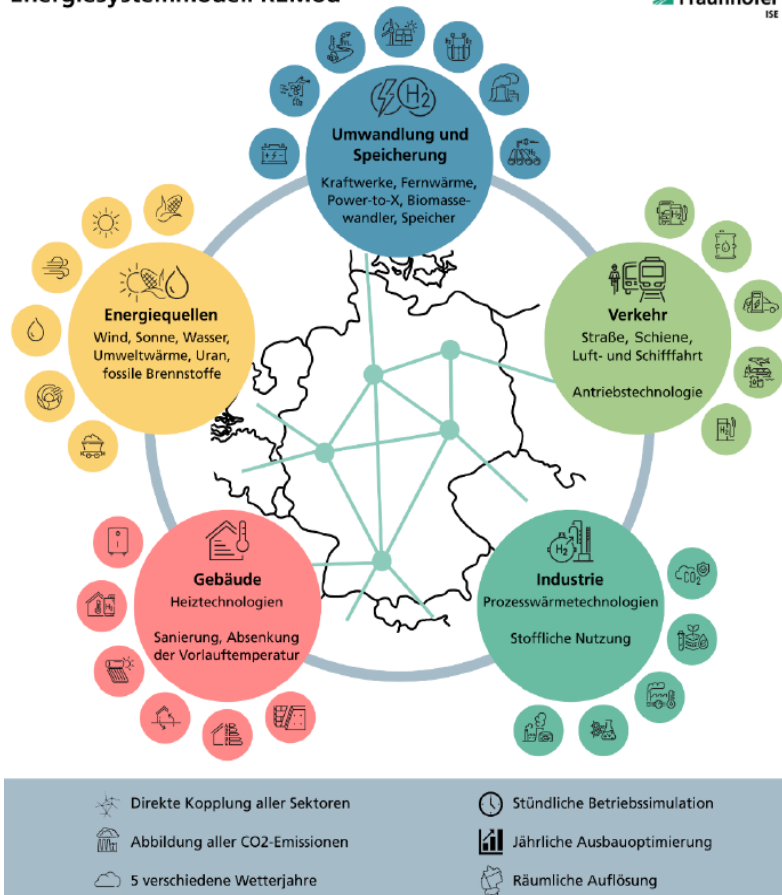


Figure 43: Schematic representation of the REMod model [ISE3].

Figure 44 shows a schematic residual load curve for Germany with a 100 percent renewable electricity supply. The hourly values of the residual load (Section 11.4) for one year are shown in descending order. Although volatile electricity production can be technically shut down at any time, this is at the price of a total economic loss of the corresponding electricity quantity. An electricity price with a meaningful control function would fall from left to right along the residual load curve of Figure 43.

On the **demand side**, flexible loads are reduced, batteries and pumped storage are discharged, combined cycle gas turbine (CCGT) generators and gas turbines are activated and other measures are taken in order to cover the residual load. The thermal power plants use green hydrogen and are designed with combined heat and power (CHP) for local heat demand.

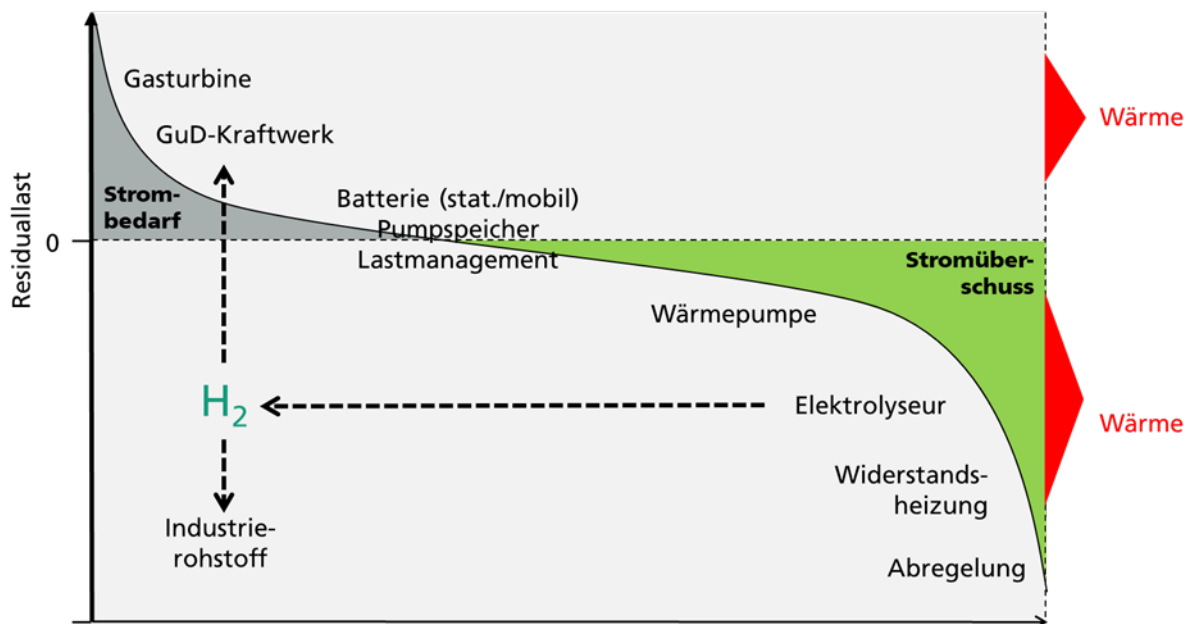


Figure 44: Schematic representation of a residual load curve for Germany with electricity supply with 100 percent RE.

On the **surplus** side, flexible loads are increased, batteries and pumped storage are charged, heat pumps, electrolyzers and resistance heaters (heating rods, electrode boiler) are activated when electricity prices increasingly fall in order to take the electricity that is not currently needed. Electrolyzers can also be operated as CHP plants and produce usable waste heat. Resistance heaters and high-temperature heat pumps can provide heat for industrial processes. The last resort is to shut down electricity production if the installed output or the grid capacities are not sufficient for a few hours of the year.

Before the occurrence of foreseeable, particularly high residual load peaks, storage and flexible loads are also operated atypically to reduce both the required maintained capacity of thermal power plants and curtailment.

In order for heat-generating converters on both sides of the curve to be operated with electricity, they require local thermal storage and heat consumers or connection to heat grids (Section 21.3.6). Generators (e.g., simple gas turbines) and consumers (e.g., resistance heaters) with particularly low performance-related investment and maintenance costs (€/W) are required for the two-sided spurs of the residual load curve. Since they are rarely in operation, they do not have to be highly efficient.

Hydrogen, produced by electrolysis, serves as an industrial raw material (direct reduction, organic chemistry, ammonia synthesis). Hydrogen and its derivatives can be converted back into electricity in gas-fired power plants and can power vehicles with the highest energy densities, e.g. aircraft. A hydrogen network is planned for Germany, and hydrogen can also be added to the existing natural gas network.

The **storage capacity** of the system must be designed for the worst case scenario of an energy generation failure (sun and wind) which lasts for several weeks, i.e., a prolonged lull in winter, possibly exacerbated by a closed snow cover. For this case, sufficient quantities of hydrogen, synthetic energy carriers and hydrogen derivatives must be kept in stock. If PV was not supported by wind during winter, then the worst case would last not weeks, but rather months, requiring many times more storage capacity.

Due to their limited capacity, stationary batteries and pumped storage fail relatively quickly as generators in continuous operation (minutes to a few hours). The same applies to vehicle batteries, which can be operated bidirectionally in the grid, but have the primary purpose of covering the mobility demand. The benefit of battery storage systems lies in the ability to frequently change between charging and discharging, which they implement more quickly and, above all, more energy efficiently compared to the electricity-to-electricity path via hydrogen as energy carrier. Additionally, many load management options only have a short-term effect in the range of a few hours.

The **power generation capacity** of the system on the left side of Figure 44 must be sufficient to take over the complete supply in the order of 100–150 GW when the hourly reserves (load management, pumped storage, battery) are exhausted. This situation occurs frequently, e.g., on low-wind nights, and can last for several weeks in a worst-case scenario.

The **power purchase capacity** of the system on the right-hand side of Figure 44 in the order of several 100 GW must be sufficient to absorb the electricity production from volatile RE minus the current electricity consumption as far as possible as soon as the hourly reserves (load management, pumped storage, battery) are exhausted. If the power demand is not sufficient during rare production peaks, the system must be regulated. This can happen, for example, on stormy nights or on sunny and very windy weekend days, when low demand coincides with very high electricity production. For these few hours of operation, it is not worthwhile to further expand the power output.

Converters allowing reversible operation operate on both sides of the curve in Figure 44 and thus achieve higher utilization. These include batteries, pumped storage, and reversible fuel cells, which carry out electrolysis in the case of excess electricity. (See 22.3.7)

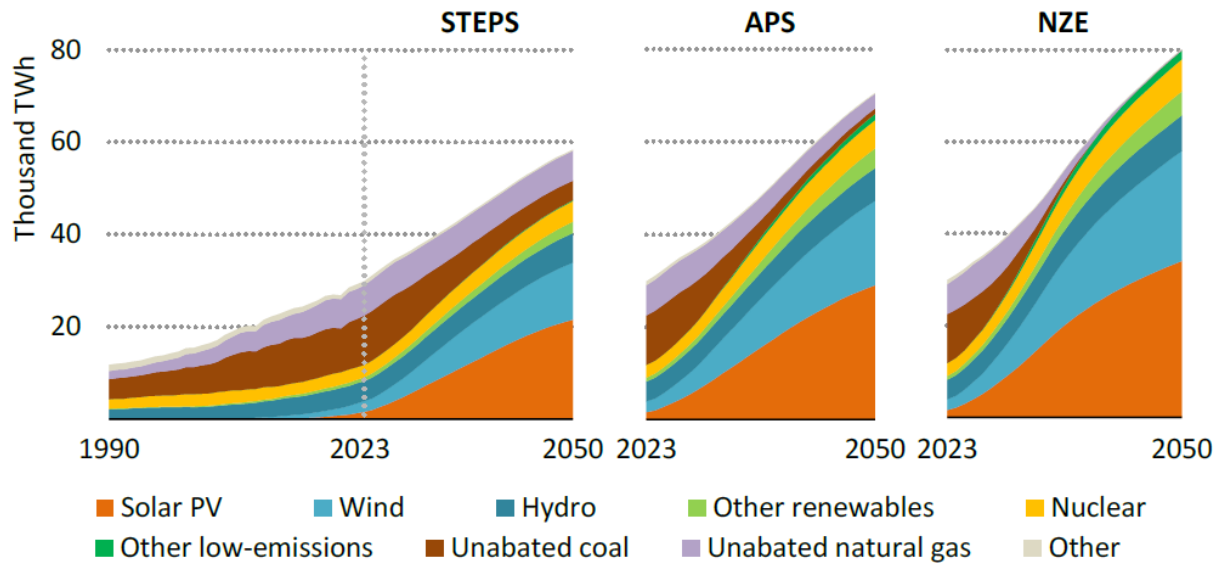


Figure 45: Development of the global electricity generation by technology and scenario [IEA2].

Taking a brief glance at global energy scenarios brings us to the International Energy Agency (IEA) that predicts that photovoltaics will play a major role in the global energy supply across all three scenarios of the global energy transition (Figure 44).

22.3 Transformation steps

For a massive, technologically, and economically controllable integration of volatile PV electricity into our energy system, there is no single patent solution, but a multitude of complementary measures. The most important steps are addressed in the following sections.

22.3.1 Keeping PV power generation constant

Elevated modules installed on fields and flat roofs have mainly been oriented towards the south and have a 25° inclination to maximize the specific annual electricity yield (Wh/W_p). For the same reason, PV has mainly been installed on pitched roofs facing south. This uniform module orientation increases the hourly spread of solar power generation around midday, especially on clear days, and the seasonal spread during the summer months. As a result, the maximum PV electricity yield in Germany is mainly reached around noon, while less power is generated in the morning and evening due to flat angles of incidence. This situation is increasing the need to stabilize power generation, for example through curtailment, load management and storage. In addition, the market value of electricity tends to drop more and more around noon. Alternative mounting methods can stabilize PV electricity generation.

Vertically mounted, bifacial modules with an east-west orientation supply significantly more electricity with direct sunlight during the mornings and afternoons than around noon. Their generation profile optimally complements the large number of existing PV power plants, most of which are south-facing (Figure 46). Another important advantage in cases of limited grid connection capacity is higher full utilization hours, which can be achieved in conjunction with a more powerful grid connection superstructure.

Uni-/bi-axial tracking systems stabilize the electricity production on clear sunny days and increase it on cloudy days by orienting the modules to the brightest point in the sky. The trackers can increase the specific annual yield by approx. 15–30 percent. Vertically oriented and moveable modules are additionally advantageous since their impact on the agricultural use of the land is minimal (Section 12).

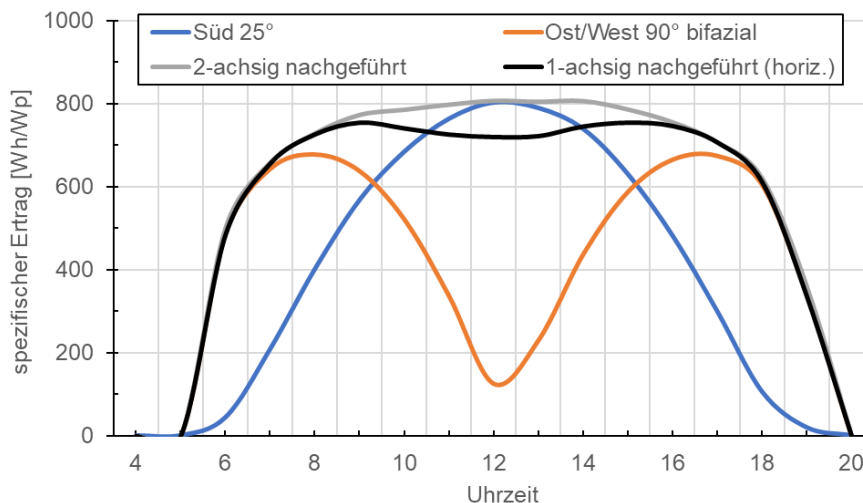


Figure 46: Electricity yield profiles of PV power plants in different mounting variants, calculated online with https://re.jrc.ec.europa.eu/pvg_tools, for a predominantly clear day in May at the Freiburg site, no shading, bifaciality factor 100 %

PV on **pitched roofs facing other directions than south** also helps extend power generation into the morning and evening, while modules facing east or west on flat roofs broaden the daily generation profile (Figure 47).

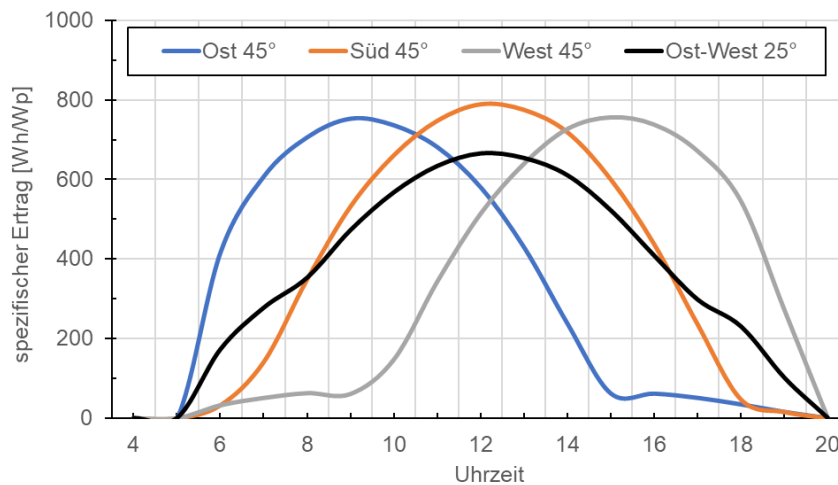


Figure 47: Electricity yield profiles of PV systems mounted in various ways on a predominantly sunny day in May, no shade, located in Freiburg, calculated online using https://re.jrc.ec.europa.eu/pvg_tools.

The very pronounced seasonal fluctuation of PV power generation can be dampened by south-facing modules, especially on vertical façades (Figure 47). Yield losses due to restricted rear ventilation were not considered in the comparison calculations.

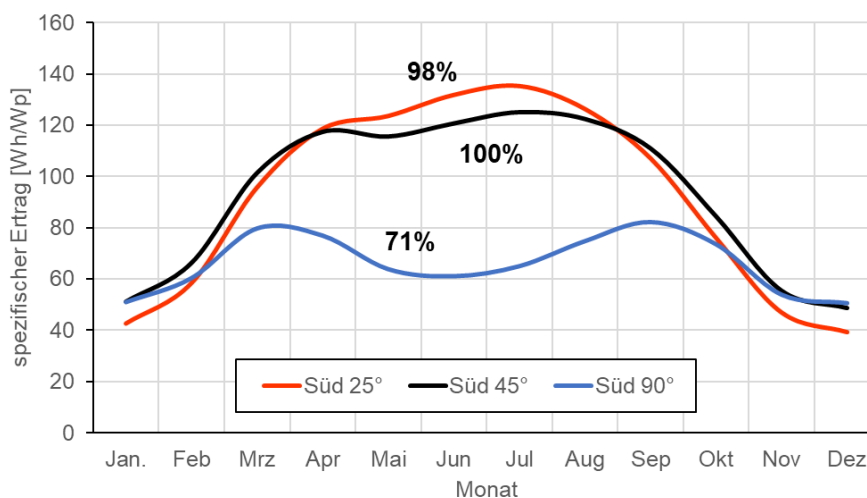


Figure 48: Electricity yield profiles of PV systems mounted in various ways, no shade, located in Freiburg (calculated online using https://re.jrc.ec.europa.eu/pvg_tools), the percentages provide the relative annual yield with respect to the 45° inclination.

The somewhat higher electricity production costs for the alternative installation variants can pay off in the context of increased self-consumption or higher market values for electricity. Feed-in tariffs that reward a higher value of electricity in the morning and evening hours, as well as in the winter months, promote the construction of systemically advantageous PV power plants that are not only optimized for maximum annual electricity yield. The measures to increase the number of full-load hours mentioned in Section 16.4 also contribute to the stabilization of PV electricity generation.

22.3.2 Expanding flexible generation capacity

The energy transition will require a dramatic increase in the generation capacity of hydrogen-compatible gas-fired and biomass power plants, with decreasing full load hours and higher market values for the electricity generated.

Gas-fired power plants are particularly well suited to cover fluctuating loads. In combination with combined heat and power (CHP), high overall efficiencies of up to 95 percent are achieved [UBA2]. Simple gas power plants based on gas engines have an investment cost (€/kW) that is a fraction of the cost of combined cycle power plants. Today, gas-fired power plants burn natural gas and biogas. They will switch from natural gas to mixed gases with increasing proportions of electrolytically produced hydrogen ("H₂-ready"). Germany has decided to phase out fossil fuels. In some coal-fired power plants, steam generation can be converted from coal to gas burners (fuel switch) so that the steam turbine and generator can continue to be used [EnBW2].

Depending on the type, biomass power plants can burn solid biomass (residual wood, waste wood), liquid biomass (vegetable oil) or biogas from agriculture or sewage treatment plants. At the end of 2024, biomass power plants with a capacity of over 9 GW_p were installed throughout Germany [ISE4]. Power plants that burn solid or liquid biomass are very easy to operate as electricity-generating plants due to the simple storage of the fuel material. There are limitations with biogas power plants that only allow limited control over the fermentation throughput. For large plants near gas grids, investments in grid connection and the processing of biogas to biomethane are often profitable. Smaller plants require local biogas storage systems and higher generator output to focus power generation on times of peak demand. The cultivation of biomass for the purpose of energy use will decrease due to low land efficiency (Section 16), and use will concentrate on agricultural residues (biogenic waste).

22.3.3 Increasing energy efficiency

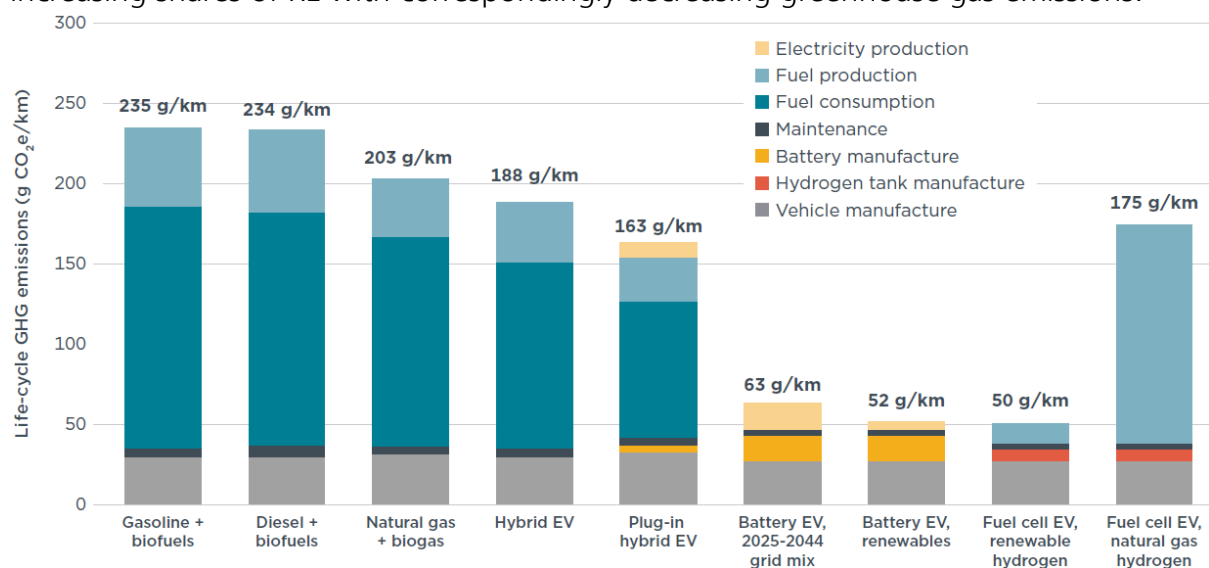
22.3.3.1 Transport

Most of the final energy consumed in traffic is converted into waste heat by combustion engines; only a small part reaches the drive train as mechanical energy. Diesel engines for passenger cars achieve an efficiency of up to approx. 42 percent at their peak, while in urban traffic the average efficiency is only approx. 20 percent due to partial load operation [Sprin]. For passenger car petrol engines, the values are even lower, at up to 37 percent at the peak and approx. 10–15 percent in urban traffic. Even in urban traffic in particular, a considerable proportion of the drive energy gained is irreversibly burnt up during braking, because vehicles with internal combustion engines can hardly recuperate via their alternator. Thus, motorized road traffic burns fossil fuels with a very low efficiency, related to the transport performance.

Electric vehicle drives use highly efficient motors with an effective efficiency of around 90 percent. Losses during charging of the vehicle battery are in the order of 15 percent and are particularly high during fast (DC) charging. Electric vehicle drives can recover a

large part of the kinetic energy; according to the manufacturer, the efficiency of recuperation for the BMW i3, for example, is around 63 percent. For reasons of energy efficiency alone, it makes sense to switch to electric drives, in addition to the considerable storage potential (Section 21.3.7.5). Figure 48 shows the greenhouse gas emissions of average mid-range cars with different drives in 2025. Preferred charging with solar or wind power (cf. Section 21.3.7.2) leads to a particularly flat course of the emission line (Elektro3). If the BEV is sold together with a small PV system of 3 kW_p nominal power, the vehicle runs on 100 percent solar power on a bi-lobal basis, with an average annual mileage of over 15000 km, a specific annual yield of 950 kWh/kW_p and 15 percent charging losses. Especially in city traffic with its typical stop-and-go operation, consumption and GHG emissions per km for BEVs decrease thanks to recuperation, while they increase for internal combustion vehicles due to braking losses and inefficient partial load operation.

For city operation, smaller batteries are usually sufficient, which further reduces GHG emissions for production and operation. In the future, the production of BEVs will use increasing shares of RE with correspondingly decreasing greenhouse gas emissions.



THE INTERNATIONAL COUNCIL ON CLEAN TRANSPORTATION [THEICCT.ORG](https://theicct.org)

Figure 49: Greenhouse gas emissions (CO₂-eq in tons) for the life cycle of a mid-range car with different drives in 2025. Average values based on sales figures in the EU

The CO₂ balance can be evaluated as follows: an electric car with a small battery, which mostly drives in town, prefers to charge with renewable electricity and covers many (necessary) km per year, has a particularly clear advantage in terms of the GHG balance. The advantage shrinks for Sunday drivers with a large battery, a high proportion of car journeys and gray charging current.

22.3.3.2 Private households

Private households use about 75 percent of the final energy consumed for heating. On average, this consumption can be halved by simple thermal insulation measures.

Stiftung Warentest has determined that a household completely equipped with old appliances consumes twice as much electricity as one that uses only efficient appliances [test]. Measures that reduce night-time electricity consumption are particularly effective when solar power (and wind power when there is a night-time lull) can only be made available through comparatively expensive storage.

22.3.4 Load management

Grid-serving load management, or demand side management, aims at a supply-oriented, temporal shift of electricity consumption. When the residual load is high (Section 10.4), consumption is reduced or stopped, but is made up for when the residual load is low. If the CO₂ price is sufficiently high to determine the merit order (Section 4.3), load management can save CO₂ emissions, because the most harmful power plants are then needed less often.

Prerequisites for load management are flexibility options through **energy or material storage** on the demand side. A washing machine can often wait a few hours or even days, while a passenger train must depart on time. For electrical systems in continuous operation, **power reserves** are necessary that enable a compensatory increase above normal power after a reduction below normal power. Grid-serving operation reduces full load hours.

A study has calculated the theoretical load management potential of 15.6 TWh for private households in the year 2024 [EON], taking into account heat pumps, electric cars, home storage units, washing machines, dryers and dishwashers. This amount of electricity corresponds to around a third of public electricity generation from gas-fired power plants and is likely to double by 2030.

The technical prerequisites and economic incentives for tapping into this potential still need to be created for the most part. Household appliances that are allowed to operate within a user-defined time interval must be technically enabled to select optimal operating times with regard to electricity and grid shortages. Dynamic electricity tariffs and electricity meters that enable time-dependent billing ("smart meters") are of crucial importance.

The current composition of electricity prices for households (Figure 10) with very high fixed costs per kWh would hardly create incentives for load management given the usual price fluctuations on the electricity exchange. According to the EU Electricity Market Directive 2019/944, end customers with smart meters should be able to choose dynamic electricity tariffs from 1.1.2021, and the German federal government's plan is to make these electricity tariffs mandatory by 2025 at the latest.

In the electricity-intensive industry, e.g., electrolytic aluminum production, there is also potential for adjusting consumption profiles. A cross-industry survey of 400 companies in Germany has shown that there is potential for increasing flexibility by around 10 to 15% of peak load, i.e. around 5 to 7 GW [MCK]. Companies that accept temporary power cuts in electricity supply announced at short notice can already receive a contractually agreed compensation payment from their transmission system operator (Ordinance on Disconnectable Loads, AbLaV).

As soon as particularly cheap daily electricity is available more frequently because the installed PV capacity grows and variable electricity tariffs are offered, flexibility on the part of industry and consumers will also increase. Self-consumption of solar power has an analogous effect to dynamic electricity tariffs because it significantly reduces the price of electricity when it is purchased directly from one's own roof. The promotion of PV self-consumption for households and businesses is a highly effective means of incentivizing load management.

As transportation and heating become increasingly electrified, the potential for load management grows significantly. Electric vehicles, heat pumps and heating rods consume large volumes of electricity and offer high flexibility potentials through integrated electricity, heat and cold storage systems.

Electric vehicles

At the end of 2025, approximately 2.1 million purely electric cars (BEV, excluding plug-in hybrids) were registered in Germany, out of a total of approximately 50 million passenger cars (de.statista.com, www.kba.de). Electric buses are already in use in many places in urban transport, as are electric vans for distribution logistics. 40-ton electric trucks for long-distance transport are increasingly arriving on the market. In 2023, passenger cars registered in Germany had an average annual mileage of 12,320 km. The total mileage of passenger cars in Germany of 591 billion km in 2023 [KBA] with a consumption of 180 Wh/km and a complete transition to e-cars corresponds to an annual electricity consumption of 106 TWh, plus approx. 15 percent charging losses. The large on-board battery capacity of electric cars (30–100 kWh), electric trucks (up to 500 kWh), and other types of electric vehicles opens up considerable potential for load management in the future by bringing forward, throttling, or postponing charging processes within a time window of hours, or up to several days in the case of private cars. The large on-board battery capacity of electric cars (on average 50–70 kWh), electric trucks (up to 500 kWh), and other types of electric vehicles opens up considerable potential for load management in the future by bringing forward, throttling, or postponing charging processes within a time window of hours, or up to several days in the case of private cars.

Supply-dependent electricity prices motivate private and business customers to choose low-cost charging times with a high share of RE in the electricity mix. Variable fuel prices are not new; petrol stations also continuously change their prices. In order to be able to supply PV electricity for direct consumption, charging stations must be located in typical daytime parking areas, e.g. at the workplace, in multi-story car parks or in public car parks.

Remote control of the charging power, taking into account the mobility needs in individual cases, allows grid operators to stabilize the grid.

The current share of electricity from renewable sources covering the grid load in Germany fluctuated between 11 and 137 percent in 2024 [ISE4]. Those who already want to supply their e-car with especially green electricity from the grid preferably charge on sunny days around midday, otherwise on windy nights. Weekends and public holidays are usually advantageous because the lower electricity consumption tends to improve the CO₂ balance. Unfortunately, many e-cars are charged at home in the evening, adding to the evening peak demand while also not using any direct PV electricity. Precise information

on the current and planned share of RE in the German electricity mix is shown in the Energy Charts [ISE4].

Heat pumps

Electric heat pumps consume electricity to provide useful heat from ambient heat (heating) or to dissipate heat into the environment (cooling). In the building sector, the efficiency of a heat pump is indicated as the annual performance factor (APF) and is around 300 percent in heating mode, depending on the technology and load. For the estimation of the maximum electrical load that can be switched on (peak load) via remote-controlled heat pumps, partly combined with heating rods, an average thermal rated output of 15 kW_{P,th} per unit is assumed. With a coefficient of performance (COP) of 3, this corresponds to a nominal electrical capacity of 5 kW_{P,el}. In normal operation, heat pumps usually work with a COP of 3–5. By the end of 2024, around 2 million heat pumps were in use [BWP], which, under the same assumptions, is equivalent to an electrical nominal capacity of 9 GW_p. Efficient operation is achieved by heat pumps with surface heating systems, mostly underfloor heating systems, which manage with low flow temperatures. Resistance heaters (heating rods) convert electricity into heat with 100 percent efficiency, but with a low exergetic efficiency when generating low-temperature heat.

Thermal storage capacity can be provided much more cheaply than electricity-to-electricity storage capacity. If the thermal storage capacity and the heat pump or heating system output are sufficiently dimensioned, the storage load can be supply-oriented, depending on the current residual load. For this purpose, heat storage units and cold storage units, e.g. of air-conditioning systems, cold stores and food markets, are preferably charged during the core time of the day or according to electricity price signals. However, if generously dimensioned thermal heat storage systems are lacking, the thermosensitivity of the electricity load increases and larger power reserves must be kept available at power plants.

Low-temperature thermal storage, especially hot-water thermal storage, enables the electricity-led, highly efficient operation of CHP plants on both sides of the residual load curve (Figure 43), as well as of heat pumps and heating rods on the electricity consumer side. The same storage unit can, for example, be charged simultaneously by heat pump and heating rod when there is a high electricity surplus, and by a CHP when there is a demand for electricity. Heat storage systems are scalable from single-family homes to multi-family homes and commercial enterprises to neighborhood supply. The relative storage losses and the specific costs decrease with the size of the storage tank. Large storage facilities of several thousand m³ or more, e.g. neighborhood storage facilities, can be operated as seasonal heat storage facilities (<http://www.saisonalspeicher.de>). They enable the transfer of useful heat from the summer to the winter half-year with its much higher heat demand. Heat storage systems increase the self-consumption of PV systems if they are charged via the heat pump and the heating rod, especially in the summer half-year. The PV system can take over the domestic hot water heating seasonally, all the more so if the PV modules are mounted with a high inclination on steep south-facing roofs or on south-facing façades. As soon as price signals become available, decentralized heat storage systems can also be selectively charged with surplus wind power from the grid.

At the point of cold generation and use, e.g. in building air conditioning or in refrigerated warehouses, cold can be stored with comparatively simple means. Another prerequisite for grid-friendly operation is a sufficiently dimensioned output of the cooling generator.

Heating rods and electrode boilers

Resistance heaters can be used to convert excess electricity into low-temperature heat for hot water and heating in buildings, and into high-temperature heat for industrial processes. When used in low and high-temperature heat storage systems, heating rods offer load flexibility potentials at a comparatively low cost of investment.

Electrolysis

With the transformation of the electricity sector and industry, the demand for green hydrogen and its derivatives will increase massively, from 55 TWh in 2023 to 290–440 TWh in 2025 [NWS]. Some of the hydrogen will be stored for reconversion into electricity (section 22.3.7.5 Hydrogen storage with reconversion), while the rest will be used for chemical reduction and synthesis processes. Electrolysis capacities operated in Germany for supplying hydrogen to industry create significant capacities for load management in conjunction with hydrogen storage facilities.

22.3.5 Balanced addition of PV and wind power capacities

Due to the weather patterns in Germany, there is a high degree of complementarity between the hourly and monthly generation of PV and wind power (11.2.2). If local and regional efforts succeed in expanding installed capacity for solar and wind power to a similar scale, the combination of the two will reduce the need for balancing power. At the national level, however, wind power plants have been built primarily in the north and solar power plants primarily in the south. Here their complementarity is limited by the capacity of the transmission grids along the north-south axis.

22.3.6 Combined heat and power generation

Low-temperature heat for space heating and hot water, as well as industrial process heat at a high temperature level, are still mainly produced by burning fossil resources and in connection with small heat storage capacities. In a renewable energy system, large amounts of useful heat are generated during the transformation of electrical energy, from the waste heat of converters.

Large heat storage capacities for low-temperature heat (Section 21.3.7.1) enable the current-controlled operation of the converters. The expansion of heat distribution networks is limited by distance-dependent transport losses to a much greater extent than in the electricity sector. For this reason, plants with electricity-heat coupling (Figure 43) must be tailored in their output and placement to local heat consumption and usable heat grids.

These can be local heating networks with heat transport between neighboring buildings or district heating networks that supply city districts or entire cities. High-temperature heat for industrial processes can be generated from the waste heat of CHP gas turbines (up to approx. 550°C) if electricity is required.

In Germany, about 39 GW of electrical CHP capacity was on the grid at the end of 2020 ([ÖKO2], statista.com), using mainly natural gas, biomass, and coal. CHP plants achieve overall efficiencies of up to 90 percent, and as gas CHPs even up to 95 percent [UBA2]. Even micro-CHP units for single-family homes can achieve electrical efficiencies of up to 25 percent and overall efficiencies of up to 90 percent [Licht1]. They use combustion or Stirling engines to generate mechanical power. As the energy transition progresses, CHP plants are being converted from fossil fuels to hydrogen and methane; some continue to burn bio methane/mass from biogenic waste.

22.3.7 Electrical energy storage

Electrical energy storage (EES) refers to storage systems that can absorb energy via electricity and release it later in the form of electricity. EES includes, in particular, stationary batteries, vehicle batteries (when stationary) and pumped-storage power plants. It will increasingly also include H₂-based systems with electrolysis, H₂ storage systems and re-conversion via thermal power plants or fuel cells. There are also experiments with systems consisting of a heating element, a high-temperature heat storage system and a thermal power plant.

Stationary batteries

At the end of 2025, the installed storage capacity of stationary batteries (large-scale, commercial and residential storage systems) was **25 GWh**, with a power of **17 GW_p** (ISE4, <https://battery-charts.rwth-aachen.de/>). On the condition that regulatory hurdles allow, stationary batteries can be used to:

- Adjust local generation profile of PV power plant to
 - Increase self-consumption to reduce costs of external electricity procurement
 - increase revenues by selling electricity
 - eliminate bottlenecks in connections and in grid capacity
- Adjust local load profile to
 - Reduce costs of external electricity procurement and make use of flexible or power-limited tariffs
 - Eliminate bottlenecks in connections/grid capacity (e.g. fast charging stations)
- Counteract grid-wide disruptions by providing
 - balancing power, instantaneous reserves
 - Emergency electricity supply
- Using electricity trading to balance generation and load across the power grid

The massive price reduction of lithium-ion batteries over the past years (e.g., by a factor of 10 [Bloom] between 2011 and 2025) establishes them as a new standard component in PV power plants of all sizes. With the help of batteries, the full utilization hours of PV power plants can be massively extended, for example doubled, by heavily overbuilding the grid connection.

With stationary batteries, households can extend their own consumption of PV electricity into the evening hours and thus massively increase it (typically doubling it, see Figure 49). If a battery has a grid-serving system management, this relieves the grid by selectively reducing the midday feed-in peak (Figure 50). Storage systems would thus enable an increased PV expansion [ISE7].

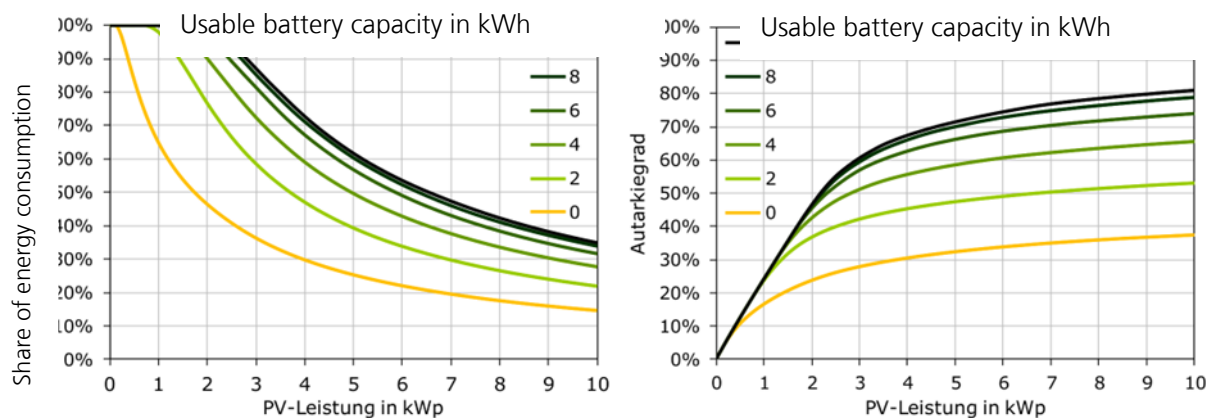


Figure 50: Effect of photovoltaic output and storage capacity on the proportion of self-consumption (left) and degree of self-sufficiency (right) for an annual electricity requirement of 4000 kWh [HTW1].

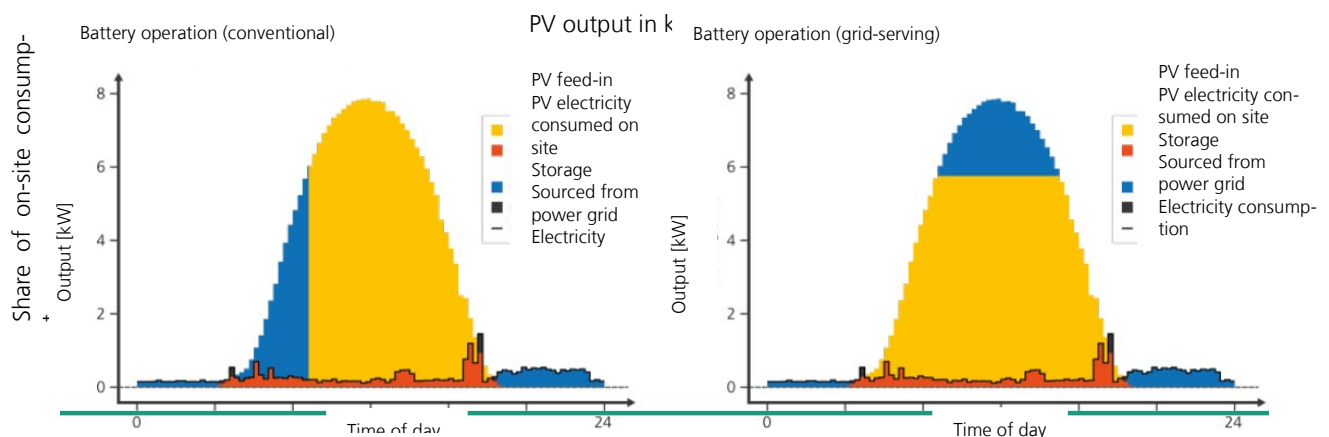


Figure 51: Comparison of conventional and grid-serving operation [ISE7].

Vehicle batteries in stationary operation

Thanks to their high consumption and storage capacity, electric vehicles already offer enormous potential as flexible loads through grid-serving charging. (Section 22.3.4).

With **bidirectional energy management** they can also be used as electricity storage units when they are connected to the grid and do not need to maintain the full charge capacity at all times.

Private cars park for an average of 23 hours per day, and the limited capacity of traffic routes alone forces most cars to be standing for most of this time. E-vehicles connected to the grid can also provide an economic benefit when stationary, unlike their predecessors with combustion engines.

New electric passenger cars (excluding hybrids) have an average battery capacity of 50–70 kWh, while larger models exceed 100 kWh. Several automakers already offer V2G (Vehicle-to-Grid) contracts. Under these contracts, customers are compensated for the periods during which their vehicle is connected to the provider's wallbox. During this time, the provider can take advantage of price differences on the Leipzig Electricity Exchange and trade electricity. This business model supports the grid as long as it does not overload the power grids.

In the expansion scenarios [ISE3] for the energy transition, it is assumed that at least 10 percent of the mobile batteries are connected to the grid at all times via bidirectional remote control. Of the projected 40 million e-cars (reference scenario), 4 million would be available for system-serving tasks. Assuming an average peak power capacity of 20 kW per charging point (wall boxes and charging stations), the bidirectional controllable capacity would be 80 GW_p. With an average storage capacity of 80 kWh per car, the total capacity of the fleet is 3.2 TWh. However, this capacity would not be available as a whole. Assuming an average battery capacity of 60 kWh for electric passenger cars, approximately 64 GWh will be on the road or parked in Germany by the end of 2024, distributed across 1.6 million vehicles. By the end of 2025, 2.1 million electric cars were registered. The federal government has set a target of at least 15 million electric cars by 2030.

Mechanical storage facilities

The currently installed **pumped storage** capacity in the German electricity grid is just under **38 GWh**, the nominal capacity is **10 GW_p**, and the average efficiency is around 70 percent, excluding electrical supply and discharge losses. For size comparison: the storage capacity mentioned corresponds mathematically to the yield of the German PV power plant park from less than one operating hour under nominal capacity.

Due to a lack of storage capacity, run-of-river power plants can hardly make any control contribution in complementary operations. Their contribution of approx. **3.8 GW_p**, nominal output [ISE4] is only slightly expandable, their electricity production (annual full load hour) is generally declining due to the increasing drought. The mechanical storage of electrical energy in compressed air storage (Compressed Air Energy Storage, CAES) is currently being tested.

High-temperature heat storage with reconversion

High-temperature heat can be stored as latent heat in liquid salt storage tanks or as sensible heat in rock fill [Siem] or steel bodies [Vatt]. If required, the heat can be used to drive

a conventional steam turbine, possibly with further utilisation of the low-temperature heat. The first pilot plants are currently being tested; the manufacturer Lumenion states an electricity-to-electricity efficiency of 25 percent.

Hydrogen storage with reconversion

For longer periods with low PV and wind power generation, our future energy system will require additional hydrogen generation, storage and reconversion capacities. Depending on production costs and geostrategic factors, part of the production will take place outside Germany. 9.4 PWh of energy in the form of hydrogen could theoretically be stored in German onshore salt caverns [Hydro], which corresponds to 3.2 times the primary energy consumption in 2024. It is already technically possible today to increase the proportion of hydrogen in the gas grid to up to 20 percent.

Hydrogen can be converted back into electricity using thermal power plants or fuel cells. Reversible high-temperature fuel cells (rSOFC, short for "reversible solid oxide fuel cell"), which can also be operated as electrolyzers, currently achieve a round-trip efficiency of up to 75% according to manufacturer specifications [REV]. During storage and retrieval, additional losses are incurred during the compression and decompression processes, depending on the required pressure. Such bidirectional converters promise a significantly higher number of full-load hours than combinations of pure electrolyzers and fuel cells.

Overview

Figure 52 shows storage technologies for PV and wind electricity. Not only the efficiency but also the costs of the nominal power of converters (€/W) and the storage capacity (€/Wh), among other things, must be considered. Water reservoirs of pumped-storage power plants serve as mechanical storage systems. Battery storage systems, on the other hand, include stationary and mobile storage systems that allow bidirectional charging.

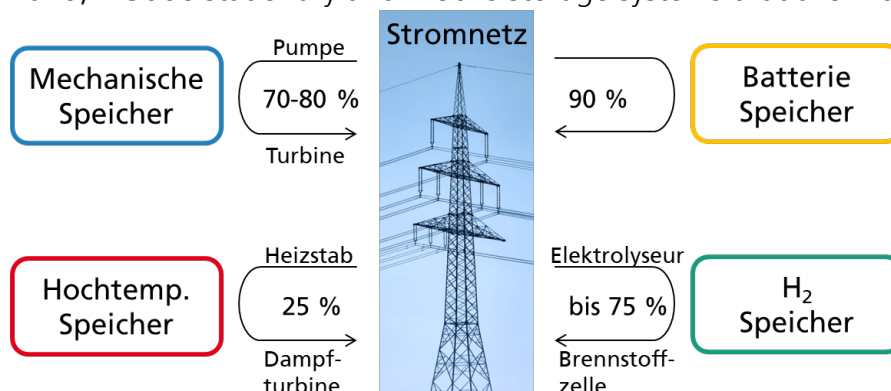


Figure 52: Technologies for electrical energy storage at efficiencies achievable today.

22.3.8 Grid expansion

Grid-bound, supra-regional energy transport takes place via the extra-high voltage grid and the gas transmission grid. In the course of the energy transition, the gas transmission

grid will increasingly transport renewable gases (hydrogen or methane). Gas transport with reconversion in the target region reduces the need for expansion of the electricity grid.

National grid expansion

The energy flows in an energy system with 100 percent renewables are fundamentally different from the situation at the turn of the millennium. PV is ideally suited for decentralized expansion close to consumption. The need for expansion of electricity and gas transmission lines can be minimized with a spatially distributed development of batteries and converters (e.g., electrolysis, gas power plants, heat pumps) that is also close to consumption. A strong concentration of wind power generation in the north or offshore, on the other hand, leads to a high demand for transmission line capacity (electricity or gas). The production of PV electricity is characterized by a high simultaneity factor. In order to avoid local grid overloads due to generation peaks, battery storage is considered an economically interesting alternative to grid expansion in some places.

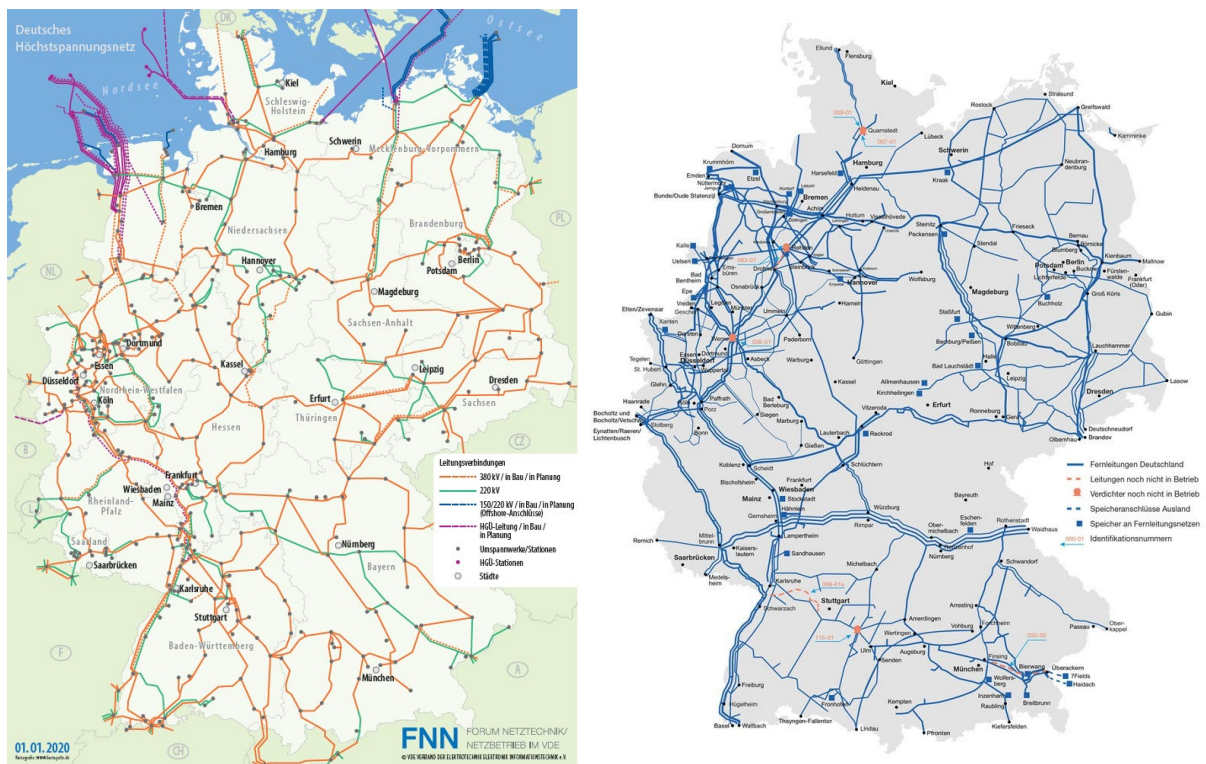


Figure 53: Extra-high voltage grid [VDE] and gas transmission grid [Fern].

Strengthening the European interconnected grid

The German electricity grid is part of the European interconnected grid. Strengthening the cross-border interconnection capacity of currently approx. 20 GW enables a better balancing of volatile PV electricity production via European electricity trading.

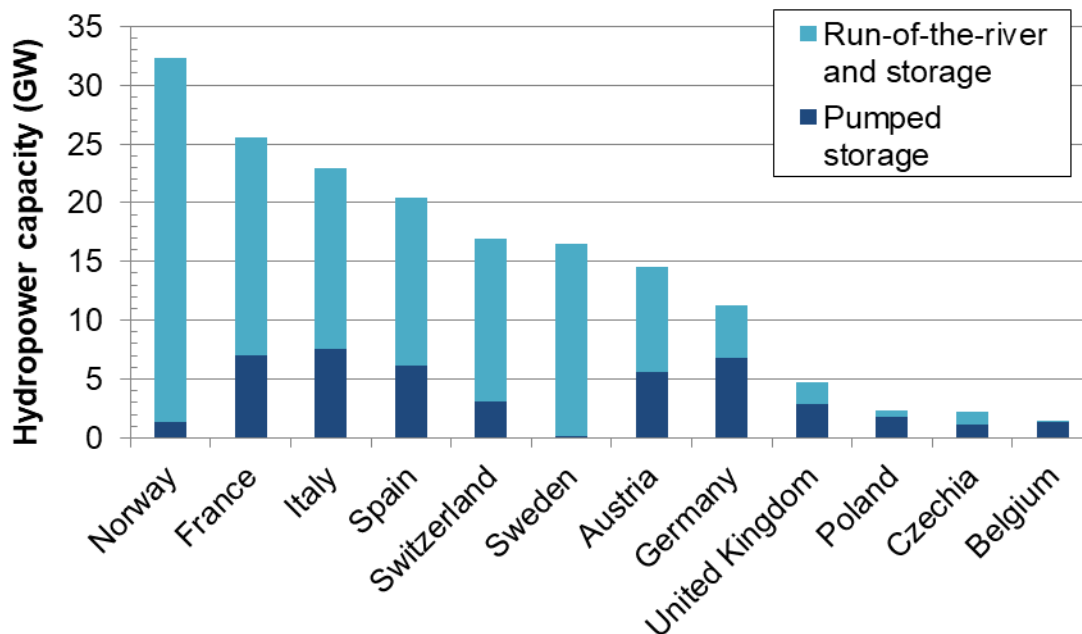


Figure 54: Installed capacity of hydropower plants in neighboring countries, figures from [IHA].

Figure 53 shows the installed capacity of run-of-river and storage hydropower plants as well as pumped storage power plants. Storage power plants can be operated as a complement to PV generation, while pumped storage plants act as efficient power-to-power storage.

22.3.9 Importing energy sources

For some applications, such as aviation and shipping, there are no alternatives to liquid fuels in the longer term. The production of synthetic fuels from PV electricity requires a lot of energy. Liquid fuels can be produced particularly cheaply in sun-rich countries and imported by sea; the logistics chain required for this is in principle in place.

22.3.10 Overview

From today's perspective, an energy system based on 100 percent RE is technically and economically feasible. Figure 54 shows the most important elements connected to the electricity grid, from generation to conversion and storage to consumption. To reduce the need for storage, the demand-side is flexibilized as much as possible.

In the **"battery"** sector (orange), stationary, centralized, or decentralized electrochemical energy storage systems are charged or discharged depending on the residual load. Mobile batteries in electric vehicles primarily serve mobility needs, but can also provide bidirectional support to the grid when connected. In most battery types, converters and storage

devices are structurally integrated, while redox flow batteries have external, independently scalable storage tanks.

In the **“heat”** sector (red), heat pumps, heating elements, and CHP power plants charge the central and decentralized heat storage devices.

In the **“gas”** sector (green), electrolyzers produce hydrogen, some of which is processed into synthetic fuels, and biomass fermenters produce methane from biogenic waste. Combined gas and steam turbines, pure gas turbines and gas engines are used to meet electricity demand. In CHP operation, these processes supply the heat sector.

In the **mechanical** sector (blue), water storage power plants are operated bidirectionally via pumps and turbines, analogously compressed air storage power plants via compressors and turbines.

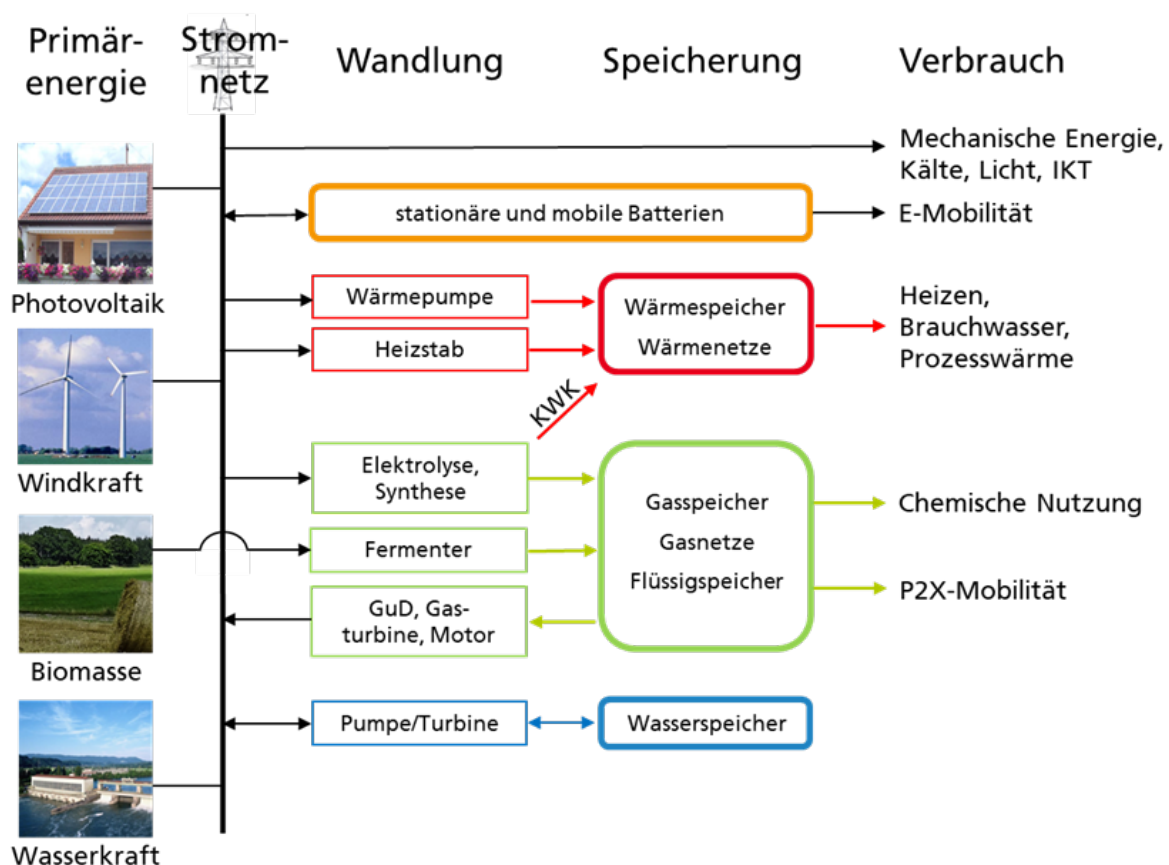


Figure 55: Simplified schematic representation of a renewable energy system with the most important grid-connected components for generation, conversion, storage, and consumption; ICT (Information and Communication Technology), CHP: Combined Heat and Power, CCGT (Combined Cycle Gas Turbine), P2X (Power-to-X, where “X” stands for H₂ synthesis products).

Figure 56 summarizes the most important expansion measures for an energy transition in the electricity sector with coupling to the transport and heat sectors.

Nennleistung	2025	2045 Szenario „Technologieoffen“
1. Primärerzeugung	108 GW _p	330 GW _p <i>PV AC, für AC/DC = 0,7</i>
	78 GW _p	308 GW _p <i>Wind, Land + See</i>
2. Gaskraftwerke	44 GW _p	146 GW _p <i>H₂-ready; auch GuD, KWK</i>
3. Batterien, stat.	17 GW _p	200 GW _p <i>bei Nennleist. = Kapazität / 2 h</i>
4. Wärmepumpen	8 GW _{pel}	75 GW _{pel} <i>bei COP = 3</i>
5. Elektrolyse	0,15 GW _p	68 GW _p <i>Kapazität 9400 TWh H₂ in Salzkavernen</i>

Figure 56: Key power generators, converters and storage units [ISE3], see also Section 22.3.7. Additional estimates and assumptions are in italics.

22.4 Does the energy transition have to wait for federal policy?

No, even if federal policy can make it easier for everyone.

As legislator, the Bundestag determines the framework for the energy transition.

In addition, there are several important actors who can move a lot in their fields of action, even independently of the regulatory framework. Action by these actors also sends clear signals to politicians. Consumers can demand renewable energies and energy efficiency when purchasing electricity and heat, when choosing their means of transport and in their overall consumption. Investors are called upon to invest in the energy transition, whether on their own roofs, in investment companies or funds. Decision-makers in commercial and industrial enterprises or in municipal utilities can examine which measures pay off sustainably and at the same time advance the energy transition. Finally, federal states, cities and municipalities can promote the energy transition through a wide range of measures, from advising stakeholders, providing investment grants and other forms of support for projects, providing land, controlling their own investment decisions, through to mandatory solar installation. Many federal states have introduced so-called solar obligations for residential and commercial construction, public buildings, and large parking lots. The solar obligation sets an example for the responsibility to make maximum use of already developed areas to cover the energy needs of their users, thereby reducing the use of open spaces. The solar obligation reminds builders of the installation of a solar system at an economically optimal time (the roof is being redone, tradespeople are nearby, wiring can be planned). Ultimately, the solar obligation defines a new normality: solar systems are as much a part of the roof as a gutter.

23 Do we need PV production in Germany?

Yes, if we want to avoid new dependencies in energy supply.

As the energy transition progresses, Germany will leave the “fossil” century behind, a century in which we spend up to 90 billion euros a year on oil and gas imports. The prices of these imports are influenced by cartels, the revenues largely finance authoritarian regimes, and often there are political costs as well as monetary ones. The energy transition offers a historic opportunity to break out of this economic and political dependency: the sun shines in Germany too, raw materials for PV production are available and technologies for solar power generation were significantly co-developed in Germany. The German PV sector, with its material producers, mechanical engineering, component manufacturers, R&D facilities, and teaching, still occupies a leading position worldwide despite the slow-down in national expansion. An energy system converted to renewables is based, among other things, on approx. 300–450 GW of installed PV capacity. Annual installations of 12–20 GW are required for the construction and increasingly for the ongoing renewal of this plant park, corresponding to approx. 40 million PV modules at a cost of several billion euros. PV production in Germany offers long-term security of supply with high environmental, social, and quality standards.

Currently, demand for solar modules is almost entirely met by imports, while demand for solar cells is met 100% by imports. This extreme dependence on imports is exacerbated by China's unprecedented monopolization of the international PV market (section 8).

Securing demand for PV products “Made in Germany” is crucial for the sustainable establishment of PV production in Germany. Pure investment subsidies have often only provided an intermittent spark. On the other hand, with secure demand, investments in factories can be made, supply chains can be established, and research and development can be financed. Countries such as India, Turkey, the USA, Mexico, and Austria are creating secure demand by making imports more expensive through import duties, setting lower limits for regional value creation, or offering more attractive conditions in certain market segments for the use of products with regional value creation.

24 Do we need a Renewable Energy Sources Act (EEG)?

Yes, whereby the energy transition legislation as a whole must come into focus.

The current market mechanisms would provide too little incentive for long-term investments in the energy transition if they were not accompanied by an EEG. The main reason is the far too low pricing of **CO₂ emissions across all sectors**, measured against the actual recovery costs and the currently foreseeable follow-up costs of the climate crisis. This de-facto subsidy for fossil energy supply must be dismantled in order to secure the market value of solar power with increasing PV expansion.

A rapid reduction in EU-ETS certificates and a broadening of the scope of emissions, combined with compensation measures, are necessary.

To advance the resilience of the electricity supply and the social acceptance of the energy transition, a broad participation and diversity of actors is beneficial, which can be realized

especially in the decentralized generation of PV electricity. Therefore, the **regulatory hurdles**, especially for small plants, must be lowered or smoothed out by a law such as the EEG. PV power plants of all sizes generally need a **grid connection** to deliver electricity that can neither be consumed nor stored on site. A legal framework must move the grid operator towards uncomplicated connection of even smaller plants.

Furthermore, PV power plants for electricity production beyond self-consumption require a long-term **purchase agreement**. Small private investors in particular can neither control the price risks on the electricity exchange nor keep up with the electricity production costs of large PV power plants. They must be able to sell surplus electricity at secured conditions. A PV power plant built today will soon be competing with PV power plants built in later years, which will supply solar power at the same time as the cost of electricity is expected to fall further. The deflationary effect is compounded by the long-term decline in the market value factor for solar power.

A long-term, guaranteed purchase agreement at fixed prices mobilizes a significant amount of equity capital among small investors. In general, it improves creditworthiness across all investment sizes, thereby opening up access to debt financing for investments. Investing in a residential photovoltaic system can ultimately trigger further steps toward electrification, such as the purchase of electric cars and heat pumps. PV systems with an option for self-consumption also encourage a load shift in electricity consumption that benefits the grid.

To better match the daily profiles of PV electricity production and electricity consumption, **transformation** incentives are necessary for the steps described in Section 22.3. They aim, among other things, at supply-oriented electricity consumption, demand-oriented electricity feed-in, and overall grid efficiency in the electrification of the heat and transport sectors.

Innovative technologies such as integrated PV (Section 11) have slightly higher LCOE in direct comparison with simple PV ground-mounted systems. However, they do not require additional land and create synergies. In order to accelerate their market entry and thus mitigate land conflicts at an early stage, they require targeted start-up support, e.g., in the form of dedicated feed-in tariffs or tender segments.

25 Do PV modules contain toxic substances?

Often yes, which is why PV modules do not belong in the residual waste.

Toxic substances can leach out of modules over long periods of time if the cover glass is broken, the edge seal is damaged, or the module is fragmented [IPV]. The leaching rate depends in particular on the pH value and temperature. Because of the above-mentioned risks, disused PV modules should not be disposed of in residual waste or landfills, and damaged modules should not be left exposed to the weather for long periods of time.

25.1 Wafer-based modules

Modules based on silicon wafers (over 90 percent market share) often still contain lead in the cell metallization (approx. 2 g lead per 60-cell module) and in the solders used (approx. 15 g lead [Li]). Lead, a toxic heavy metal, is soluble in certain strongly acidic or basic environments, and lamination in the module does not permanently prevent mass transfer [IPV]. In wafer-based modules, lead can be completely substituted by harmless materials at low additional cost. Some module manufacturers use backsheet foils that contain fluoropolymers, e.g., polyvinyl fluoride.

25.2 Thin-film modules

Thin-film modules based on CdTe (approx. 5 percent market share) contain cadmium in salt form; it cannot be substituted in this technology. The metallic cadmium and cadmium oxide are classified as very toxic, CdTe as harmful to health. There are alternative thin-film technologies based on amorphous silicon or copper indium selenide (CIS) that contain no or very little Cd.

CIS solar cells contain selenium, which can have a toxic effect, especially as an oxide (e.g., after fires), depending on the amount absorbed. Some manufacturers explain the conformity of their CIS solar modules with the RoHS Directive (Restriction of certain Hazardous Substances) and the EU chemicals regulation REACH (Registration, Evaluation, Authorization and Restriction of Chemicals). For a differentiated assessment, reference is made to independent investigations of the respective module type.

25.3 Solar glass

Common solar modules require a glass with a very low absorption (solar glass quality) as a front panel. Some glass manufacturers purify the molten glass and increase the light transmission by adding antimony (Sb). When this glass is disposed of in landfills, antimony can enter the groundwater. Alternative refining processes without antimony addition are available.

25.4 Take-back and recycling

The recycling of PV modules, including collection, material separation, and reprocessing, is technically feasible, but under current market conditions cannot be financed momentarily by its proceeds alone. A regulatory framework is essential for defining and achieving recycling targets.

In June 2010, PV producers launched a cross-manufacturer recycling system (PV Cycle). The version of the European WEEE Directive (Waste Electrical and Electronic Equipment Directive) that came into force on 13 August 2012 had to be implemented in all EU countries by the end of February 2014. It obliges producers to take back PV modules free of

charge and return them to the recycling system. In October 2015, the Act on the Marketing, Return and Environmentally Sound Disposal of Electrical and Electronic Equipment (Electrical and Electronic Equipment Act — ElektroG) came into force in Germany. It classifies PV modules as large appliances and regulates take-back obligations and financing. The proportion of recovery (collection rate) must be at least 85 percent and the proportion of preparation for reuse and recycling at least 80 percent (recycling rate).

Recycling centers accept small quantities of old solar modules free of charge, and there are industrial processes for separating and recycling the materials. The aluminum frame, cables, and connection boxes are simply removed and the metals recycled. Mechanical separation processes are then used to separate the glass, connectors, silicon, and plastic. The glass fraction is used to manufacture glass containers or glass wool. Trials are currently underway to process the glass on an industrial scale for higher-value melting applications, including new solar glass. The copper is recovered from the solder-coated connectors in the melt. The silicon fraction is reused as a metallurgical raw material. Here, too, attempts are being made to serve higher-value applications, including new solar cells. Silver accounts for less than one-tenth of one percent by weight but makes up almost half of the material value. There are processes for recovering silver in pure form (by the company Solar Materials). Whether these methods are used depends largely on the current price of silver. The plastic fraction is often thermally recycled.

Figure 56 shows the mass fractions of the most important materials in PV modules and the relative material value of the recycled materials. In a white paper, the Deutsche Umwelthilfe (German Environmental Aid) highlights potential for improvement in the reuse and recycling of PV modules [DUH].

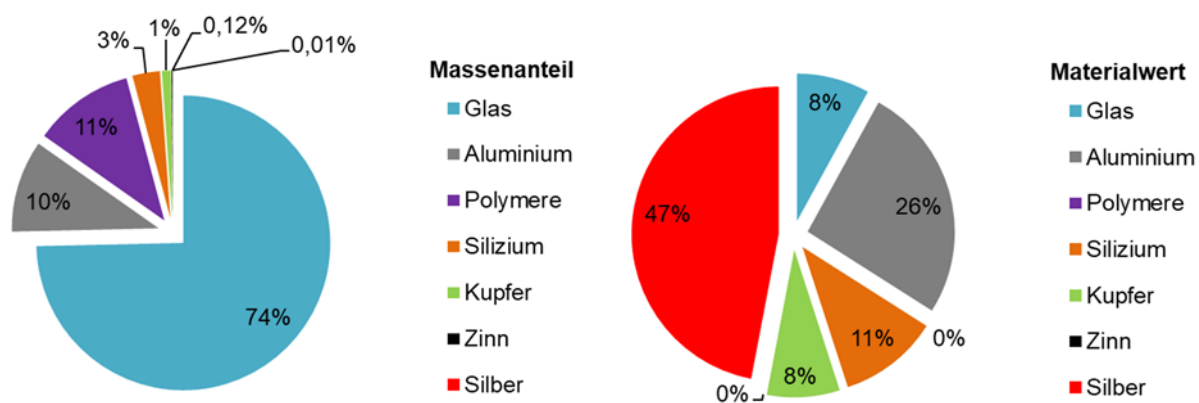


Figure 57: PV Module: Share of mass and value of the recovered materials [Peters]

26 Are there sufficient raw materials available for the production of PV modules?

Wafer-based modules do not require raw materials for which limited availability is foreseeable. The main components by weight are glass, aluminum, polymers and silicon, with silicon and aluminum among the most important components of the earth's crust by weight. Most critical is the consumption of silver for the production of solar cells. The PV

industry consumes about 1500 t of silver per year worldwide, which corresponds to just under 6 percent of the production volume in 2020. The silver for solar cell metallization can be technically substituted by copper to the greatest possible extent, and some manufacturers are already using this technology.

For thin-film modules, which account for about 5 percent of the world market, CdTe technology dominates. There are contradictory statements about the long-term availability of tellurium and indium for CdTe modules.

27 Do PV systems increase the risk of fire?

27.1 Can defective PV systems cause a fire?

Yes, like all electrical installations.

Certain defects in current-carrying components of a PV system can lead to the formation of arcs or critical “hotspots” in individual solar cells. If there is flammable material in the immediate vicinity, for example roofing felt or wood, a fire can occur. The source current characteristics of the solar cells can stabilize a fault current compared to alternating current installations. The current can only be stopped by interrupting the circuit or shading all modules. For this reason, PV systems must be installed with special care.

Particularly with roof-integrated PV systems, the close proximity of solar cells and cables to combustible materials requires special care. Some cases have occurred in which the combination of these factors led to a fire. The source of the fire was usually faulty cabling and connections.

Here is a quote from a press release by Fraunhofer ISE from 2013: “Compliance with the existing rules by qualified specialists is the best fire protection. 0.006 percent of photovoltaic systems have caused a fire with major damage to date. In the last 20 years, there have been 350 fires in which the solar system was involved, in 120 of which it was the cause of the fire. In 75 cases the damage was greater, in 10 of these cases a building burned down.

Important characteristics of photovoltaic systems: they operate on direct current, and standard PV modules cannot simply be turned off. As long as light falls on the modules, they generate an electrical voltage. If, for example, a substandard or improperly installed connector comes loose during operation, the current flow can be maintained via an arc, which, in the worst case, can trigger a fire. On the European market, many manufacturers now offer inverters with built-in arc detection. In the event of a fault, they interrupt the direct current before a fire can start.

Compared to other technical systems, photovoltaic systems do not pose a noticeably higher fire risk. There are also sufficient rules for electrical safety — it is important that they are observed. Fires often occur when inexperienced installation crews install systems on a piecework basis. If the solar plugs are attached with combination pliers instead of special tools, or if incompatible plugs are used, then the weak spot is pre-programmed. System operators must not save money in the wrong place.

In addition to technical improvements, it is therefore also important to have control regulations. At present, the installer of a system can confirm that it has been properly installed. One recommendation of the experts is therefore to make acceptance by an independent third-party mandatory. It is also under discussion to prescribe a recurring safety inspection for private photovoltaic systems, as is mandatory for commercial systems every four years.” [ISE6]

27.2 Do PV systems endanger firefighters?

Yes, but this is true for many live lines.

When fighting fires from the outside, a minimum distance of a few meters protects firefighters from electric shocks; this safety distance is generally given for roof systems. The greatest risk for firefighters when fighting fires from the inside arises when they enter rooms where live, scorched cables of the PV system come into contact with water or the firefighting force itself. To reduce this risk, the industry is working on emergency switches that disconnect the modules from the descending DC cable via safety relays while they are still near the roof.

So far, no firefighters in Germany have been injured fighting fires caused by PV electricity. A case report that went through the press had confused solar thermal collectors with PV modules. No PV system had been installed on the house in question at all. “Initial uncertainties could be eliminated through comprehensive training measures for the fire brigades. As with any electrical installation, depending on the type of jet, water can also be used to safely extinguish photovoltaic systems from a distance of one to five meters. All claims that the fire brigade did not extinguish a burning residential building because of photovoltaics turned out to be false in the research carried out so far.” [ISE6]

27.3 Do PV modules hinder direct firefighting access via the roof?

Yes, the second “roof skin” created by the PV modules impedes the success of the extinguisher because the water simply runs off. From the fire brigade’s point of view, however, an object exposed to fire in this way can usually no longer be saved, i.e., the damage is already largely present and irreversible even before the PV system impedes the extinguishing activity.

27.4 Do toxic emissions occur when PV modules burn?

With regard to CdTe modules, a dispersion calculation by the Bavarian State Office for the Environment states that in the event of a fire, a serious danger to the surrounding neighborhood and the general public can be safely ruled out [LFU1]. For CIS modules, reference is made to independent investigations of the respective product.

In the case of wafer-based modules, the backsheet foils may contain fluoropolymers, which are not toxic themselves but can decompose at high temperatures in the event of

a fire. In a study, the Bavarian State Office for the Environment concludes that when plastics containing fluorine burn, the hazard potential is not largely determined by hydrogen fluoride, but by the other fire gases [LFU2].

28 Appendix: Technical terms

28.1 Module efficiency

Unless otherwise stated, module efficiency refers to nominal efficiency. It is determined under standardized conditions ("STC", standard test conditions) as the ratio of the electrical power output to the irradiated power on the module surface. The standard conditions stipulate a module temperature of 25° C, vertical irradiation of 1000 W/m² and a specific irradiation spectrum. In real operation, the conditions usually deviate significantly, so that in many cases the STC efficiency is not achieved.

28.2 Nominal power of a PV power plant

The nominal power of a power plant [kW_p, MW_p] is the idealized DC power of the module array under STC conditions, i.e. the product of the generator area, standard irradiation (1000 W/m²) and the nominal efficiency of the modules. During operation, the generation capacity of a PV power plant is usually below 70 percent of its nominal capacity.

28.3 Annual full load hours (annual full operating hours)

Number of hours a PV power plant would need to generate at its nominal capacity in order to reach its actual annual yield with frequent partial load operation and energy standstill at night (see Section 15.4).

28.4 Annual utilization rate (capacity factor)

Ratio of annual full load hours and the 8,760 hours in a 365-day-year.

28.5 Specific yield

The specific yield [kWh/kW_p] of a PV power plant is the ratio of the user yield (alternating current yield) over a certain period of time, often a year, and nominal power. The specific annual yield corresponds to the number of full load hours.

28.6 System efficiency

The system efficiency of a PV system is the ratio of the useful yield (alternating current yield) and the total irradiation on the generator surface. The nominal module efficiency is included in the system efficiency.

28.7 Performance Ratio

The term “performance ratio (PR)” refers to the ratio of the useful yield (alternating current yield) and the idealized yield (product of the total irradiation on the generator surface and the nominal module efficiency) of a system over a certain period of time, usually one year. PR is often used to compare the efficiency of grid-connected PV systems at different locations with various PV technologies.

New, carefully planned systems achieve annual PR values between 80 and 90 percent.

28.8 Base load, medium load, peak load, grid load and residual load

“The power demand fluctuates depending on the time of day. As a rule, maxima occur during the day and the minimum at night between 0 and 6 o’clock. The course of the power demand is described as a load curve or load pattern. In classical energy technology, the load curve is divided into three areas:

- (i) The base load
- (ii) The medium load
- (iii) The peak load

The base load describes the load band that is almost constant over 24 hours. It is covered by so-called base load power plants such as nuclear power plants, lignite-fired power plants and currently also run-of-river power plants.

The medium load describes predictable, closed power blocks that cover most of the daily demand in addition to the base load. The medium load is covered by so-called medium load power plants such as hard coal-fired power plants and methane-fired combined cycle power plants. Oil-fired power plants are also used in rare cases. The peak load covers the remaining power demand, which is usually the daily maximum. The peak load is covered by so-called peak-load power plants such as gas turbine power plants and pumped-storage power plants. These can be run at nominal power within a very short time and thus compensate for load fluctuations and cover peak loads [ISET].

The residual load results from the net load minus the feed-in from renewable energies.

28.9 Electricity generation and consumption

Figure 57 shows the energy path from the primary energy source, e.g., solar radiation (irradiance [W/m^2]), wind or natural gas (energy density during combustion [J/kg]), to the useful energy that is important to the end user.

Gross electricity generation is the total amount of electricity produced, equal to the difference between primary energy consumption and conversion losses in electricity production. Large gas turbines show conversion losses of 60–65 percent. PV power plants show conversion losses of 80–85 percent, with virtually free and unlimited primary energy. **Net electricity generation** (not shown in the figure) is the difference between

gross electricity generation and self-consumption. **Gross electricity consumption** equals gross electricity generation minus import balance and storage. Gross electricity consumption accounts for the power plants' self-consumption. The self-consumption of fossil and nuclear power plants is about 7 percent of their gross power generation; for PV-only power plants it is marginal. Battery storage systems are increasingly used in connection with PV power plants ("behind-the-meter"). Their storage losses are included in the self-consumption of such PV-battery power plants. Battery storage systems that only exchange energy with the power grid ("front-of-meter") are balanced in the same way as pumped-storage power plants. Storage losses of Li-ion batteries amount to approx. 5–10 percent of the amount of electricity stored, plus the losses in the battery management system. For pumped-storage power plants it is around 25 percent. When hydrogen is used to store electricity using stationary electrolyzers and fuel cells, for example, the losses are in the order of 50 percent.

Grid losses, especially line and transformer losses, amount to just under 6 percent in the German electricity grid. The decentralized nature of PV installations reduces grid losses for PV electricity.

The amount of electricity purchased from the end user is the net consumption (final energy). The efficiency of the devices determines the conversion losses to the final useful energy, e.g., power or light.

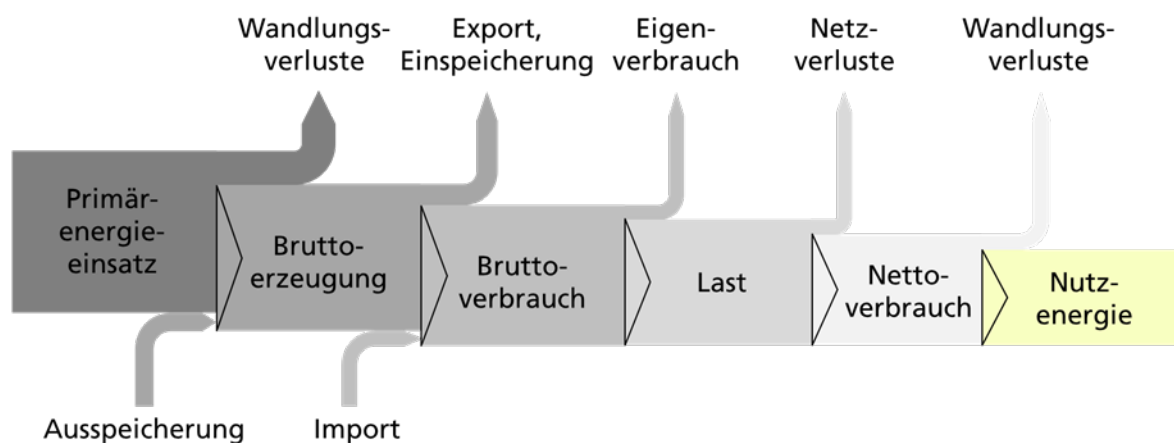


Figure 58: Electricity generation and consumption terms.

29 Appendix: Abbreviations

AC	Alternating Current
BEV	Battery Electric Vehicle
BMU	German Federal Ministry for the Environment, Nature Conservation and Nuclear Safety
BSW	German Solar Industry Association
BMWK	Federal Ministry for Economic Affairs and Climate Action
CCGT	Gas and steam generators
CHP	Combined heat and power, the simultaneous generation of mechanical (ultimately electrical) energy and usable heat
CHP plant	Combined heat and power plant – a plant that uses combustion engines or gas turbines to generate electrical energy and heat
COP	Coefficient of Performance (heat pumps)
DC	Direct Current
EEG	Act on Granting Priority to Renewable Energy Sources (Renewable Energy Sources Act, EEG)
ESC	Energy supply company
GHG	Greenhouse Gas
ICT	Information and communications technology
IEA	International Energy Agency
PHEV	Plug-in hybrid electric vehicle
PV	Photovoltaics
RE	Renewable energy
W_p	Watt "peak", unit for nominal output of a PV module, a module array or a power plant

30 Appendix: Conversion tables [EEBW]

Vorsätze und Vorzeichen

k	Kilo	10 ³	thousand
M	Mega	10 ⁶	million (m)
G	Giga	10 ⁹	billion (bn)
T	Tera	10 ¹²	trillion (tn)
P	Peta	10 ¹⁵	quadrillion (qa)

Conversions

		PJ	GWh	Mio. t	tce	Mio. t	toe
1 PJ	petajoule	1	277,78		0,034	0,024	
1 GWh	gigawatt hours	0,0036	1		0,00012	0,000086	
1 Mio. t	tce million tonnes of coal equivalent	29,31	8.141	1		0,70	
1 Mio. t	toe million tonnes of oil equivalent	41,87	11.630		1,43		1

Typical Fuel Properties

	Density	Heating value	Heating value	Heating value	Heating value
	[kg/l]	[kWh/kg]	[kWh/l]	[MJ/kg]	[MJ/l]
Biodiesel	0,88	10,3	9,1	37,1	32,6
Bioethanol	0,79	7,4	5,9	26,7	21,1
Rapeseed oil	0,92	10,4	9,6	37,6	34,6
Diesel	0,84	12,0	10,0	43,1	35,9
Petrol	0,76	12,2	9,0	43,9	32,5

Typical Properties of Solid and Gaseous Energy Yields

	Density	Heating value	Heating value	Heating value	Heating value
	[kg/l] bzw. [kg/m ³]	[kWh/kg]	[kWh/l] bzw. [kWh/m ³]	[MJ/kg]	[MJ/l] bzw. [MJ/m ³]
Hard coal	-	8,3 - 10,6	-	30,0 - 38,1	-
Brown coal	-	2,6 - 6,2	-	9,2 - 22,2	-
Natural gas H (in m ³)	0,76	11,6	8,8	41,7	31,7
Heating oil EL	0,86	11,9	10,2	42,8	36,8
Biogas (in m ³)	1,20	4,2 - 6,3	5,0 - 7,5	15,0 - 22,5	18,0 - 27,0
Wooden pellets	0,65	4,9 - 5,4	3,2 - 3,5	17,5 - 19,5	11,4 - 12,7

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